CHAPTER

29

HYDRAULIC POWER

CHAPTER 29
HYDRAULIC POWER

Subject/Page	Date	COC
29-EFFECTIVE PAGES		
1 thru 2	Jun 15/2026	
29-HOW TO USE THE FIM		
1	Feb 15/2013	
2	Feb 15/2013	
3	Feb 15/2013	
4	Feb 15/2013	
5	Feb 15/2013	
6	Feb 15/2013	
29-FAULT CODE INDEX		
101	Feb 15/2013	
102	Feb 15/2013	
29-MAINT MSG INDEX		
101	Sep 15/2023	
102	Sep 15/2023	
103	Feb 15/2013	
104	BLANK	
29-10 TASKS		
201	Oct 15/2017	
202	Oct 15/2017	
203	Oct 15/2017	
204	Feb 15/2023	
205	Feb 15/2023	
206	Feb 15/2023	
207	Feb 15/2023	
208	Feb 15/2023	
209	Feb 15/2023	
210	Feb 15/2023	
211	Feb 15/2023	
212	Feb 15/2023	
213	Feb 15/2023	
214	Feb 15/2026	
215	Feb 15/2026	
216	Feb 15/2026	
217	Oct 15/2023	
218	Feb 15/2026	
219	Feb 15/2026	
220	Feb 15/2026	
29-10 TASKS (cont)		
221	Feb 15/2026	
222	Oct 15/2023	
223	Oct 15/2023	
224	Oct 15/2023	
225	Oct 15/2023	
226	Oct 15/2023	
227	Oct 15/2023	
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243	Oct 15/2025	
244	Feb 15/2026	
245	Feb 15/2026	
246	Feb 15/2026	
247	Feb 15/2026	
248	BLANK	
29-11 TASKS		
201	Jun 15/2013	
202	Feb 15/2024	
203	Jun 15/2018	
204	Oct 15/2020	
205	Oct 15/2020	
206	BLANK	
29-20 TASKS		
201	Oct 15/2017	
29-20 TASKS (cont)		
202	Oct 15/2017	
203	Feb 15/2013	
204	Feb 15/2013	
205	Oct 15/2017	
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207	Oct 15/2017	
208	Oct 15/2017	
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211	Feb 15/2026	
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214	Oct 15/2017	
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225	Oct 15/2017	
226	Oct 15/2017	
227	Feb 15/2026	
228	Sep 15/2023	
229	Feb 15/2020	
230	BLANK	
29-30 TASKS		
201	Oct 15/2017	
202	Jun 15/2021	
203	Jun 15/2021	
204	Oct 15/2017	
205	Jun 15/2021	
206	Jun 15/2021	

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

29-EFFECTIVE PAGES

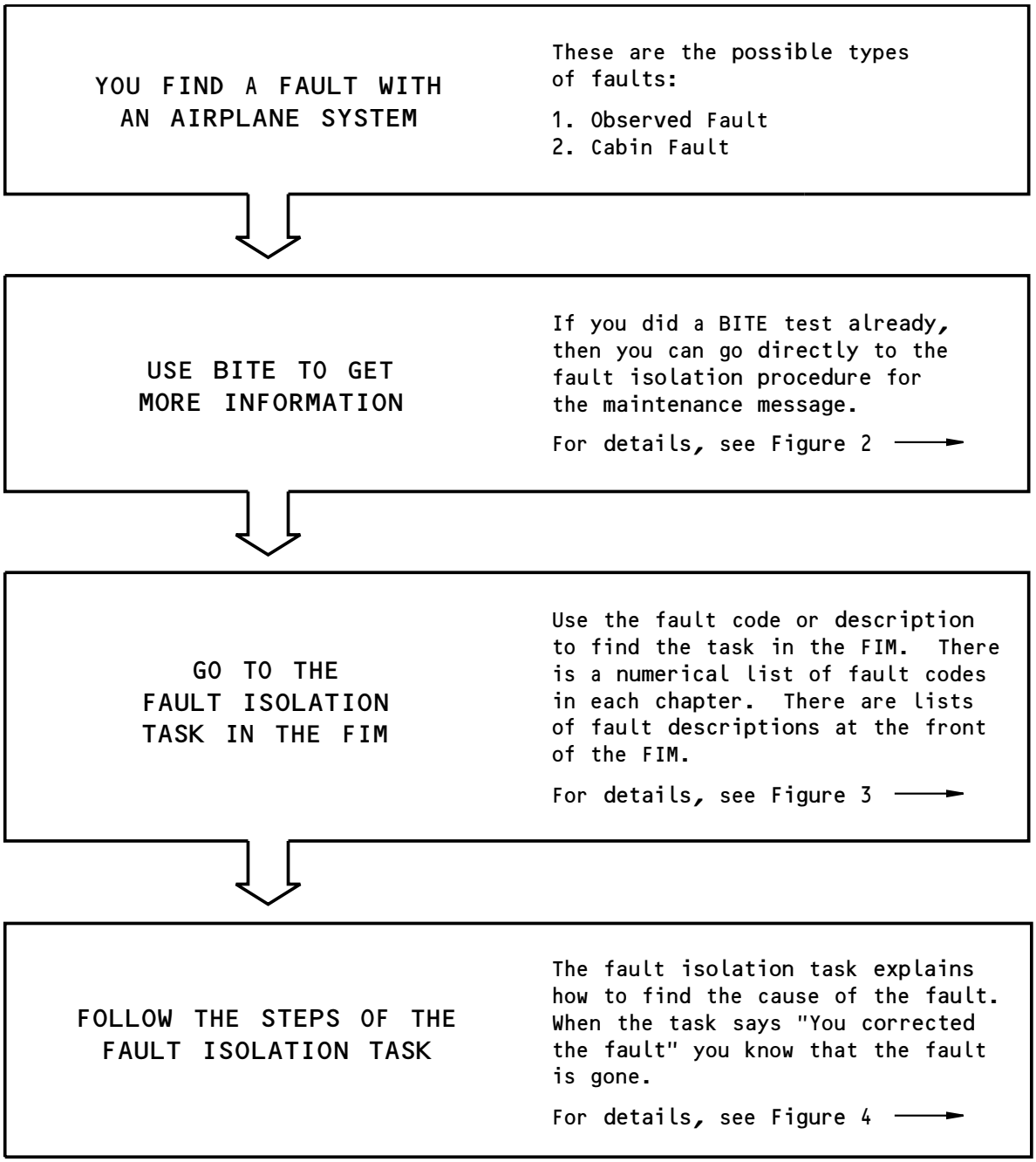
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Subject/Page	Date	COC	Subject/Page	Date	COC	Subject/Page	Date	COC
29-30 TASKS (cont)								
207	Jun 15/2024							
208	Oct 15/2025							
209	Oct 15/2025							
210	Jun 15/2024							
211	Oct 15/2025							
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238	BLANK							

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29-EFFECTIVE PAGES

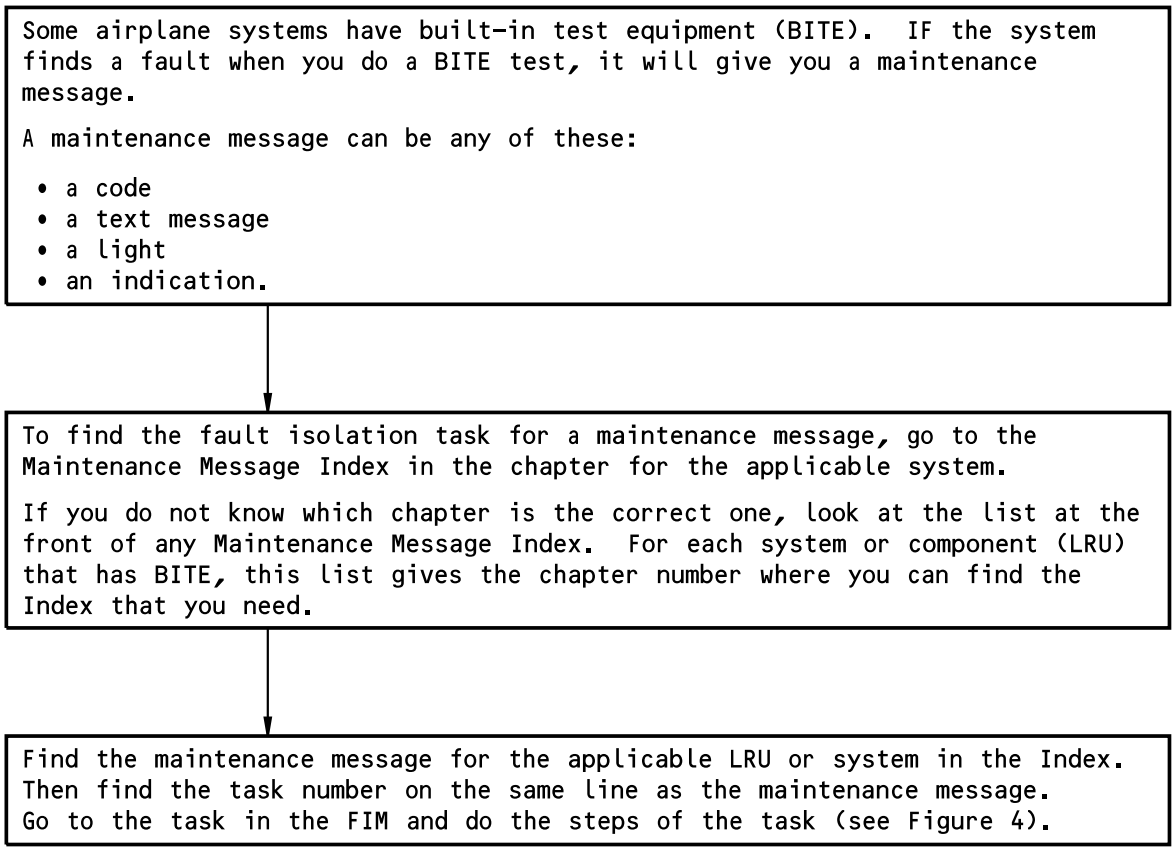
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G04902 S0000148576_V1

Basic Fault Isolation Process
Figure 1

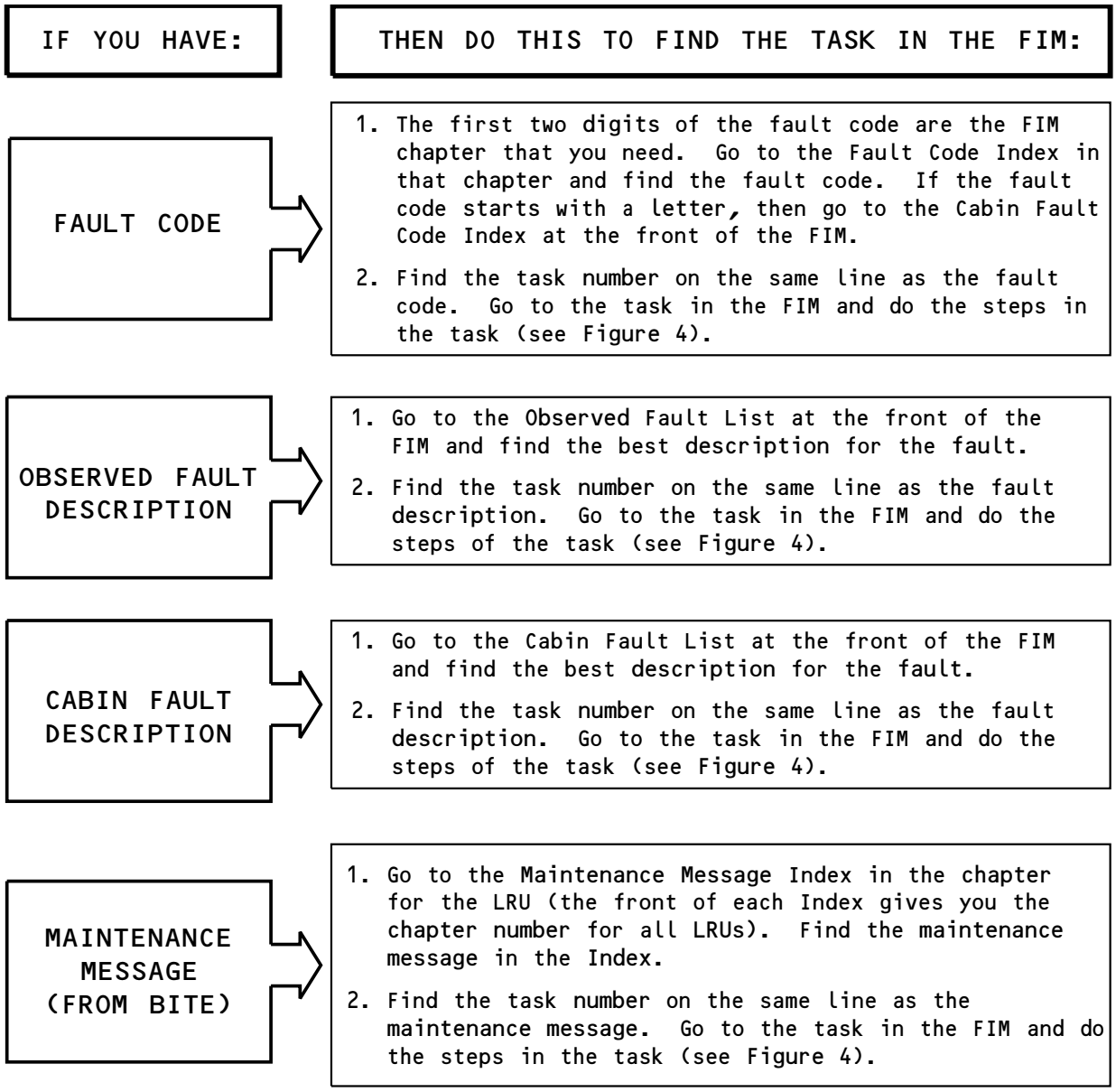
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G04950 S0000148578_V1

Getting Fault Information from BITE
Figure 2

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G04979 S0000148579_V2

Finding the Fault Isolation Task in the FIM
Figure 3

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ASSUMED CONDITIONS AT START OF TASK

- External electrical power is ON
- Hydraulic power and pneumatic power are OFF
- Engines are shut down
- No equipment in the system is deactivated

POSSIBLE CAUSES

- The list of possible causes has the most likely cause first and the least likely cause last.
- You can use the maintenance records of your airline to determine if the fault occurred before. Compare the list of possible causes to the past maintenance actions. This will help prevent repetition of the same maintenance actions.

INITIAL EVALUATION PARAGRAPH

- The primary purpose of the Initial Evaluation paragraph at the start of the task is to help you find out if you can detect the fault right now:
  - If you cannot detect the fault right now, then the task cannot isolate the fault and the Initial Evaluation paragraph will say that there was an intermittent fault.
  - If you have an intermittent fault, you must use your judgement (and follow your airline's policy) to decide which maintenance action to take. Then monitor the airplane to see if the fault happens again on subsequent flights.
- The Initial Evaluation paragraph can also help you find out which Fault Isolation Procedure to use to isolate and correct the fault.

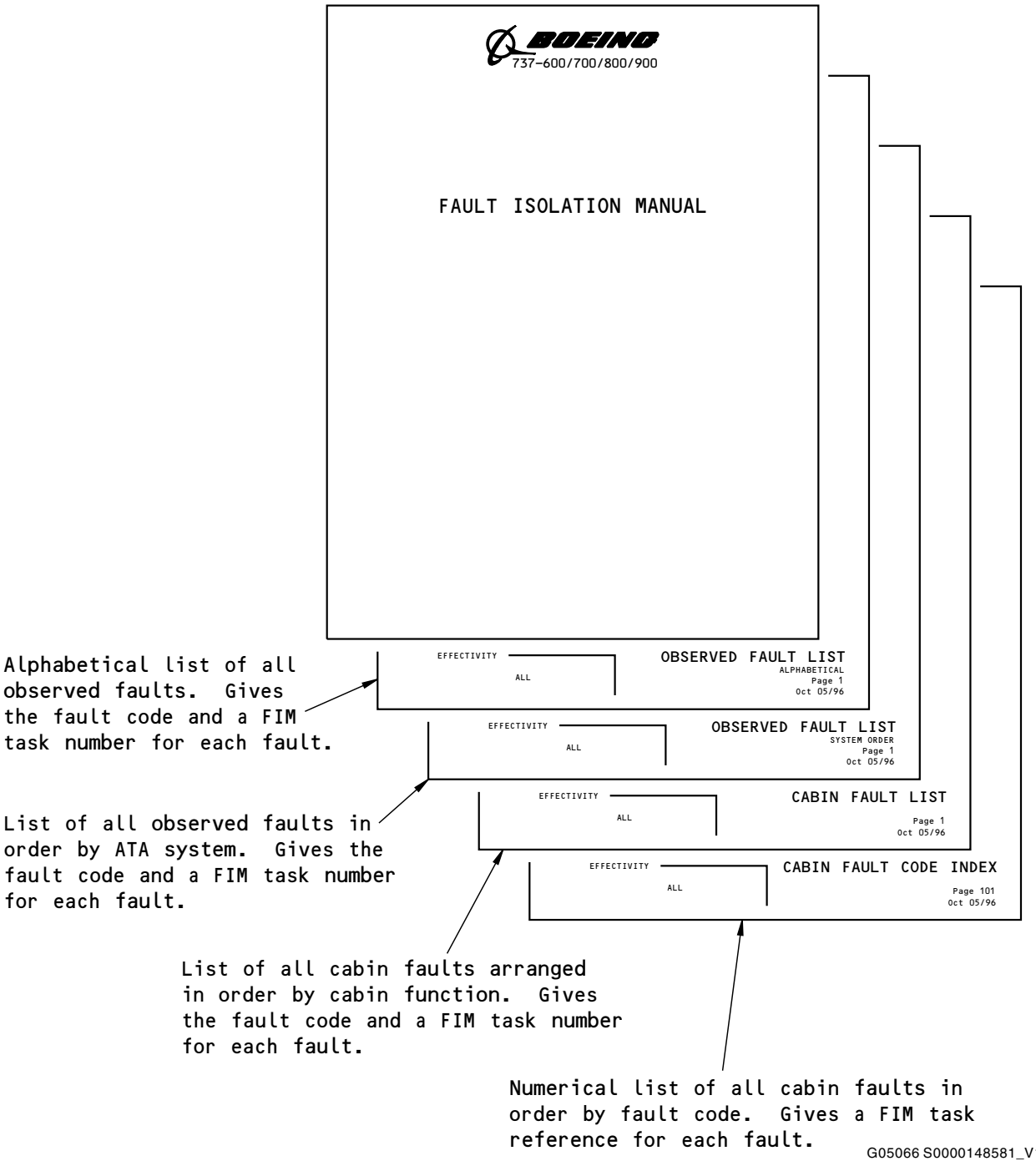
FAULT ISOLATION STEPS

- The FIM task steps are presented in a specified order. The "If... then" statements will guide you along a logical path. But if you do not plan to follow the FIM task exactly, make sure that you read it before you start to isolate the fault. Some FIM procedures start with important steps that have an effect on the other steps in the procedure.
- When you are at the endpoint of the path, the step says "...you corrected the fault." Complete the step and exit the procedure.

G05009 S0000148580_V3

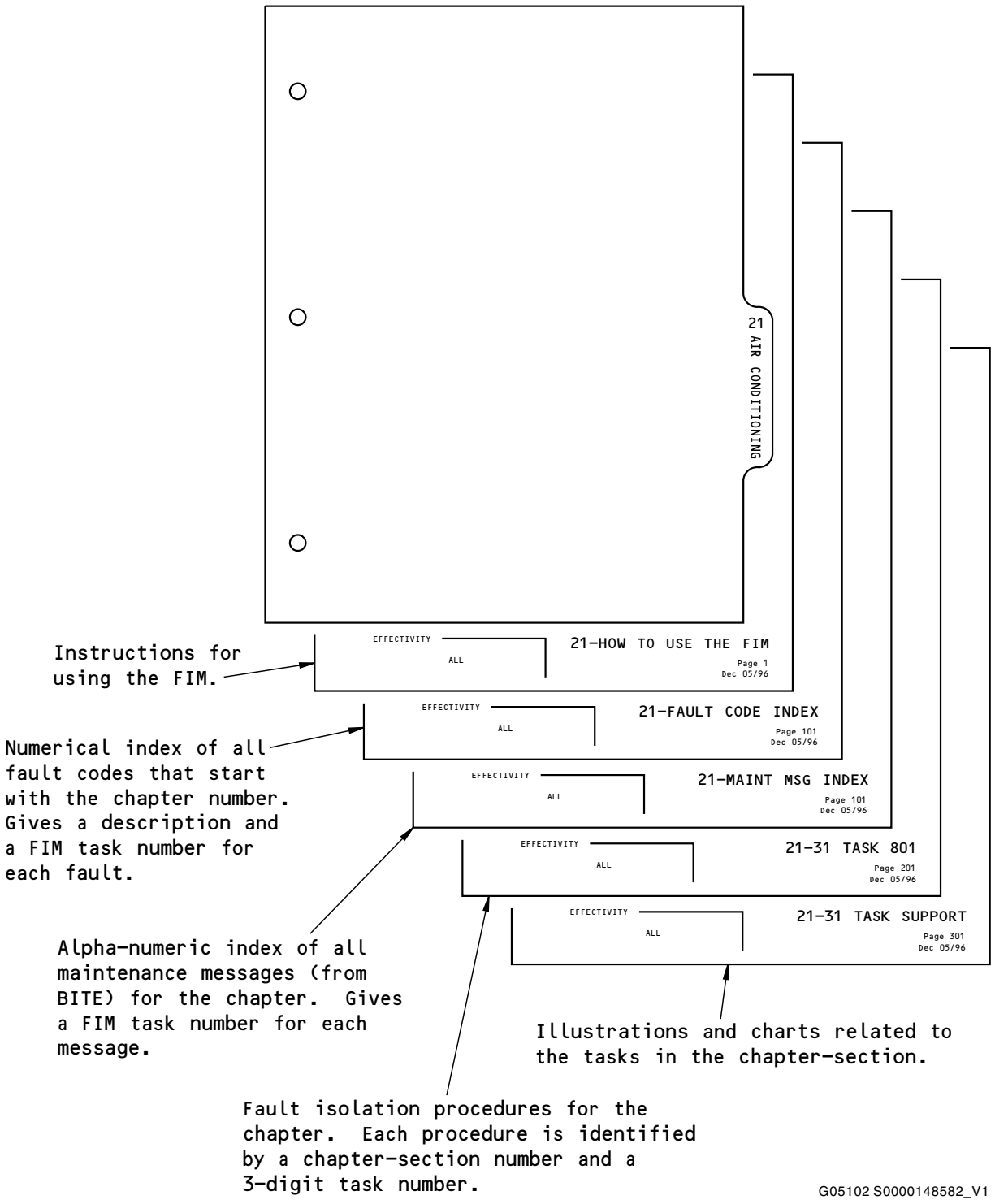
Doing the Fault Isolation Task
Figure 4

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Subjects at Front of FIM
Figure 5

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Subjects in Each FIM Chapter
Figure 6

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
291 010 51	ENG 1 LOW PRESSURE light: light off when the electric motor-driven pump is operated.	29-10 TASK 801
291 010 52	ENG 2 LOW PRESSURE light: light off when the electric motor-driven pump is operated.	29-10 TASK 802
291 020 00	Electrical power: power to the electric motor-driven pump is interrupted during normal operation - system A.	29-10 TASK 803
291 025 00	Electrical power: power to the electric motor-driven pump is interrupted during normal operation - system B.	29-10 TASK 804
291 040 00	Hydraulic fluid: fluid leakage.	29-10 TASK 812
291 050 00	Hydraulic fluid: Comes out of the reservoir vent line.	29-10 TASK 805
291 060 00	Hydraulic fluid: Comes out of the reservoir vent line drain mast.	29-10 TASK 805
291 070 00	Hydraulic fluid: Transfers from system A to system B.	29-10 TASK 806
291 080 00	Hydraulic fluid: Transfers from system B to system A.	29-10 TASK 807
291 090 00	Hydraulic pressure indication: above 3200 psi - system A.	29-10 TASK 808
291 100 00	Hydraulic pressure indication: above 3200 psi - system B.	29-10 TASK 809
291 110 00	Hydraulic pressure indication: Below 2800 psi - system A.	29-10 TASK 810
291 120 00	Hydraulic pressure indication: Below 2800 psi - system B.	29-10 TASK 811
291 131 00	Hydraulic pressure indication: momentarily below 2900 psi - system A.	29-10 TASK 813
291 132 00	Hydraulic pressure indication: momentarily below 2900 psi - system B.	29-10 TASK 813
291 133 00	Hydraulic pressure indication: momentarily below 2900 psi - systems A and B.	29-10 TASK 813
291 201 51	Fire handle in the pulled position while engine turns - engine 1.	29-11 TASK 801
291 201 52	Fire handle in the pulled position while engine turns - engine 2.	29-11 TASK 801
292 010 00	Standby hydraulic pump: does not operate.	29-20 TASK 801
292 040 00	PTU: does not operate.	29-20 TASK 804
293 020 00	Hydraulic pressure indication: Zero - system A.	29-30 TASK 803
293 025 00	Hydraulic pressure indication: Zero - system B.	29-30 TASK 804
293 030 00	ELEC 1 OVERHEAT light: light on.	29-30 TASK 805
293 040 00	ELEC 2 OVERHEAT light: light on.	29-30 TASK 806
293 050 00	Hydraulic fluid quantity indication: error or blank - system A.	31-62 TASK 801
293 060 00	Hydraulic fluid quantity indication: error or blank - system B.	31-62 TASK 801
293 070 00	Hydraulic fluid quantity indication: flight compartment display and hydraulic reservoir gauge do not read the same - system A.	31-62 TASK 801
293 080 00	Hydraulic fluid quantity indication: flight compartment display and hydraulic reservoir gauge do not read the same - system B.	31-62 TASK 801

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
293 090 00	Hydraulic fluid quantity indication: flight compartment display shows zero, reservoir has hydraulic fluid - system A.	31-62 TASK 801
293 100 00	Hydraulic fluid quantity indication: flight compartment display shows zero, reservoir has hydraulic fluid - system B.	31-62 TASK 801
293 110 00	Hydraulic fluid quantity indication: flight compartment display moves erratically or jumps between readings - system A.	31-62 TASK 801
293 120 00	Hydraulic fluid quantity indication: flight compartment display moves erratically or jumps between readings - system B.	31-62 TASK 801
293 130 00	Hydraulic fluid quantity indication: Shows RF (76%) - system A.	29-30 TASK 815
293 140 00	Hydraulic fluid quantity indication: Shows RF (76%) - system B.	29-30 TASK 815
293 150 00	Hydraulic fluid quantity indication: does not return to the same level after operation of a "heavy use" system.	29-30 TASK 808
293 160 00	Hydraulic reservoir gauge: Shows zero or below, reservoir has hydraulic fluid - system A.	29-30 TASK 808
293 170 00	Hydraulic reservoir gauge: Shows zero or below, reservoir has hydraulic fluid - system B.	29-30 TASK 808
293 180 00	STANDBY HYD LOW QUANTITY light: light on.	29-30 TASK 809
293 190 00	ELEC 1 LOW PRESSURE light: light on.	29-30 TASK 810
293 200 00	ELEC 2 LOW PRESSURE light: light on.	29-30 TASK 811
293 210 51	ENG 1 LOW PRESSURE light: light on.	29-30 TASK 812
293 210 52	ENG 2 LOW PRESSURE light: light on.	29-30 TASK 813
293 220 00	STANDBY HYD LOW PRESSURE light: light on.	29-30 TASK 814

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LRU/SYSTEM	SHORT NAME	CHAPTER
Air Data Inertial Reference System	ADIRS	34
Air Traffic Controller Transponder - 1 (Left)	ATC XPDR - 1 (L)	34
Air Traffic Controller Transponder - 2 (Right)	ATC XPDR - 2 (R)	34
Airborne Vibration Monitor System Signal Conditioner	AVM SIG COND	77
Antiskid Control Unit	ANTISKID	32
Attendant Control Panel	ACP	23
Automatic Direction Finder Receiver - 1	ADF RECVR - 1	34
Automatic Direction Finder Receiver - 2	ADF RECVR - 2	34
Autothrottle Computer	A/T COMPUTER	22
Auxiliary Power Unit	APU	49
Auxiliary Power Unit Generator Control Unit	APU GCU	24
Bus Power Control Unit	BPCU	24
Cabin Pressure Controller	CAB PRESS CON	21
Cabin Temperature Controller	CAB TEMP CONT	21
Cargo Electronic Unit - Lower Aft	CEU - LWR AFT	26
Cargo Electronic Unit - Lower Forward	CEU - LWR FWD	26
Cargo Electronic Unit - Main Aft	CEU - MAIN AFT	26
Cargo Electronic Unit - Main Forward	CEU - MAIN FWD	26
Cargo Fire Control Panel	CFCP	26
Common Display System	CDS	31
Compartment Overheat Detection Control Module	WING/BODY OHT	26
Digital Flight Control System	DFCS	22
Distance Measurement Equipment Interrogator	DME INTRROGTR	34
Electrical Meters, Battery, and Galley Power Module	P5-13	24
Electronic Engine Controller - 1	ENGINE - 1	73
Electronic Engine Controller - 2	ENGINE - 2	73
Emergency Locator Transmitter	ELT	23
Engine Accessory Unit	EAU	78
Engine Accessory Unit/TR DEPLOY ENG 1	EAU/TR DPLOY-ENG 1	78
Engine Accessory Unit/TR DEPLOY ENG 2	EAU/TR DPLOY-ENG 2	78
Engine Accessory Unit/TR STOW ENG 1	EAU/TR STOW-ENG 1	78
Engine Accessory Unit/TR STOW ENG 2	EAU/TR STOW-ENG 2	78
Engine and Auxiliary Power Unit Fire Detection Control Module	ENG/APU FIRE	26
Enhanced Digital Flight Control Computer-A	EDFCC-A	22
Enhanced Digital Flight Control System	EDFCS	22

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LRU/SYSTEM	SHORT NAME	CHAPTER
Flap/Slat Electronics Unit	FSEU	27
Flight Data Acquisition Unit	FDAU	31
Flight Management Computer System	FMCS	34
Fuel Quantity Indicating System	FQIS	28
Generator Control Unit - 1	GCU - 1	24
Generator Control Unit - 2	GCU - 2	24
Ground Proximity Computer	GROUND PROX	34
High Frequency Transceiver	HF XCVR	23
Low Limit (35 Degree F) Controller - Left	35 DEG CONT L	21
Low Limit (35 Degree F) Controller - Right	35 DEG CONT R	21
Multi-Mode Receiver	MMR	34
Nitrogen Generation System BITE Display Unit	NGS	47
Pack Flow Temperature Controller	PFTC	21
Pack/Zone Temperature Controller - Left	PACK/ZN CON - L	21
Pack/Zone Temperature Controller - Right	PACK/ZN CON - R	21
Proximity Switch Electronics Unit	PSEU	32
Radio Altimeter Receiver/Transmitter	RADIO ALTIMTR	34
Stall Management Yaw Damper Computer - 1	SMYD - 1	27
Stall Management Yaw Damper Computer - 2	SMYD - 2	27
Supplemental Proximity Switch Electronics Unit	SPSEU	32
Traffic Alert and Collision Avoidance System Computer	TCAS COMPUTER	34
VHF Omnidirectional Ranging Marker Beacon Receiver	VOR/MKR RCVR	34
Very High Frequency Transceiver	VHF XCVR	23
Waste Tank Logic Control Module	WASTE TANK	38
Weather Radar Receiver/Transmitter	WEATHER RADAR	34
Window Heat Control Unit - Left Forward	WHCU - L FWD	30
Window Heat Control Unit - Left Side	WHCU - L SIDE	30
Window Heat Control Unit - Right Forward	WHCU - R FWD	30
Window Heat Control Unit - Right Side	WHCU - R SIDE	30
Window Heat Control Unit 1 - Left Forward and Right Side	WHCU1 - L FWD/R SIDE	30
Window Heat Control Unit 2 - Right Forward and Left Side	WHCU2 - R FWD/L SIDE	30

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PSEU	29-20001 NO LGTV PRI PWR	29-20 TASK 807
PSEU	29-20002 NO LGTV SEC PWR	29-20 TASK 808
PSEU	29-22001 ENG RUN L FAULT	29-20 TASK 809
PSEU	29-22002 HYD QTY B FAULT	29-20 TASK 810
PSEU	29-22003 HYD QTY BITE FLT	29-20 TASK 811
PSEU	29-22004 ALT NOSE STRG FLT	29-20 TASK 812
PSEU	29-22005 LGTV SET FAULT	29-20 TASK 813
PSEU	29-22105 LGTV SET FAULT	29-20 TASK 813
PSEU	29-23001 ENG NOT RUNNING	29-20 TASK 809
PSEU	29-23002 ALT NOSE STRG SEL	29-20 TASK 812
PSEU	29-24001 PRI LGTV FAIL	29-20 TASK 805
PSEU	29-24002 SEC LGTV FAIL	29-20 TASK 806
PSEU	29-24003 LGTV RESET	29-20 TASK 813

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. ENG 1 LOW PRESSURE Light Off When EMDP is Operating - Fault Isolation

A. Description

(1) The ENG 1 LOW PRESSURE light comes on when the engine-driven pump (EDP) for the hydraulic system A is below normal.

(2) This is normal operation of the ENG 1 LOW PRESSURE light:

(a) When the system A EDP operates, the ENG 1 LOW PRESSURE light is off.

(b) When the pressure of the system A EDP is below normal, the ENG 1 LOW PRESSURE light will come on.

(c) When the system A EDP does not operate, the ENG 1 LOW PRESSURE light will come on.

(3) The EDP low pressure switch connects to ground when the EDP pressure decreases to less than 1200 psi.

(4) The EDP check valve prevents hydraulic backflow through the pressure filter. It also isolates the EDP low pressure switch from the system pressure.

B. Possible Causes

(1) Hydraulic system A pressure module EDP check valve

C. Circuit Breaker

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	12	C00318	INDICATOR MASTER DIM SECT 6

D. Related Data

(1) (SSM 29-33-11)

(2) (SSM 33-18-35)

(3) (WDM 29-33-11)

(4) (WDM 33-18-35)

E. Initial Evaluation

(1) Do this check of the ENG 1 LOW PRESSURE light, L5:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system A with the EMDP. To pressurize it, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(b) Make sure the ELEC 2 LOW PRESSURE light is off.

(c) Make sure the ENG 1 LOW PRESSURE light is on.

(d) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) If the ENG 1 LOW PRESSURE light is not on, then do the Fault Isolation Procedure below.

(f) If the ENG 1 LOW PRESSURE light comes on and stays on, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Replace the EDP check valve on the system A pressure module. These are the tasks:

- Hydraulic Systems A and B Pressure Module Check Valve Removal, AMM TASK 29-11-71-000-803
- Hydraulic Systems A and B Pressure Module Check Valve Installation, AMM TASK 29-11-71-400-803

(a) Do this check of the ENG 1 LOW PRESSURE light, L5:

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Pressurize the hydraulic system A with the EMDP. To pressurize it, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

2) Make sure the ENG 1 LOW PRESSURE light is on.

3) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

4) If the ENG 1 LOW PRESSURE light stays on, then you corrected the fault.

——— END OF TASK ———

802. ENG 2 LOW PRESSURE Light Off When EMDP is Operating - Fault Isolation

A. Description

(1) The ENG 2 LOW PRESSURE light comes on when the engine-driven pump (EDP) for the hydraulic system B is below normal.

(2) This is normal operation of the ENG 2 LOW PRESSURE light:

(a) When the system B EDP operates, the ENG 2 LOW PRESSURE light is off.

(b) When the pressure of the system B EDP is below normal, the ENG 2 LOW PRESSURE light will come on.

(c) When the system B EDP not operates, the ENG 2 LOW PRESSURE light will come on.

(3) The EDP low pressure switch connects to ground when the EDP pressure decreases to less than 1200 psi.

(4) The EDP check valve prevents hydraulic backflow through the pressure filter. Then also isolate the EDP low pressure switch from the system pressure.

B. Possible Causes

(1) Hydraulic system B pressure module EDP check valve

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	11	C00317	INDICATOR MASTER DIM SECT 5

D. Related Data

(1) (SSM 29-33-11)

(2) (SSM 33-18-35)

(3) (WDM 29-33-11)

(4) (WDM 33-18-35)

E. Initial Evaluation

(1) Do this check of the ENG 2 LOW PRESSURE light, L6:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system B with the EMDP. To pressurize it, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(b) Make sure the ELEC 1 LOW PRESSURE light is off and ENG 2 LOW PRESSURE light is on.

(c) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(d) If the ENG 2 LOW PRESSURE light is not on, then do the Fault Isolation Procedure below.

(e) If the ENG 2 LOW PRESSURE light comes on and stays on, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Replace the EDP check valve on the system B pressure module. These are the tasks:

- Hydraulic Systems A and B Pressure Module Check Valve Removal, AMM TASK 29-11-71-000-803
- Hydraulic Systems A and B Pressure Module Check Valve Installation, AMM TASK 29-11-71-400-803

(a) Do this check of the ENG 2 LOW PRESSURE light, L6:

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WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Pressurize the hydraulic system B with the EMDP. To pressurize it, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.
2) Make sure the ENG 2 LOW PRESSURE light is on.
3) If the ENG 2 LOW PRESSURE light stays on, then you corrected the fault.
4) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

——— END OF TASK ———

803. Electrical Power to the System A EMDP Problem - Fault Isolation

A. Description

(1) When the system A ELEC 2 switch is ON, 28v DC energizes the system A Electric Motor-Driven Pump (EMDP) relay which closes three contacts. These contacts send 115v AC power to the system A EMDP. When the system A EMDP switch is in the OFF position, the system A EMDP relay has no power and the contacts open to stop the EMDP.

(2) The system A EMDP relay is connected to ground through an internal jumper in the EMDP. This enables control of the EMDP from the hydraulic control panel. An open circuit in this path will disable the relay and prevent electrical power from reaching the EMDP. The ELEC 2 LOW PRESSURE light will come on if the system A EMDP is unable to operate.

(3) The ground fault detector for the system A EMDP is in the P92 power distribution panel in the electrical/electronic (EE) compartment. The trip coil of the ground fault detector energizes when a ground fault occurs.

(4) When the trip coil energizes, the system A EMDP relay has no power and the relay contacts open to stop the 115v AC power to the system A EMDP.

(5) Push the reset switch on the ground fault detector to move the contacts to their normal position.

B. Possible Causes

(1) Wiring
(2) System A EMDP relay, R317
(3) System A EMDP ground fault detector, M1105
(4) Hydraulic Control Module, P5-8
(5) System A EMDP, M1103

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	8	C00767	ELEC HYD PUMP CONTROL SYS A

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	3	C00881	ELEC HYD PUMP SYS A

D. Related Data

(1) (SSM 29-11-12)

(2) (WDM 29-11-12)

E. Initial Evaluation

(1) Do this check of the system A EMDP, M1103:

(a) Make sure that these circuit breakers are closed:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	8	C00767	ELEC HYD PUMP CONTROL SYS A
F	3	C00881	ELEC HYD PUMP SYS A

(b) Push the reset switch on the system A ground fault detector.

NOTE: The system A ground fault detector is in the power distribution panel, P92, in the EE compartment.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(c) Pressurize the hydraulic system A with the EMDP. To pressurize it, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(d) If the electrical power to the system A EMDP is interrupted, then do the Fault Isolation Procedure below.

(e) If the electrical power to the system A EMDP is not interrupted, then there is an intermittent fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

F. Fault Isolation Procedure

(1) Do this check of the system A ground fault detector, M1105:

(a) Disconnect connector D2664 from the system A EMDP, M1103.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION DO NOT RESET GROUND FAULT DETECTOR BEFORE DISCONNECTING ELECTRICAL CONNECTOR FROM PUMP MOTOR, RISK OF FIRE EXISTS.

(b) Push the reset switch on the system A ground fault detector, M1105.

NOTE: The system A ground fault detector is in the power distribution panel, P92, in the EE compartment.

(c) Install a jumper wire between pin 4 and pin 5 of connector D2664.

(d) Do a check for 200 Alternating Current Volts (VAC) between pins 1, 3, and 6 of connector D2664.

(e) If there is not 200 VAC between pins 1, 3, and 6 of connector D2664, then do these steps:

1) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	3	C00881	ELEC HYD PUMP SYS A

2) Remove the jumper wire.

3) Do a check for an open circuit between these pins of connector D2664 for the EMDP and the relay for the system A EMDP, R317:

NOTE: The R317 relay for the system A EMDP is in the power distribution panel, P92, in the EE compartment.

D2664		R317
pin 1	. . . . .	term L3
pin 3	. . . . .	term L2
pin 6	. . . . .	term L1
pin 4	. . . . .	term X2

4) Make sure there is continuity from pin 5 of connector D2664 to ground.

5) If you find a problem with the wiring, then repair the wiring.

6) Make sure there is continuity between pin 4 and pin 5 at the system A EMDP connector.

7) If you do not find continuity between pin 4 and pin 5 at the EMDP connector, then replace the system A EMDP, M1103. These are the tasks:

- Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
- Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001

8) If you do not find a problem with the wiring, do these steps (WDM 29-11-12):

a) Check for 28V DC from terminal X1 of the SYS A EMDP Relay, R317, to ground.

b) If there is 28V DC from terminal X1 to ground, replace the relay.

c) If there is not 28V DC from terminal X1 to ground, do these steps:

<1> Disconnect connector D642 from the hydraulic pumps panel, P5-8.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<2> Check for 28V DC from pin 14 of connector D642 to ground.

<a> If there is 28V DC from pin 14 to ground, replace the hydraulic pumps panel, P5-8. These are the tasks:

- Removal of the Hydraulic Pumps Module, P5-8, AMM TASK 29-11-91-000-801
- Installation of the Hydraulic Pumps Module, P5-8, AMM TASK 29-11-91-400-801

<b> If there is not 28V DC from pin 14 to ground, replace the system A ground fault detector, M1105. Reconnect connector D642 to the hydraulic pumps panel, P5-8.

9) Re-connect connector D2664.

10) Do the Repair Confirmation at the end of this task.

(f) If there is 200 VAC at pins 1, 3, and 6 of connector D2664, then remove the jumper wire and continue.

(2) Replace the system A EMDP, M1103. These are the tasks:

- Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
- Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Put the HYD PUMPS A ELEC 2 switch, on the forward overhead panel P5, in the OFF position.

(2) Do this check of the system A EMDP, M1103:

(a) Make sure that these circuit breakers are closed:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	8	C00767	ELEC HYD PUMP CONTROL SYS A
F	3	C00881	ELEC HYD PUMP SYS A

(b) Push the reset switch on the system A ground fault detector.

NOTE: The system A ground fault detector is in the power distribution panel, P92, in the EE compartment.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(c) Pressurize the hydraulic system A with the EMDP. To pressurize it, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(d) If the electrical power to the system A EMDP is not interrupted, then you corrected the fault. Do this step to complete the task:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

——— END OF TASK ———

804. Electrical Power to the System B EMDP Problem - Fault Isolation

A. Description

(1) When the system B ELEC 1 switch is ON, 28v DC energizes the system B EMDP relay which closes three contacts. These contacts send 115v AC power to the system B EMDP. When the system B EMDP switch is in the OFF position, the system B EMDP relay has no power and the contacts open to stop the EMDP.

(2) The system B EMDP relay is connected to ground through an internal jumper in the EMDP. This enables control of the EMDP from the hydraulic control panel. An open circuit in this path will disable the relay and prevent electrical power from reaching the EMDP. The ELEC 1 LOW PRESSURE light will come on if the system B EMDP is unable to operate.

(3) The ground fault detector for the system B EMDP is in the P91 power distribution panel in the EE compartment. The trip coil of the ground fault detector energizes when a ground fault occurs.

(4) When the trip coil energizes, the system B EMDP relay has no power and the relay contacts open to stop the 115v AC power to the system B EMDP.

(5) Push the reset switch on the ground fault detector to move the contacts to their normal position.

B. Possible Causes

(1) Wiring

(2) System B EMDP relay, R318

(3) System B EMDP ground fault detector, M1106

(4) Hydraulic Control Module, P5-8

(5) System B EMDP, M1104

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	8	C00768	ELEC HYD PUMP CONTROL SYS B
F	3	C00882	ELEC HYD PUMP SYS B

D. Related Data

(1) (SSM 29-11-12)

(2) (WDM 29-11-12)

E. Initial Evaluation

(1) Do this check of the system B EMDP, M1104:

(a) Make sure that these circuit breakers are closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	8	C00768	ELEC HYD PUMP CONTROL SYS B

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
F	3	C00882	ELEC HYD PUMP SYS B

(b) Push the reset switch on the system B ground fault detector.

NOTE: The system B ground fault detector is in the power distribution panel, P91, in the EE compartment.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(c) Pressurize the hydraulic system B with the EMDP. To pressurize it, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(d) If the electrical power to the system B EMDP is interrupted, then do the Fault Isolation Procedure below.

(e) If the electrical power to the system B EMDP is not interrupted, then there is an intermittent fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

F. Fault Isolation Procedure

(1) Do this check of the system B ground fault detector, M1106:

(a) Disconnect connector D2666 from the system B EMDP, M1104.

(b) Push the reset switch on the system B ground fault detector, M1106.

NOTE: The system B ground fault detector is in the power distribution panel, P91, in the EE compartment.

(c) Install a jumper wire between pin 4 and pin 5 of connector D2666.

(d) Do a check for 200 VAC between pins 1, 3, and 6 of connector D2666.

(e) If there is not 200 VAC between pins 1, 3, and 6 of connector D2666, then do these steps:

1) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
F	3	C00882	ELEC HYD PUMP SYS B

2) Remove the jumper wire.

3) Do a check for an open circuit between these pins of connector D2666 for the EMDP and the relay for the system B EMDP, R318:

NOTE: The R318 relay for the system B EMDP is in the power distribution panel, P91, in the EE compartment.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D2666		R318
pin 1	. . . . .	term L1
pin 3	. . . . .	term L2
pin 6	. . . . .	term L3
pin 4	. . . . .	term X2

4) Make sure there is continuity from pin 5 of connector D2666 to ground.
5) If you find a problem with the wiring, then repair the wiring.
6) Make sure there is continuity between pin 4 and pin 5 at the system B EMDP connector.
7) If you do not find continuity between pin 4 and pin 5 at the EMDP connector, then replace the system B EMDP, M1104. These are the tasks:
   - Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
   - Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001
8) If you do not find a problem with the wiring, do these steps (WDM 29-11-12):
   a) Check for 28V DC from terminal X1 of the SYS B EMDP Relay, R318, to ground.
   b) If there is 28V DC from terminal X1 to ground, replace the relay.
   c) If there is not 28V DC from terminal X1 to ground, do these steps:
      <1> Disconnect connector D2486 from the hydraulic pumps panel, P5-8.
      <2> Check for 28V DC from pin 14 of connector D2486 to ground.
         <a> If there is 28V DC from pin 14 to ground, replace the hydraulic pumps panel, P5-8. These are the tasks:
            - Removal of the Hydraulic Pumps Module, P5-8, AMM TASK 29-11-91-000-801
            - Installation of the Hydraulic Pumps Module, P5-8, AMM TASK 29-11-91-400-801
         <b> If there is not 28V DC from pin 14 to ground, replace the system B ground fault detector, M1106. Reconnect connector D2486 to the hydraulic pumps panel, P5-8.
9) Re-connect connector D2666.
10) Do the Repair Confirmation at the end of this task.

(f) If there is 200 VAC between pins 1, 3, and 6 of connector D2666, then remove the jumper wire and continue.

(2) Replace the system B EMDP, M1104. These are the tasks:
- Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
- Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001

(a) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Repair Confirmation

(1) Put the HYD PUMPS B ELEC 1 switch, on the forward overhead panel P5, in the OFF position.

(2) Do this check of the system B EMDP, M1104:

(a) Make sure that these circuit breakers are closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	8	C00768	ELEC HYD PUMP CONTROL SYS B
F	3	C00882	ELEC HYD PUMP SYS B

(b) Push the reset switch on the system B ground fault detector.

NOTE: The system B ground fault detector is in the power distribution panel, P91, in the EE compartment.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(c) Pressurize the hydraulic system B with the EMDP. To pressurize it, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(d) If the electrical power to the system B EMDP is not interrupted, then you corrected the fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

——— END OF TASK ———

805. Hydraulic Fluid Comes Out of the Reservoir Vent Line/Drain Mast - Fault Isolation

A. Description

(1) Hydraulic fluid overflow and excessive servicing can occur under this condition:

(a) Large ambient temperature changes between departure and arrival gates. If the hydraulic reservoirs are filled at the cold location before dispatch back to the warm location, at the warm location the reservoirs would have been overfilled. This can cause fluid to be overflow.

1) If the ground ambient is 20 degrees F or lower upon arrival at the gate, it is recommended to fill the reservoir to just above "REFILL" - the minimum amount to satisfy dispatch requirements (AMM TASK 12-12-00-610-801).

B. Possible Causes

(1) Over-service of Reservoir

(2) Relief valve

(3) Hydraulic fluid quantity transmitter/indicator, T436 or T184

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802 PRE SB 737-29-1106

(4) Reservoir pressurization system

OMR ALL

(5) Hydraulic fluid transfer

C. Fault Isolation Procedure

(1) Do a check of the fluid level in the system A and system B reservoir.

(a) If the reservoir fluid level is more than full, then do these steps:

1) Lower the fluid level. To lower it, do this task: Hydraulic Reservoir Servicing, AMM TASK 12-12-00-610-801.

2) Pressurize the applicable hydraulic reservoir. To pressurize it, do this task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801 or Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807

3) If hydraulic fluid does not come out of the reservoir vent line/drain mast, then you corrected the fault. Do this step to complete this task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(b) If the reservoir level is not more than full, then continue.

(2) Do a check for the relief valve:

(a) Pressurize the applicable hydraulic reservoir. To pressurize it, do this task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801 or Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807.

(b) Listen to the applicable relief valve.

(c) If the applicable relief valve does not open between 60-65 psi, then do these steps:

1) Replace the relief valve.

2) Pressurize the applicable hydraulic reservoir. To pressurize it, do this task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801 or Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807

3) If hydraulic fluid does not come out of the reservoir vent line/drain mast, then you corrected the fault. Do this step to complete this task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(d) If the relief valve opens between 60-65 psi, then continue.

(3) Do a check of the fluid quantity transmitter/indicator on the reservoir. To do the check, do this task: Hydraulic Fluid Quantity Transmitter/Indicator Inspection, AMM TASK 29-33-12-200-801.

(a) If the fluid quantity transmitter/indicator is not satisfactory, then do these steps:

1) Replace the fluid quantity transmitter/indicator, T436 or T184. These are the tasks:

- Hydraulic Fluid Quantity Transmitter/Indicator Removal, AMM TASK 29-33-12-000-801
- Hydraulic Fluid Quantity Transmitter/Indicator Installation, AMM TASK 29-33-12-400-801

2) Pressurize the applicable hydraulic reservoir. To pressurize it, do this task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801 or Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) If hydraulic fluid does not come out of the reservoir vent line/drain mast, then you corrected the fault. Do this step to complete this task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(b) If the fluid quantity transmitter/indicator is satisfactory, then continue.

OMR 701-799, 802 PRE SB 737-29-1106

(4) Do this check of the hydraulic reservoir pressurization module:

(a) Do a check to make sure that the vent caps are not clogged. To check the vent caps, do this task: Vent Cap - Check, AMM TASK 29-09-06-211-801.

(b) If the vent cap was clogged, then do these steps:

1) Pressurize the applicable hydraulic reservoir. To pressurize it, do this task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807

2) If hydraulic fluid does not come out of the reservoir vent line/drain mast, then you corrected the fault. Do this step to complete this task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(c) If the vent cap was not clogged, then continue.

OMR ALL

(5) Do a check for hydraulic fluid transfer:

(a) If there is hydraulic fluid transfer from A to B, then, do this task: Hydraulic Fluid Transfers from System A to System B - Fault Isolation, 29-10 TASK 806.

(b) If there is hydraulic fluid transfer from B to A, then, do this task: Hydraulic Fluid Transfers from System B to System A - Fault Isolation, 29-10 TASK 807.

(c) Pressurize the applicable hydraulic reservoir. To pressurize it, do this task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801 or Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807

(d) If hydraulic fluid does not come out of the reservoir vent line/drain mast, then you corrected the fault. Do this step to complete this task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

——— END OF TASK ———

806. Hydraulic Fluid Transfers from System A to System B - Fault Isolation

A. Description

(1) Hydraulic system design allows fluid transfer to occur from system A to system B under this condition:

(a) When parking brakes are set with system A pump(s) only pressurized and then released with system A depressurized. In this case, fluid is taken from system A when the brakes are set and returned to system B when the brakes are released. Approximately 5 fluid ounces can be transferred per cycle.

1) To prevent hydraulic fluid transfer under the above condition, follow the normal procedures in the pre-flight and post-flight Flight Operations Manual.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Valve Position (in W/W)	Valve Port Identifier	System Connection
Left/Forward	INA	Norm (Sys B)
Left/Forward	INB	Alt (Sys A)
Left/Aft	INA	Alt (Sys A)
Left/Aft	INB	Norm (Sys B)
Right/Forward	INA	Norm (Sys B)
Right/Forward	INB	Alt (Sys A)
Right/Aft	INA	Alt (Sys A)
Right/Aft	INB	Norm (Sys B)

B. Possible Causes

(1) Brake Shuttle Valve (BSV) internal leakage

(2) Left thrust reverser shuttle valve internal leakage.

(3) Landing Gear Transfer Valve (LGTV) internal leakage

C. Fault Isolation Procedure

(1) Do this check for leakage through the BSV:

(a) Make sure that hydraulic power is removed from system A. This is the task:

- Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Pressurize hydraulic system B. This is the task:

- Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(c) Apply brakes fully one time and release.

(d) Remove hydraulic power from hydraulic system B. This is the task:

- Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(e) Remove pressure from the hydraulic system B reservoir only. This is the task:

- Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-802 or Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-808.

OMR 701-799, 802 PRE SB 737-29-1106

(f) Install the lockout pin in the Depressurization Valve for hydraulic system B reservoir.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106

(g) Remove the pneumatic reservoir pressurization line to the system B reservoir.

1) Install a cap on the system B reservoir pressurization line.

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Disconnect the hydraulic lines from the INA ports of the two Forward BSVs.

(i) Disconnect the hydraulic lines from the INB ports of the two Aft BSVs.

WARNING KEEP ALL PERSONNEL AWAY FROM DISCONNECTED HYDRAULIC LINES. PRESSURIZED HYDRAULIC FLUID CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(j) Pressurize the hydraulic system A. This is the task:

- Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(k) Look at the disconnected system A ports of the two Forward BSVs.

(l) Look at the disconnected system B ports of the two Aft BSVs.

(m) If there is leakage at the BSVs, then do these steps:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

2) Replace the applicable BSV that leak. These are the tasks:

- Brake Shuttle Valve Removal, AMM TASK 32-41-95-000-801
- Brake Shuttle Valve Installation, AMM TASK 32-41-95-400-801

OMR 701-799, 802 PRE SB 737-29-1106

3) Remove the lockout pin from the Depressurization Valve for the hydraulic system B reservoir.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106

4) Remove the cap from the system B reservoir pressurization line.

a) Reconnect the reservoir pressurization line to the system B reservoir.

OMR ALL

5) If there is no hydraulic fluid transfer from system A to system B after two or three airplane flights, then you corrected the problem.

(n) If there is no leakage at the four BSVs, then continue.

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(o) If not already pressurized, pressurize hydraulic system A. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(p) Apply brakes and hold. Do not set the parking brake.

NOTE: A small amount of hydraulic fluid may come out of port B while you apply the brake. This is normal and should not be confused with leakage.

(q) Look at the disconnected system A ports of the two Forward BSVs.

(r) Look at the disconnected system B ports of the two Aft BSVs.

(s) If there is leakage at the BSV(s), then do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

2) Replace the applicable leakage BSV. These are the tasks:

- Brake Shuttle Valve Removal, AMM TASK 32-41-95-000-801
- Brake Shuttle Valve Installation, AMM TASK 32-41-95-400-801

OMR 701-799, 802 PRE SB 737-29-1106

3) Remove the lockout pin from the Depressurization Valve for the hydraulic system B reservoir.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106

4) Remove the cap from the system B reservoir pressurization line.

a) Reconnect the reservoir pressurization line to the system B reservoir.

OMR ALL

5) If there is no hydraulic fluid transfer from system A to system B after two or three airplane flights, then you corrected the problem.

(t) If there is no leakage at the four BSVs, reconnect the system A and B ports then continue.

OMR 701-799, 802 PRE SB 737-29-1106

(u) Remove the lockout pin from the Depressurization Valve for the hydraulic system B reservoir.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106

(v) Remove the cap from the system B reservoir pressurization line.

1) Reconnect the reservoir pressurization line to the system B reservoir.

OMR ALL

(2) Do this check for high pressure fluid transfer through the left thrust reverser shuttle valve:

(a) Make sure that the hydraulic power is removed from system A and B. This is the task:

- Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(b) Remove pressure from the hydraulic reservoirs. This is the task:

- Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-802 or Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-808.

(c) Disconnect the standby system line from the left thrust reverser shuttle valve.

(d) Pressurize the hydraulic reservoirs. This is the task:

- Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801 or Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807.



WARNING KEEP ALL PERSONNEL AWAY FROM DISCONNECTED HYDRAULIC LINES. PRESSURIZED HYDRAULIC FLUID CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(e) Pressurize hydraulic system A. This is the task:

- Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(f) Look at the disconnected standby port of the left thrust reverser shuttle valve.

(g) If you find leakage at the left thrust reverser shuttle valve, then do these steps:

1) Remove hydraulic power from system A. This is the task:

- Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

2) Replace the left thrust reverser shuttle valve. These are the tasks:

- Thrust Reverser Shuttle Valve Removal, AMM TASK 78-34-08-000-801-F00,
- Thrust Reverser Shuttle Valve Installation, AMM TASK 78-34-08-400-801-F00.

3) If there is no hydraulic fluid transfer from system A to system B after three subsequent flights, then you corrected the fault.

(h) If you do not find leakage at the left thrust reverser shuttle valve, then continue.

1) Remove the hydraulic power from system A. This is the task:

- Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(3) Replace the LGTV, V123. These are the tasks:

- Landing Gear Transfer Valve Removal, AMM TASK 32-31-71-020-801
- Landing Gear Transfer Valve Installation, AMM TASK 32-31-71-400-801

(a) If there is no hydraulic fluid transfer from system A to system B after two or three airplane flights, then you corrected the problem.

——— END OF TASK ———

807. Hydraulic Fluid Transfers from System B to System A - Fault Isolation

A. Description

(1) Hydraulic system design allows fluid transfer to occur from system B to system A under these abnormal or infrequent operating conditions:

(a) When parking brakes are set with system B pump(s) pressurized and then released with system B depressurized and system A pump(s) pressurized. In this case, fluid is taken from system B when the brakes are set and returned to system A when the brakes are released. Approximately 5 fluid ounces can be transferred per cycle.

1) To prevent hydraulic fluid transfer under these conditions, follow the normal procedures in the pre-flight and post-flight Flight Operations Manual.

(b) When the left thrust reverser is operated by the standby hydraulic system. During thrust reverser deployment, fluid is taken from the standby system reservoir (and the B system reservoir through the fluid balance line), to fill the left thrust reverser actuators, and then is returned to the A system reservoir during reverser stowing. Approximately 0.50 gallons of fluid can be transferred per thrust reverser cycle.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) This transfer occurs infrequently and only during abnormal conditions.

(c) When landing gears are cycled, the airplane is in AIR mode and N2 on engine 1 is less than 50%. In this condition, the LGTV shuttles during gear retraction when the system B fluid is used to fill the actuators. After gear retraction, the LGTV shuttles back to its normal position, so that subsequent extension of the landing gear returns the fluid to system A. Approximately 0.50 gallon of fluid will be transferred per cycle. LGTV operation is controlled through the Proximity Sensor Electronic Unit (PSEU).

1) Fluid transfer will occur under specific flight conditions and during cycling of the landing gear during maintenance (airplane on jacks). To prevent this, you must open the LGTV primary and secondary circuit breakers before you cycle the landing gear.

Valve Position (in W/W)	Valve Port Identifier	System Connection
Left/Forward	INA	Norm (Sys B)
Left/Forward	INB	Alt (Sys A)
Left/Aft	INA	Alt (Sys A)
Left/Aft	INB	Norm (Sys B)
Right/Forward	INA	Norm (Sys B)
Right/Forward	INB	Alt (Sys A)
Right/Aft	INA	Alt (Sys A)
Right/Aft	INB	Norm (Sys B)

B. Possible Causes

(1) BSV internal leakage

(2) Proximity Sensor Electronics Unit (PSEU) logic

(3) LGTV internal leakage

(4) Left Trust Reverser Shuttle Valve internal leakage

C. Fault Isolation Procedure

(1) Do this check for leakage through the BSVs:

(a) Make sure that hydraulic power is removed from system B. This is the task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

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(b) Pressurize hydraulic system A. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(c) Apply brakes fully one time and release.

(d) Remove hydraulic power from the hydraulic system A. This is the task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Remove pressure from the hydraulic system A reservoir only. This is the task: Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-802 or Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-808.

OMR 701-799, 802 PRE SB 737-29-1106

(f) Install the lockout pin in the Depressurization Valve for the hydraulic system A reservoir.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106

(g) Remove the pneumatic reservoir pressurization line to the system A reservoir.

1) Install a cap on the system A reservoir pressurization line.

OMR ALL

(h) Disconnect the hydraulic lines from the INA ports of the two Aft BSVs.

(i) Disconnect the hydraulic lines from the INB ports of the two Forward BSVs.

WARNING KEEP ALL PERSONNEL AWAY FROM DISCONNECTED HYDRAULIC LINES. PRESSURIZED HYDRAULIC FLUID CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(j) Pressurize the hydraulic system B. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(k) Look at the disconnected system A ports of the two Aft BSVs.

(l) Look at the disconnected system B ports of the two Forward BSVs.

(m) If there is leakage at the BSV(s), then do these steps:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

2) Replace the applicable leakage BSV(s). These are the tasks:

- Brake Shuttle Valve Removal, AMM TASK 32-41-95-000-801
- Brake Shuttle Valve Installation, AMM TASK 32-41-95-400-801

OMR 701-799, 802 PRE SB 737-29-1106

3) Remove the lockout pin from the Depressurization Valve for the hydraulic system A reservoir.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106

4) Remove the cap from the system A reservoir pressurization line.

a) Reconnect the reservoir pressurization line to the system A reservoir.

OMR ALL

5) If there is no hydraulic fluid transfer from system B to system A after two or three airplane flights, then you corrected the problem.

(n) If there is no leakage at the four BSVs, then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

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(o) If not already pressurized, pressurize hydraulic system B. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(p) Set the parking brake.

NOTE: A small amount of the hydraulic fluid may come out of port A while you set the brake. This is normal and should not be confused with leakage.

(q) Look at the disconnected system A ports of the two Aft BSVs.

(r) Look at the disconnected system B ports of the two Forward BSVs.

(s) If there is leakage at the BSV(s), then do these steps:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

2) Replace the applicable leakage BSV(s). These are the tasks:

- Brake Shuttle Valve Removal, AMM TASK 32-41-95-000-801
- Brake Shuttle Valve Installation, AMM TASK 32-41-95-400-801

OMR 701-799, 802 PRE SB 737-29-1106

3) Remove the lockout pin from the Depressurization Valve for the hydraulic system A reservoir.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106

4) Remove the cap from the system A reservoir pressurization line.

a) Reconnect the reservoir pressurization line to the system A reservoir.

OMR ALL

5) If there is no hydraulic fluid transfer from system B to system A after two or three airplane flights, then you corrected the problem.

(t) If there is no leakage at the four BSVs, reconnect the system A and B pressure lines to the four brake shuttle ports then do this step and continue:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

OMR 701-799, 802 PRE SB 737-29-1106

(u) Remove the lockout pin from the Depressurization Valve for the hydraulic system A reservoir.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106

(v) Remove the cap from the system A reservoir pressurization line.

1) Reconnect the reservoir pressurization line to the system A reservoir.

OMR ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) Do this check of the LGTV, V123:

(a) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(b) Look for maintenance message 29-23001 ENG NOT RUNNING or 29-22001 ENG RUN L FAULT.

(c) If maintenance message 29-23001 ENG NOT RUNNING or 29-22001 ENG RUN L FAULT shows on the PSEU Display, then do this task: Left Engine Running Signal to PSEU Problem - Fault Isolation, 29-20 TASK 809.

(d) If maintenance message 29-23001 ENG NOT RUNNING or 29-22001 ENG RUN L FAULT does not show on the PSEU Display, then continue.

(3) Do this check for high pressure fluid transfer through the Left Trust Reverser Shuttle Valve:

(a) Make sure that hydraulic power is removed from systems A and B. This is the task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(b) Remove pressure from the hydraulic reservoirs. This is the task: Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-802 or Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-808.

(c) Disconnect the standby system line from the Left Trust Reverser Shuttle Valve.

(d) Pressurize the hydraulic reservoirs. This is the task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801 or Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807.

WARNING

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(e) Pressurize hydraulic system A. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(f) Look at the disconnected standby port of the Left Trust Reverser Shuttle Valve.

(g) If there is leakage at the Left Trust Reverser Shuttle Valve, then do these steps:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

2) Replace the Left Trust Reverser Shuttle Valve. These are the tasks:

- Thrust Reverser Shuttle Valve Removal, AMM TASK 78-34-08-000-801-F00
- Thrust Reverser Shuttle Valve Installation, AMM TASK 78-34-08-400-801-F00

3) If there is no hydraulic fluid transfer from system A to system B after two or three airplane flights, then you corrected the problem.

(h) If there is no leakage at the Left Trust Reverser Shuttle Valve, then do this step and continue:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(4) Replace the LGTV, V123. These are the tasks:

- Landing Gear Transfer Valve Removal, AMM TASK 32-31-71-020-801
- Landing Gear Transfer Valve Installation, AMM TASK 32-31-71-400-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If there is no hydraulic fluid transfer from system B to system A after two or three airplane flights, then you corrected the problem.

——— END OF TASK ———

808. System A Hydraulic Pressure Above 3200 psi Problem - Fault Isolation

A. Description

(1) The system pressure transmitter monitors the system pressure and sends the information to the flight compartment. The system pressure transmitter sends pressure signals between 0 and 4000 psi.

(2) The hydraulic system A pressure transmitter is on the hydraulic system A pressure module.

B. Possible Causes

(1) Hydraulic system A EDP or EMDP

(2) System A pressure transmitter, T424

C. Related Data

(1) (SSM 29-32-11)

(2) (WDM 29-32-11)

D. Initial Evaluation

(1) Do this check of the system A pressure:

WARNING

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(a) Pressurize the hydraulic system A with the EMDP. To pressurize it, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(b) If the system A hydraulic pressure indication on the center instrument panel, P2, is more than 3200 psi, then do the Fault Isolation Procedure - EMDP.

(c) If the system A hydraulic pressure indication on the center instrument panel, P2, is between 2800 and 3200 psi, then do these steps:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

WARNING

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2) Pressurize the hydraulic system A with an EDP. To pressurize it, do this task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) If the system A hydraulic pressure indication on the center instrument panel, P2, is greater than 3200 psi, then do the Fault Isolation Procedure - EDP.

4) If the system A hydraulic pressure indication on the center instrument panel, P2, is between 2800 and 3200 psi, then there was an intermittent fault. Do this step to complete the task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

E. Fault Isolation Procedure - EMDP

(1) Remove power from hydraulic system A. To remove power, do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805

(2) Do this check of the hydraulic system A pressure.

(a) Pressurize the hydraulic system A with the EDP. To pressurize it, do this task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804

(b) If the system A hydraulic pressure indication on the center instrument panel, P2, is more than 3200 psi, then do these steps:

1) Replace the system A pressure transmitter, T424. These are the tasks:

- System Pressure Transmitter Removal, AMM TASK 29-31-12-000-801
- System Pressure Transmitter Installation, AMM TASK 29-31-12-400-801

WARNING

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2) Pressurize the hydraulic system A. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

3) If the system A hydraulic pressure indication on the center instrument panel, P2, is between 2800 and 3200 psi, then you corrected the fault. Do this step to complete the task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

4) If the system A hydraulic pressure indication on the center instrument panel, P2, is greater than 3200 psi, then do these steps:

a) Do this task: CDS BITE Procedure, 31-62 TASK 801.

b) If the CDS BITE test shows a hydraulic oil pressure fault or a DEU fault, then go to the fault isolation task for the applicable maintenance message to correct the fault.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

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c) Pressurize the hydraulic system A. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

d) If the system A hydraulic pressure indication on the center instrument panel, P2, is between 2800 and 3200 psi, then you corrected the fault.

e) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(c) If the system A hydraulic pressure indication on the center instrument panel, P2, is between 2800 and 3200 psi, then continue.

(3) Replace the hydraulic system A EMDP. These are the tasks:

- Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
- Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system A with an EMDP. To pressurize it, do this task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

(b) If the system A hydraulic pressure indication on the pilots center panel, P2, is between 2800 and 3200 psi, then you corrected the fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

F. Fault Isolation Procedure - EDP

(1) Replace the hydraulic system A EDP. These are the tasks:

- Hydraulic Systems A and B Engine-Driven Pump (EDP) Removal, AMM TASK 29-11-11-000-801-001
- Hydraulic Systems A and B Engine-Driven Pump (EDP) Installation, AMM TASK 29-11-11-400-801-001

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

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(a) Pressurize the hydraulic system A with an EDP. To pressurize it, do this task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

(b) If the system A hydraulic pressure indication on the pilots center panel, P2, is between 2800 and 3200 psi, then you corrected the fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

——— END OF TASK ———

809. System B Hydraulic Pressure Above 3200 psi Problem - Fault Isolation

A. Description

(1) The system pressure transmitter monitors the system pressure and sends the information to the flight compartment. The system pressure transmitter sends pressure signals between 0 and 4000 psi.

(2) The hydraulic system B pressure transmitter is on the hydraulic system B pressure module.

B. Possible Causes

(1) Hydraulic system B EDP or EMDP

(2) System B pressure transmitter, T423

C. Related Data

(1) (SSM 29-32-11)

(2) (WDM 29-32-11)

D. Initial Evaluation

(1) Do this check of the system B pressure:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system B with the EMDP. To pressurize it, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(b) If the system B hydraulic pressure indication on the center instrument panel, P2, is greater than 3200 psi, then do the Fault Isolation Procedure - EMDP.

(c) If the system B hydraulic pressure indication on the center instrument panel, P2, is between 2800 and 3200 psi, then do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

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2) Pressurize the hydraulic system B with an EDP. To pressurize it, do this task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

3) If the system B hydraulic pressure indication on the instrument center panel, P2, is greater than 3200 psi, then do the Fault Isolation Procedure - EDP.

4) If the system B hydraulic pressure indication on the center instrument panel, P2, is between 2800 and 3200 psi, then there was an intermittent fault. Do this step to complete the task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

E. Fault Isolation Procedure - EMDP

(1) Remove power from hydraulic system B. To remove power, do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805

(2) Do this check of the hydraulic system B pressure.

(a) Pressurize the hydraulic system B with the EDP. To pressurize it, do this task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804

(b) If the system B hydraulic pressure indication on the center instrument panel, P2, is greater than 3200 psi, then do these steps:

1) Replace the system B pressure transmitter, T423. These are the tasks:

- System Pressure Transmitter Removal, AMM TASK 29-31-12-000-801
- System Pressure Transmitter Installation, AMM TASK 29-31-12-400-801

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

2) Pressurize the hydraulic system B. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

3) If the system B hydraulic pressure indication on the center instrument panel, P2, is between 2800 and 3200 psi, then you corrected the fault. Do this step to complete the task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) If the system B hydraulic pressure indication on the center instrument panel, P2, is greater than 3200 psi, then do these steps:

a) Do this task: CDS BITE Procedure, 31-62 TASK 801.

b) If the CDS BITE test shows a hydraulic oil pressure fault or a DEU fault, then go to the fault isolation task for the applicable maintenance message to correct the fault.

WARNING

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c) Pressurize the hydraulic system B. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

d) If the system B hydraulic pressure indication on the center instrument panel, P2, is between 2800 and 3200 psi, then you corrected the fault.

e) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(c) If the system B hydraulic pressure indication on the center instrument panel, P2, is between 2800 and 3200 psi, then continue.

(3) Replace the hydraulic system B EMDP. These are the tasks:

- Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
- Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001

WARNING

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(a) Pressurize the hydraulic system B with an EMDP. To pressurize it, do this task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

(b) If the system B hydraulic pressure indication on the pilots center panel, P2, is between 2800 and 3200 psi, then you corrected the fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

F. Fault Isolation Procedure - EDP

(1) Replace the hydraulic system B EDP. These are the tasks:

- Hydraulic Systems A and B Engine-Driven Pump (EDP) Removal, AMM TASK 29-11-11-000-801-001
- Hydraulic Systems A and B Engine-Driven Pump (EDP) Installation, AMM TASK 29-11-11-400-801-001

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

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(a) Pressurize the hydraulic system B with an EDP. To pressurize it, do this task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

(b) If the system B hydraulic pressure indication on the center instrument panel, P2, is between 2800 and 3200 psi, then you corrected the fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

END OF TASK

810. System A Hydraulic Pressure Below 2800 psi Problem - Fault Isolation

A. Description

(1) The system pressure transmitter monitors the system pressure and sends the information to the flight compartment. The system pressure transmitter sends pressure signals between 0 psi (0 kPa) and 4000 psi (27,579 kPa).

(2) The hydraulic system A pressure transmitter is on the hydraulic system A pressure module.

(3) The low pressure light comes on if system pressure is below 1200 psi (8274 kPa).

B. Possible Causes

(1) Hydraulic system A EDP or EMDP

(2) Internal leakage

(3) System A Pressure Transmitter, T424

(4) EDP quick disconnect pressure and supply hose not installed correctly

(5) EDP quick disconnect case drain hose not installed correctly

(6) System A Pressure Module Relief Valve is open

C. Related Data

(1) SSM 29-32-11

(2) WDM 29-32-11

D. Initial Evaluation

(1) Do this check of the system A pressure:

(a) Examine the case drain filters for the EMDP. This is the task: EMDP Case Drain Filter Element and Bowl Inspection, AMM TASK 29-11-41-200-801.

1) If debris are found replace the case drain filters. These are the tasks:

- EMDP Case Drain Filter Element Removal, AMM TASK 29-11-41-000-801
- EMDP Case Drain Filter Element Installation, AMM TASK 29-11-41-400-801

(b) Examine the case drain filters for the Engine Driven Pump (EDP). This is the task: EDP Case Drain Filter Element and Bowl Inspection, AMM TASK 29-11-51-200-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If debris are found replace the case drain filters. These are the tasks:
   - EDP Case Drain Filter Element Removal, AMM TASK 29-11-51-000-801
   - EDP Case Drain Filter Element Installation, AMM TASK 29-11-51-400-801

(c) Examine the case drain filter module check valve for debris, blockage, or contamination.

1) If debris, blockage, or contamination are found, replace the case drain filter module check valve. These are the tasks:
   - EDP Case Drain Filter Module Removal, AMM TASK 29-11-51-000-802
   - EDP Case Drain Filter Module Installation, AMM TASK 29-11-51-400-802

(d) Examine the pressure module filters. This is the task: Hydraulic Systems A and B Pressure Module Filter Element Removal, AMM TASK 29-11-71-000-802.

1) If debris are found replace the pressure module filters. This is the task: Hydraulic Systems A and B Pressure Module Filter Element Installation, AMM TASK 29-11-71-400-802.

2) If no debris are found, reinstall the removed pressure module filters. This is the task: Hydraulic Systems A and B Pressure Module Filter Element Installation, AMM TASK 29-11-71-400-802.

(e) Do this check of the System A Pressure Module Relief Valve:

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1) Pressurize the hydraulic system A. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

2) Read the temperature at the System A Pressure Module Relief Valve. This is the task: Thermal Imaging Checks, AMM TASK 20-10-75-700-801.

   a) If there is a significant temperature on the System A Pressure Module Relief Valve and it is similar to that of the connected return line, then replace the System A Pressure Module Relief Valve. This is the task:

      NOTE: The relief valve is open and porting fluid back to the reservoir.

      - Hydraulic Systems A and B Pressure Module Relief Valve Removal, AMM TASK 29-11-71-000-804
      - Hydraulic Systems A and B Pressure Module Relief Valve - Installation, AMM TASK 29-11-71-400-804

   b) If there is no significant temperature on the System A Pressure Module Relief Valve and return line, then do this step and continue:

      <1> Remove hydraulic power from hydraulic system A. This is the task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(f) Make sure that the FLT CONTROL and SPOILER switches, on the P5 panel, are in the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

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(g) Pressurize the hydraulic system A with the EMDP. This is the task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

1) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then do the Fault Isolation Procedure - EMDP.

2) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then continue.

(h) Put the FLT CONTROL and SPOILER switches, on the P5 panel, in the ON position.

1) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then do the Fault Isolation Procedure - Internal Leakage.

2) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then continue.

(i) Put the FLT CONTROL and the SPOILER switches, on the P5 panel, in the OFF position.

(j) Remove hydraulic power from hydraulic system A. This is the task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

WARNING

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(k) Pressurize the hydraulic system A with the EDP. This is the task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

1) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then there was an intermittent fault. Do this step to complete the task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

2) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then do the Fault Isolation Procedure - EDP.

E. Fault Isolation Procedure - EMDP

(1) Remove hydraulic power from hydraulic system A. This is the task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(2) Do this check of hydraulic system A pressure.

(a) Pressurize the hydraulic system A with the EDP. This is the task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then do these steps:

1) Replace the hydraulic system A Pressure Transmitter, T424. These are the tasks:
   - System Pressure Transmitter Removal, AMM TASK 29-31-12-000-801
   - System Pressure Transmitter Installation, AMM TASK 29-31-12-400-801

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2) Pressurize the hydraulic system A. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.
3) Make sure the FLT CONTROL A switch on the flight controls panel is in the ON position.
4) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem. Do this step to complete the task:
   a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.
5) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then continue.
6) Do this task: CDS BITE Procedure, 31-62 TASK 801.
7) If the CDS BITE test shows a hydraulic oil pressure fault or a DEU fault, then go to the fault isolation task for the applicable maintenance message to correct the problem.

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   a) Pressurize the hydraulic system A. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.
   b) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem.
   c) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.
8) If the CDS BITE test does not show a hydraulic oil pressure fault or a DEU fault, then do the Fault Isolation Procedure - Internal Leakage below.

(c) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(3) Replace the hydraulic system A EMDP. These are the tasks:

- Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
- Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001

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(a) Pressurize the hydraulic system A with the EMDP. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) If the system A hydraulic pressure indication on the pilots center panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

F. Fault Isolation Procedure - Internal Leakage

(1) Do a check for internal leakage of the ailerons, elevator, elevator feel, rudder, autopilot, and flight spoilers (AMM PAGEBLOCK 29-00-00/601).

(a) Replace any component that has an internal leakage more than the maximum allowable limit.

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(b) Pressurize the hydraulic system A with the EMDP. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(c) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

G. Fault Isolation Procedure - EDP

(1) Do a check of the EDP case drain quick disconnect hose where it attaches to the pump.

(a) Make sure that the case drain quick disconnect hose is hand tight and the lock tooth is engaged.

(b) If the lock tooth is correctly engaged, then continue.

(2) Do a check of the EDP pressure and supply quick disconnect hoses where they attach to the pump.

(a) Make sure that the supply hose is connected to the pump.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the pressure quick disconnect hose is hand tight and the lock tooth is engaged.

(c) If the lock tooth is correctly engaged, then continue.

(3) Do a check of the EDP case drain quick disconnect hose where it attaches to the pylon.

(a) Make sure that the case drain quick disconnect hose is hand tight and the lock tooth is engaged.

(b) If the lock tooth is correctly engaged, then continue.

(4) Do a check of the EDP pressure and supply quick disconnect hoses where they attach to the pylon.

(a) Make sure that each quick disconnect hose is hand tight and each lock tooth is engaged.

(b) If each lock tooth is correctly engaged, then continue.

(5) Do a check for leakage and proper connection of the hydraulic pressure line, hydraulic case drain line, and hydraulic supply line quick-release fittings on the EDP and at the pylon firewall.

(a) Examine the hydraulic hoses for signs of kinking and/or damage.

(b) Pressurize the hydraulic system A with the EDP. This is the task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

(c) Correct any leakages as necessary.

(d) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(6) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then continue.

(7) Replace the hydraulic system A EDP. These are the tasks:

- Hydraulic Systems A and B Engine-Driven Pump (EDP) Removal, AMM TASK 29-11-11-000-801-001
- Hydraulic Systems A and B Engine-Driven Pump (EDP) Installation, AMM TASK 29-11-11-400-801-001



WARNING

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(a) Pressurize the hydraulic system A with an EDP. This is the task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

(b) If the system A hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

811. System B Hydraulic Pressure Below 2800 psi Problem - Fault Isolation

A. Description

(1) The system pressure transmitter monitors the system pressure and sends the information to the flight compartment. The system pressure transmitter sends pressure signals between 0 psi (0 kPa) and 4000 psi (27,579 kPa).

(2) The hydraulic system B pressure transmitter is on the hydraulic system B pressure module.

(3) The low pressure light comes on if system pressure is below 1200 psi (8274 kPa).

B. Possible Causes

(1) Hydraulic system B EDP or EMDP

(2) Internal leakage

(3) System B Pressure Transmitter, T423

(4) EDP quick disconnect pressure and supply hose not installed correctly

(5) EDP quick disconnect case drain hose not installed correctly

(6) System B Pressure Module Relief Valve is open

C. Related Data

(1) SSM 29-32-11

(2) WDM 29-32-11

D. Initial Evaluation

(1) Do this check of the system B pressure:

(a) Examine the case drain filters for the EMDP. This is the task: EMDP Case Drain Filter Element and Bowl Inspection, AMM TASK 29-11-41-200-801.

1) If debris are found replace the case drain filters. These are the tasks:

- EMDP Case Drain Filter Element Removal, AMM TASK 29-11-41-000-801
- EMDP Case Drain Filter Element Installation, AMM TASK 29-11-41-400-801

(b) Examine the case drain filters for the EDP. This is the task: EDP Case Drain Filter Element and Bowl Inspection, AMM TASK 29-11-51-200-801.

1) If debris are found replace the case drain filters. These are the tasks:

- EDP Case Drain Filter Element Removal, AMM TASK 29-11-51-000-801
- EDP Case Drain Filter Element Installation, AMM TASK 29-11-51-400-801

(c) Examine the case drain filter module check valve for debris, blockage, or contamination.

1) If debris, blockage, or contamination are found, replace the case drain filter module check valve. These are the tasks:

- EDP Case Drain Filter Module Removal, AMM TASK 29-11-51-000-802
- EDP Case Drain Filter Module Installation, AMM TASK 29-11-51-400-802

(d) Examine the pressure module filters. This is the task: Hydraulic Systems A and B Pressure Module Filter Element Removal, AMM TASK 29-11-71-000-802.

1) If debris are found replace the pressure module filters. This is the task: Hydraulic Systems A and B Pressure Module Filter Element Installation, AMM TASK 29-11-71-400-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) If no debris are found, reinstall the removed pressure module filters. This is the task: Hydraulic Systems A and B Pressure Module Filter Element Installation, AMM TASK 29-11-71-400-802.

(e) Do this check of the System B Pressure Module Relief Valve:

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1) Pressurize the hydraulic system B. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

2) Read the temperature at the System B Pressure Module Relief Valve. This is the task: Thermal Imaging Checks, AMM TASK 20-10-75-700-801.

a) If there is a significant temperature on the System B Pressure Module Relief Valve and it is similar to that of the connected return line, then replace the System B Pressure Module Relief Valve. This is the task:

NOTE: The relief valve is open and porting fluid back to the reservoir.

- Hydraulic Systems A and B Pressure Module Relief Valve Removal, AMM TASK 29-11-71-000-804
- Hydraulic Systems A and B Pressure Module Relief Valve - Installation, AMM TASK 29-11-71-400-804

b) If there is no significant temperature on the System B Pressure Module Relief Valve and return line, then do this step and continue:

<1> Remove hydraulic power from hydraulic system B. This is the task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(f) Make sure that the FLT CONTROL B and SPOILER B switches, on the P5 panel, are in the OFF position.

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(g) Pressurize the hydraulic system B with the EMDP. This is the task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

1) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then do the Fault Isolation Procedure - EMDP.

2) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then continue.

(h) Put the FLT CONTROL B and SPOILER B switches, on the P5 panel, in the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then do the Fault Isolation Procedure - Internal leakage.

2) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then continue.

(i) Put the FLT CONTROL B and the SPOILER B switches, on the P5 panel, in the OFF position.

(j) Remove hydraulic power from hydraulic system B. This is the task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

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(k) Pressurize the hydraulic system B with the EDP. This is the task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

1) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then there was an intermittent fault. Do this step to complete the task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

2) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then do the Fault Isolation Procedure - EDP.

E. Fault Isolation Procedure - EMDP

(1) Remove hydraulic power from hydraulic system B. This is the task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(2) Do this check of hydraulic system B pressure.

(a) Pressurize the hydraulic system B with the EDP. This is the task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

(b) If the system B hydraulic pressure indication on the pilot's center panel, P2, is less than 2800 psi (19,305 kPa), then do these steps:

1) Replace the hydraulic system B Pressure Transmitter, T423. These are the tasks:

- System Pressure Transmitter Removal, AMM TASK 29-31-12-000-801
- System Pressure Transmitter Installation, AMM TASK 29-31-12-400-801

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

2) Pressurize the hydraulic system B. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Make sure the FLT CONTROL B switch on the flight controls panel is in the ON position.

4) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem. Do this step to complete the task:

   a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

5) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then continue.

6) Do this task: CDS BITE Procedure, 31-62 TASK 801.

7) If the CDS BITE test shows a hydraulic oil pressure fault or a DEU fault, then go to the fault isolation task for the applicable maintenance message to correct the problem.

WARNING

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   a) Pressurize the hydraulic system B. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

   b) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem.

   c) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

8) If the CDS BITE test does not show a hydraulic oil pressure fault or a DEU fault, then do the Fault Isolation Procedure - Internal Leakage below.

(c) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then continue.

(3) Replace the hydraulic system B EMDP. These are the tasks:

- Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
- Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001

WARNING

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(a) Pressurize the hydraulic system B with the EMDP. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

F. Fault Isolation Procedure - Internal Leakage

(1) Do a check for internal leakage of the ailerons, elevator, elevator feel, rudder, autopilot, and flight spoilers (AMM PAGEBLOCK 29-00-00/601).

(a) Replace any component that has an internal leakage more than the maximum allowable limit.

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(b) Pressurize the hydraulic system B with the EMDP. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(c) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

G. Fault Isolation Procedure - EDP

(1) Do a check of the EDP case drain quick disconnect hose where it attaches to the pump.

(a) Make sure that the case drain quick disconnect hose is hand tight and the lock tooth is engaged.

(b) If the lock tooth is correctly engaged, then continue.

(2) Do a check of the EDP pressure and supply quick disconnect hoses where they attach to the pump.

(a) Make sure that the supply hose is connected to the pump.

(b) Make sure that the pressure quick disconnect hose is hand tight and the lock tooth is engaged.

(c) If the lock tooth is correctly engaged, then continue.

(3) Do a check of the EDP case drain quick disconnect hose where it attaches to the pylon.

(a) Make sure that the case drain quick disconnect hose is hand tight and the lock tooth is engaged.

(b) If the lock tooth is correctly engaged, then continue.

(4) Do a check of the EDP pressure and supply quick disconnect hoses where they attach to the pylon.

(a) Make sure that each quick disconnect hose is hand tight and each lock tooth is engaged.

(b) If each lock tooth is correctly engaged, then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(5) Do a check for leakage and proper connection of the hydraulic pressure line, hydraulic case drain line, and hydraulic supply line quick-release fittings on the EDP and at the pylon firewall.

(a) Examine the hydraulic hoses for signs of kinking and/or damage.

(b) Pressurize the hydraulic system B with the EDP. This is the task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

(c) Correct any leakages as necessary.

(d) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(6) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then continue.

(7) Replace the hydraulic system B EDP. These are the tasks:

- Hydraulic Systems A and B Engine-Driven Pump (EDP) Removal, AMM TASK 29-11-11-000-801-001
- Hydraulic Systems A and B Engine-Driven Pump (EDP) Installation, AMM TASK 29-11-11-400-801-001

WARNING

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(a) Pressurize the hydraulic system B with an EDP. This is the task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804

(b) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

——— END OF TASK ———

812. Hydraulic Fluid Leakage Problem - Fault Isolation

A. Description

(1) This fault isolation procedure is used to find if the hydraulic system is in good condition. It will help you to find which parts are worn.

(2) You can use this procedure to find if the parts have too much external leakage or the general internal condition of the hydraulic system. You can also use it for trouble shooting the system.

(a) When you are trouble-shooting the hydraulic system, it is only necessary for you to do these procedures for the system (A, B, or standby) with the problem. When you can, use standard tools to measure heat, vibration, or sound. Before you get near parts that move, operate them to make sure they can not hit you.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Possible Causes

(1) Damage or worn parts

C. Fault Isolation Procedure

(1) Do a hydraulic system external leakage check. To do it, do this task: Hydraulic System External Leakage Check, AMM TASK 29-00-00-790-801.

NOTE: You can feel for hot tubes or actuators, or listen for fluid leaks to find which parts have the problem.

(a) If the hydraulic fluid leakage rate is more than the external leakage limits, then repair the cause of the leak (AMM TASK 29-00-00-790-801).

(b) If the hydraulic fluid leakage rate is not more than the external leakage limits, then continue.

(2) Do a hydraulic system internal leakage check (AMM PAGEBLOCK 29-00-00/601).

(a) If the hydraulic fluid leakage rate is more than the internal leakage limits, then repair the cause of the leak.

——— END OF TASK ———

813. System A or System B Hydraulic Pressure Momentarily Below 2800 psi Problem - Fault Isolation

A. Description

(1) When selecting landing gear or high lift devices, hydraulic system pressures may drop for up to 10 seconds during normal operation. Momentary hydraulic pressure drops lasting from 10 to 40 seconds may be indications of a fault.

(2) The low pressure light illuminates when the pressure drops below 1200 psi (8274 kPa).

(3) Hydraulic system pressure drop during high flow demand is usually caused by cavitation of the EDP. Cavitation may be caused by a restriction in the EDP supply line or by reservoir pressurization system problems.

(4) The landing gear runs off of System A pressure, and the wing high lift devices run off of System B pressure. Each system is tested separately in this procedure.

(5) A hydraulic pump can be damaged if it operates during a momentary loss of pressure. If pressure is lost for an extended time, then check the applicable pressure and case drain filters. A contaminated filter can be an indication of a damaged pump.

B. Possible Causes

(1) If there is a momentary loss of, or no recovery of hydraulic pressure after ending high demand activities, the following are possible causes, in order of probability:

(a) Symptoms that occur during descent, but which can not be duplicated on the ground, may be caused by water trapped in the reservoir pressurization system freezing during cruise. These symptoms are considered intermittent faults.

(b) Reservoir pressurization system faults:

OMR 701-799, 802 PRE SB 737-29-1106

1) Reservoir pressurization module filter plugged

2) Reservoir pressurization orifice assembly is corroded or plugged

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106

3) Air pressure filter assembly plugged

4) Secondary check valve is corroded or plugged

5) Cross fitting assembly is corroded or plugged

OMR ALL

6) Reservoir pressurization system leakage

(c) Engine-driven pump supply line kinks or obstructions

(d) Hose liner collapse

(e) Quick-disconnect broken

1) Center body of the quick disconnect may have broken loose, obstructing flow

(f) Faulty check valves in the nose landing gear steering lines (System A only)

(g) Faulty engine-driven pump

(h) EDP quick disconnect pressure and supply hose not installed correctly

(i) EDP quick disconnect case drain hose not installed correctly

C. Related Data

(1) SSM 29-00-00

(2) WDM 29-10-01

(3) SSM 29-33-11

(4) WDM 29-33-11

D. Initial Evaluation

(1) Do the following steps to configure the airplane and prepare for an initial evaluation:

(a) Make sure that the FLT CONTROL and SPOILER switches, on the P5 panel, are in the OFF position.

WARNING

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(b) Pressurize the hydraulic system A and system B with the EDP. This is the task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

(c) Make sure the system A and system B hydraulic system pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then continue.

(d) If the system A or system B hydraulic pressure indication on the Center Instrument Panel, P2, is less than 2800 psi (19,305 kPa), then, do this task: System A Hydraulic Pressure Below 2800 psi Problem - Fault Isolation, 29-10 TASK 810 or, do this task: System B Hydraulic Pressure Below 2800 psi Problem - Fault Isolation, 29-10 TASK 811, as appropriate.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, is between 2800 psi (19,305 kPa) and 3200 psi (22,063 kPa), then continue with Test System B Hydraulic Pressure Recovery.

NOTE: There is no test of the system A hydraulic pressure recovery. The test requires operating high-flow systems while running the EDP. Proper testing of system A would retract the landing gear, which would require that the airplane be on jacks. This is not practical while running the EDP. If momentary hydraulic pressure drops have only been noted in system A, go to the Fault Isolation below.

(2) Test System B Hydraulic Pressure Recovery

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(a) Make sure that the FLT CONTROL and SPOILER switches, on the P5 panel, are in the ON position.

(b) Deploy the leading edge flaps and slats, and trailing edge flaps.

(c) Monitor the system B hydraulic pressure indication on the Center Instrument Panel, P2, during the deployment time period.

(d) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, dropped during the deployment of the leading edge flaps and slats, and trailing edge flaps and did not recover, then do the Fault Isolation Procedure - Pressurization Faults.

(e) If the system B hydraulic pressure indication on the Center Instrument Panel, P2, dropped during the deployment of the leading edge flaps and slats, and trailing edge flaps, and then recovered to normal pressures within 10 seconds, then there was an intermittent fault in system B. Trapped water which freezes in the reservoir pressurization system during flight is the most likely cause of the intermittent fault.

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(f) Stow the leading edge flaps and slats, and trailing edge flaps.

(g) Continue with Restore the Airplane to Its Usual condition to complete this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Restore the Airplane to Its Usual Condition

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(1) Make sure that the FLT CONTROL and SPOILER switches, on the P5 panel, are in the OFF position.

(2) Remove hydraulic power from hydraulic system A and system B. This is the task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

F. Fault Isolation Procedure - Pressurization Faults

(1) Do a check of the EDP case drain quick disconnect hose where it attaches to the pump.

(a) Make sure that the case drain quick disconnect hose is hand tight and the lock tooth is engaged.

(b) If the lock tooth is correctly engaged, then continue.

(2) Do a check of the EDP pressure and supply quick disconnect hoses where they attach to the pump.

(a) Make sure that the supply hose is connected to the pump.

(b) Make sure that the pressure quick disconnect hose is hand tight and the lock tooth is engaged.

(c) If the lock tooth is correctly engaged, then continue.

(3) Do a check of the EDP case drain quick disconnect hose where it attaches to the pylon.

(a) Make sure that the case drain quick disconnect hose is hand tight and the lock tooth is engaged.

(b) If the lock tooth is correctly engaged, then continue.

(4) Do a check of the EDP pressure and supply quick disconnect hoses where they attach to the pylon.

(a) Make sure that each quick disconnect hose is hand tight and each lock tooth is engaged.

(b) If each lock tooth is correctly engaged, then continue.

OMR 701-799, 802 PRE SB 737-29-1106

(5) Do a check to see if the reservoir pressurization module filter is plugged.

(a) Pressurize the hydraulic reservoirs. This is the task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807

(b) If air does not flow continuously from the vent caps between the reservoir pressurization module and the reservoirs, then replace the reservoir pressurization module filter element. These are the tasks:

- Hydraulic Reservoir Pressurization Module Filter - Removal, AMM TASK 29-09-06-000-806
- Hydraulic Reservoir Pressurization Module Filter - Installation, AMM TASK 29-09-06-400-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802 PRE SB 737-29-1106 (Continued)

(c) If air flows continuously from the vent caps between the reservoir pressurization module and the reservoirs, then continue.

(6) Do a check of the reservoir pressurization orifice tee for corrosion or plugged passages. This is the task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807.

(a) Do a check for air flow from the vent caps between the reservoir pressurization module and the reservoirs.

(b) Do a check for air pressure shown on the reservoir pressurization gauges.

(c) If there is no air flow, or the gauges show no pressure, then clean or replace the tee.

(d) If there is air flow then continue.

(7) Do a check for reservoir pressurization system leakage. This is the task: Hydraulic Reservoir Pressurization - Leakage Test, AMM TASK 29-09-00-860-809.

(a) If there is leakage, then repair the leaks as necessary.

(b) If there is no leakage, then continue.

(8) Symptoms that occur during descent, but cannot be reproduced on the ground, may be caused by water trapped in the reservoir pressurization system. The trapped water can freeze during cruise, causing a blockage during descent but not on the ground.

(a) Do a check of and clean the vent caps, and remove the reservoir pressurization module filter bowl and drain water, as necessary. This is the task: Vent Cap - Check, AMM TASK 29-09-06-211-801.

(b) If the vent caps were clean and not clogged, the fault was an intermittent fault. This completes the fault isolation task.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-29-1106

(9) Do a check to see if the air pressure filter assembly is plugged. This is the task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.

(a) If air does not flow continuously from the vents between the air pressure filter assembly and the reservoirs, then replace the air pressure filter assembly filter element. These are the tasks:

NOTE: Do only the steps to remove and install the filter element. It is not necessary to remove the filter assembly to replace the filter element.

- Air Pressure Filter Assembly - Removal, AMM TASK 29-09-01-000-801
- Air Pressure Filter Assembly - Installation, AMM TASK 29-09-01-400-801

(b) If air flows continuously from the vents between the air pressure filter assembly and the reservoirs, then continue.

(10) Do a check of the secondary check valve for corrosion or plugged passages. This is the task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.

(a) Do a check for air flow from the vents between the air pressure filter assembly and the reservoirs.

(b) If air does not flow continuously from the vents, then clean or replace the secondary check valve. These are the tasks:

- Secondary Check Valve - Removal, AMM TASK 29-09-01-000-802
- Secondary Check Valve - Installation, AMM TASK 29-09-01-400-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) If air flows continuously from the vents, then continue.

(11) Do a check of the cross fitting assembly for corrosion or plugged passages. This is the task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801.

(a) Do a check for air flow from the vents between the cross fitting assembly and the reservoirs.

(b) Do a check for air pressure shown on the reservoir pressurization gauges.

(c) If there is no air flow, or the gauges show no pressure, then clean or replace the cross fitting assembly. These are the tasks:

- Cross Fitting Assembly - Removal, AMM TASK 29-09-04-000-801
- Cross Fitting Assembly - Installation, AMM TASK 29-09-04-400-801

(d) If air flows continuously from the vents, then continue.

(12) Do a check for reservoir pressurization system leakage. This is the task: Hydraulic Reservoir Pressurization System - Leakage Test, AMM TASK 29-09-00-860-803.

(a) If there is leakage, then repair the leaks as necessary.

(b) If there is no leakage, then continue.

(13) Symptoms that occur during descent, but cannot be reproduced on the ground, may be caused by water trapped in the reservoir pressurization system. The trapped water can freeze during cruise, causing a blockage during descent but not on the ground.

(a) Do a check of and clean the vents, and remove the air pressure filter assembly filter element and drain water as necessary.

If the vents were clean and not clogged, the fault was an intermittent fault. This completes the fault isolation task.

OMR ALL

(14) Do a check for hose-liner collapse in the engine-driven pump supply hose.

(a) Inspect for kinked hoses.

(b) Remove the EDP supply hoses and inspect the liner. To remove the hose, do the following steps:

1) Prepare for the hose removal.

a) For the applicable hydraulic system, do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

b) Release pressure from the applicable hydraulic reservoir. This is the task: Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-802 or Hydraulic Reservoirs Depressurization, AMM TASK 29-09-00-860-808.

c) For engine 1, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	15	C00779	HYD SYS ENG PUMP DEPRESS VALVE 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

d) For engine 2, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	17	C00780	HYD SYS ENG PUMP DEPRESS VALVE 2

e) For the left fan cowl panel, do this task: Open the Fan Cowl Panels, AMM TASK 71-11-02-010-801-F00.

2) Remove the hose.

a) Disconnect the electrical connector, DP1204, from the receptacle on the pump, under the hose.

b) Put a container, (1 gal) under the pump to catch spills.

c) Disconnect the hydraulic supply line.

3) Inspect the hose liner for separation or collapse.

NOTE: Look for kinked hoses.

4) Inspect the hose quick-disconnect fitting for broken or obstructed passages.

NOTE: Look for the quick-disconnect halves being completely screwed together, or the center-body of the disconnect being broken.

5) If the hose is separated, collapsed, broken or obstructed, then replace the hose.

6) If the hose is not faulty, then continue.

7) Install the hose.

a) Lubricate the hydraulic supply hose union to pump o-ring with MCS 352B fluid, D00054.

b) Install the hydraulic supply hose union in the forward port of the pump.

c) Tighten the union to 855 in-lb (97 N·m) - 945 in-lb (107 N·m).

d) Lubricate the hydraulic supply hose union to hose o-ring with MCS 352B fluid, D00054.

e) Connect the hydraulic supply hose to the hydraulic supply hose union.

f) Tighten the hydraulic supply line hose to fitting joint to 1520 in-lb (172 N·m) - 1680 in-lb (190 N·m).

8) Put the airplane back to its usual condition.

a) For the left fan cowl panel, do this task: Close the Fan Cowl Panels, AMM TASK 71-11-02-410-801-F00.

b) For engine 1, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	15	C00779	HYD SYS ENG PUMP DEPRESS VALVE 1

c) For engine 2, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	17	C00780	HYD SYS ENG PUMP DEPRESS VALVE 2

d) This completes the fault isolation task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

9) Do a check for faulty check valves in the nose landing gear steering lines (system A only). If leaky, these check valves can allow flow to go directly from pressure to return (Landing gear up to down, or down to up), creating a high flow demand and possible reduced system pressure. Do the following steps:

a) Install the landing gear down lock pins. This is the task: Landing Gear Downlock Pins Installation, AMM TASK 32-00-01-480-801.

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b) Pressurize the hydraulic system A and system B with the EMDP. This is the task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

c) Select landing gear DOWN.

d) Do a check of the steering gear check valves.

e) If the check valves are hot, replace them.

f) Select landing gear Up.

g) If the check valves are hot, replace them.

h) If the check valves are not hot, then continue.

i) Replace the EDP. This is the task: Hydraulic Systems A and B Engine-Driven Pump (EDP) Removal, AMM TASK 29-11-11-000-801-001.

j) This completes the fault isolation task.

G. Repair Confirmation

(1) Repeat the Initial Evaluation tests above.

(a) If the hydraulic pressure in each system momentarily drops and then recovers, or does not drop at all, then you corrected the problem.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Engine Driven Pump Operates (engine windmills) with the Supply Shutoff Valve Closed (fire handle pulled) - Fault Isolation

A. Description

(1) The engine fire switches isolate the engine if there is an engine fire. Each engine fire switch controls the supply shutoff valve for its related Engine Driven Pump (EDP).

(2) When you pull the engine fire switch, the Engine Driven Pump (EDP) supply shutoff valve moves to the closed position. This stops the supply of hydraulic fluid to the EDP.

B. Possible Causes

(1) Because the supply of hydraulic fluid to EDP stops when you move the engine fire switch up, case drain flow does not flow through the EDP. If the engine continues to turn (windmill) with the supply shutoff valve in the closed position, the EDP may be damaged.

C. Initial Evaluation

(1) If you move either engine No. 1 or 2 fire switches to the up position with the corresponding engine driven pump (EDP) operating, then do this check:

(a) If the supply shutoff valve was closed for less than five minutes, no maintenance action is required and shutoff periods are not accumulative.

(b) If the supply shutoff valve was closed for more than five minutes, make sure that the EDP has operated for at least one minute with supply fluid available before you check the filter elements. This is to permit fluid to reach the filter that may have metal contamination, possibly generated during fluid shutoff, then do the Fault Isolation Procedure below:

D. Fault Isolation Procedure

(1) Check the case drain filter element for metal contamination. To check it, do this task: EDP Case Drain Filter Element Removal, AMM TASK 29-11-51-000-801.

(2) Check the pressure filter for metal contamination. To check it, do this task: Hydraulic Systems A and B Pressure Module Filter Element Removal, AMM TASK 29-11-71-000-802.

(3) If no metal contamination is found in either filter, then do the steps that follow:

(a) Install a new EDP case drain filter element. To install it, do this task: EDP Case Drain Filter Element Installation, AMM TASK 29-11-51-400-801.

(b) Install a new pressure filter element. To install it, do this task: Hydraulic Systems A and B Pressure Module Filter Element Installation, AMM TASK 29-11-71-400-802.

(c) Continue to operate the EDP and recheck the EDP case drain filter element and the pressure filter element in 200 hours.

(4) If you find metal contamination in either filter, then do the steps that follow:

(a) Replace the EDP. To replace it, these are the tasks:

- Hydraulic Systems A and B Engine-Driven Pump (EDP) Removal, AMM TASK 29-11-11-000-801-001
- Hydraulic Systems A and B Engine-Driven Pump (EDP) Installation, AMM TASK 29-11-11-400-801-001

(b) Flush the hydraulic lines between the EDP and the filters. To flush the lines, do this task: Hydraulic System A or B Flushing, AMM TASK 29-11-00-170-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

802. Reservoir Fluid Quantity Problems - Fault Isolation

A. Description

(1) This procedure can be used to look for hydraulic leaks inside the fuel tanks or engine struts when the following conditions occur:

(a) A hydraulic system has required repeated servicing because of a low reservoir condition.

(b) The hydraulic fluid level decreases quickly and the external hydraulic leakage is within the maintenance limits.

(2) This procedure is applicable only after the following check has been unsuccessful in isolating the source of hydraulic fluid leaks:

(a) External hydraulic leakage check (Hydraulic System External Leakage Check, AMM TASK 29-00-00-790-801).

(3) This procedure is not intended to troubleshoot for the transfer of hydraulic fluid from one system to another system.

(4) The presence of hydraulic fluid in the fuel tanks can be from leakage at the hydraulic fluid heat exchangers, the associated heat exchanger tubing or various flight control or brake system pressure hydraulic tubing that is contained in the tanks.

(5) The heat exchanger for hydraulic system A is in fuel tank 1, in the left wing. The heat exchanger for hydraulic system B is in fuel tank 2, in the right wing.

(6) The presence of hydraulic fluid in the engine struts can occur if leakage from the strut upper spar hydraulic lines occurs due to failure in a tube/line or loose connection.

B. Possible Causes

(1) Hydraulic fluid leakage into fuel tank

(2) Hydraulic fluid leakage on the strut upper spar

(a) Hydraulic fluid leakage into the strut torque box dry bay.

C. Related Data

(1) AMM 29-11-00, I/C

D. Fault Isolation Procedure - Hydraulic Fluid Leakage on the Strut Upper Spar

(1) Use these steps to find the sequence in which strut must be examined for leaks:

(a) If the hydraulic system A has the hydraulic quantity problem, do this task: AMM TASK 05-41-04-210-811.

1) Visually examine strut upper spar and look for loose/leaking hydraulic line fittings/connections or indication of hydraulic fluid leakage.

NOTE: Indication of hydraulic leak are wet/coked hydraulic fluid, damaged primer, or clogged drain line and/or inlet in the vapor barrier.

a) Using torque-loosen-torque procedure, tighten hydraulic line fitting/connections found loose or leaking (AMM TASK 54-51-01-400-801 and AMM TASK 20-10-51-000-802).

2) If hydraulic fluid is found on upper spar, do a visual inspection of torque box through side access panels 433AL/433AR, looking for hydraulic fluid (coked, wet).

a) If hydraulic fluid is found in the torque box, do this task: AMM TASK 05-51-22-210-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) Visually examine strut upper spar access panels (433AT, 433BT, 433CT, 433DT) for damage in the form of loose panel installation, disbonded seal/gasket, damaged latch hardware, or otherwise anything that may contribute to leakage into the torque box (AMM TASK 54-53-01-000-801).

NOTE: Damage to access panel is likely the cause of breach into the torque box.

c) If damage is found, replace defective access panel.

(b) If the hydraulic system B is the system with the hydraulic quantity problem, then inspect the right strut. Do this task: AMM TASK 05-41-04-210-815.

1) Visually examine strut upper spar and look for loose/leaking hydraulic line fittings/connections or indication of hydraulic fluid leakage.

NOTE: Indication of hydraulic leak are wet/coked hydraulic fluid, damaged primer, or clogged drain line and/or inlet in the vapor barrier.

a) Using torque-loosen-torque procedure, tighten hydraulic line fitting/connections found loose or leaking (AMM TASK 54-51-01-400-801 and AMM TASK 20-10-51-000-802).

2) If hydraulic fluid is found on upper spar, do a visual inspection of torque box through side access panels 443AL/443AR, looking for hydraulic fluid (coked, wet).

a) If hydraulic fluid is found in the torque box, do this task: AMM TASK 05-51-22-210-801.

b) Visually examine strut upper spar access panels (433AT, 433BT, 433CT, 433DT) for damage in the form of loose panel installation, disbonded seal/gasket, damaged latch hardware, or otherwise anything that may contribute to leakage into the torque box (AMM TASK 54-53-01-000-801).

NOTE: Damage to access panel is likely the cause of breach into the torque box.

c) If damage is found, replace defective access panel.

(2) If hydraulic fluid leak is not resolved through inspection of hydraulic lines, then do Fault Isolation Procedure - Hydraulic Fluid Leakage into the Fuel Tank.

(3) If the hydraulic fluid leakage is resolved, then do Repair Confirmation.

E. Fault Isolation Procedure - Hydraulic Fluid Leakage into the Fuel Tank

(1) Use these guidelines to determine the order in which the fuel tanks should be sampled or inspected for leaks:

(a) If hydraulic system B is the system with the hydraulic quantity problem, then inspect the fuel tanks for hydraulic leaks in this order:

1) Fuel tank 2, right wing.

2) Fuel tank 1, left wing- inspection is required if fault was not isolated in the inspection of the right tank.

(b) If the hydraulic system A is the system with the hydraulic quantity problem, then inspect the fuel tanks for hydraulic leaks in this order:

1) Fuel tank 1, left wing.

2) Fuel tank 2, right wing- inspection is required if fault was not isolated in the inspection of the left tank.

3) Center tank - inspection is required if fault was not isolated in the inspection of the left or right tanks.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) If the hydraulic fluid level decreases slowly, then do the following steps:

(a) Get a fuel sample from the applicable tank(s). To get a fuel sample, do this task: Fuel System Sumping, AMM TASK 12-11-00-680-801

(b) Send the samples to a certified testing facility to determine if hydraulic fluid is present.

NOTE: A standard test such as the ASTM D7111, or an equivalent test procedure, you can use to find the phosphorous contents in the fuel sample.

1) If a visual examination of the fuel samples reveals contamination, or a decreasing hydraulic fluid level cannot otherwise be explained, then the technician may use his experience and judgement, if airline policy allows, to continue this procedure without confirmation of hydraulic fluid contamination of the fuel.

(3) If the hydraulic fluid level decreases quickly, or if hydraulic fluid contamination of the fuel is confirmed or suspected, then continue.

(4) Defuel the applicable fuel tank. To defuel the tank, do this task: Fuel Tank Defueling, AMM TASK 28-26-00-650-801.

WARNING CAREFULLY DO ALL OF THE SAFETY PROCEDURES TO PREPARE TO GO INTO THE FUEL TANK. IF YOU DO NOT OBEY THE SAFETY PROCEDURES, YOU CAN CAUSE AN EXPLOSION. AN EXPLOSION WILL CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(5) Do this Task: Fuel Tank - Purging and Tank Entry, AMM TASK 28-11-00-910-802.

(6) Enter the applicable fuel tank and do this step

(a) Wipe all traces of fuel from the heat exchanger, all parts of the tubing attached to the heat exchanger and all other tubing that penetrates the fuel tank walls.

(7) Exit the applicable fuel tank and close the access door to the tank.

(8) Pressurize the applicable hydraulic system with a portable hydraulic cart. Do this task: Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, AMM TASK 29-11-00-860-802.

(9) Remove the pressure from the portable hydraulic cart. Do this task:Bleed the Hydraulic Systems, AMM TASK 29-00-00-870-801.

(10) Enter the applicable fuel tank and look for leaks at the heat exchanger, all parts of the tubing attached to the heat exchanger and all other tubing that penetrates the fuel tank walls.

(11) If leaks are found, do these steps:

(a) Repair any leaks found at the tubing connections to the heat exchanger or any other tubing in the fuel tank.

(b) If leaks are found in the heat exchanger, replace the heat exchanger. Do these tasks:Heat Exchanger Removal, AMM TASK 29-11-04-000-801 and Heat Exchanger Installation, AMM TASK 29-11-04-400-801.

(c) Contact the applicable engine manufacturer with the calculated hydraulic fluid contamination level and length of time the airplane was operated with the leak to get further evaluation of specific condition and case-by-case support.

NOTE: Hydraulic fluid stays in solution in fuel at concentrations up to 5% at temperatures more than -22°F (-30°C). The affect of hydraulic fluid contamination is dependant upon the concentration of the hydraulic fluid and the time of exposure.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) If the contamination was more than 1 parts per million or if the hydraulic fluid was in the tank for more than three days, do this task: Engine Fuel Feed Manifold - Leak Test, AMM TASK 28-22-15-710-801.

NOTE: Hydraulic fluid contamination of less than 1 parts per million will not cause damage to the airplane fuel system seals or the fuel tank sealant for time periods less than three days. If the contamination is more than 1 parts million or if the hydraulic fluid was in the fuel tank for more than three days, some seals in the fuel system and fuel pumps could be affected.

(e) If the hydraulic fluid leak is in the left main tank, the APU is affected if it operated during the time period when the hydraulic fluid leakage occurred.

NOTE: The left main tank supplies fuel to the APU.

1) If the APU is affected, do this task: APU Operates with Hydraulic Contaminated Fuel - Fault Isolation, 49-10 TASK 811.

(12) If leaks are not found, then look for hydraulic leaks in the appropriate fuel tanks as listed in the guidelines at the start of this fault isolation procedure.

F. Repair Confirmation

(1) Return the aircraft to service.

(2) Monitor the hydraulic reservoirs for indication of excessive fluid loss.

(a) If the hydraulic fluid loss has stopped or is within specification, you have corrected the fault.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. Standby Hydraulic EMDP Does Not Operate - Fault Isolation

A. Description

(1) The standby hydraulic system electric motor-driven pump (EMDP) supplies hydraulic pressure to the standby hydraulic system.

(2) The standby hydraulic system EMDP has an electric motor that mechanically connects to the EMDP. The motor operates with three-phase, 400 hertz, 115 VAC power.

(3) The EMDP is rated at 4.2 gpm at 2700 psi. The standby hydraulic system EMDP is in the right aft wing to body fairing.

B. Possible Causes

(1) Standby hydraulic pump relay, R643

(2) Standby hydraulic System EMDP, M242

(3) Flight controls panel, P5-3

(4) Wiring problem

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

D. Related Data

(1) (SSM 29-22-11)

(2) (WDM 29-22-11)

E. Initial Evaluation

(1) Do this check of the standby hydraulic system EMDP, M242:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the standby hydraulic system. To pressurize it, do this task: Standby Hydraulic System Pressurization, AMM TASK 29-21-00-000-801.

(b) If the standby hydraulic system EMDP does not operate, then do the Manual Mode Fault Isolation Procedure below.

(c) If the standby hydraulic system EMDP operates, then continue with the initial evaluation.

(2) Do these steps to check the automatic operation of the standby hydraulic system EMDP:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do this task to extend the Trailing Edge Flaps: Extend the Trailing Edge Flaps, AMM TASK 27-51-00-860-803

NOTE: The position of the flaps is not important as long as they are extended.

(b) Do this task: Standby Hydraulic System Power Removal, AMM TASK 29-21-00-000-802.

(c) Do this task to put the airplane in air mode:

Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801

(d) Put the System A Flight Control Switch on the Flight Controls Panel, P5-3, to the ON position.

(e) Put the System B Flight Control Switch on the Flight Controls Panel, P5-3, to the ON position.

(f) Put the Alternate Flaps Switch on the Flight Controls Panel, P5-3, to the OFF position.

(3) If the standby hydraulic system EMDP operates, then there was an intermittent fault. Do this task to put the airplane back to its usual condition:

Return the Airplane Systems Back to Their Normal On Ground Condition, AMM TASK 32-09-00-840-802

(4) If the standby hydraulic system EMDP does not operate, then do the Automatic Mode Fault Isolation Procedure below.

F. Manual Mode Fault Isolation Procedure

(1) Do a check of the standby hydraulic system EMDP, M242:

(a) Disconnect connector D812 from the standby hydraulic system EMDP, M242.

(b) Do a check for 115 VAC from pins A, B, and C of connector D812 to structure ground (WDM 29-22-11).

(c) If there is 115 VAC at pins A, B, and C of connector D812, then do these steps:

1) Replace the standby hydraulic system EMDP, M242. These are the tasks:

- Standby Hydraulic System Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-21-21-000-801
- Standby Hydraulic System Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-21-21-400-801

2) Do the Repair Confirmation at the end of this task.

(d) If there is not 115 VAC at pin A, B, or C of connector D812, then do these steps:

1) Re-connect connector D812.

2) Replace the standby hydraulic pump relay, R643 in the power distribution panel 2, P92.

NOTE: The power distribution panel 2, P92, is in the electrical/electronic compartment.

3) Do the Repair Confirmation at the end of this task.

4) If the Repair Confirmation is not satisfactory, then continue.

5) Disconnect connector D11740 from the power distribution panel, P92.

6) Disconnect connector D812 from the standby hydraulic system EMDP, M242.

7) Do a wiring check between these pins of connector D812 at the standby hydraulic system EMDP, M242 and connector D11740 at the panel P92 (WDM 29-22-11):

NOTE: You only need to do the wiring check at the pin that does not have 115 VAC.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D812		D11740
pin A	. . . . .	pin A
pin B	. . . . .	pin B
pin C	. . . . .	pin C

8) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-connect connector D812.
   c) Re-connect connector D11740.
   d) Do the Repair Confirmation at the end of this task.

(e) If there is not 115 VAC at pins A, B, and C of connector D812, then continue.
   1) Re-connect connector D812.
   2) Re-connect connector D11740.

(2) Do this wiring check of the standby hydraulic pump relay, R643:
   (a) Disconnect connector D11448 from the power distribution panel, P92.

   NOTE: The power distribution panel, P92 is in the electrical/electronic compartment.

   (b) Do a check for 28 VDC at pin 25 of connector D11448 to structure ground (WDM 29-22-11).
   (c) If there is 28 VDC at pin 25 of connector D11448, then do these steps:
      1) Replace the standby hydraulic pump relay, R643 in the power distribution panel, P92.
      2) Re-connect connector D11448.
      3) Do the Repair Confirmation at the end of this task.
      4) If the Repair Confirmation is not satisfactory, then continue.
      5) Do a check for 115 VAC from the load terminal of this circuit breaker, C1449, to structure ground (WDM 29-22-11):
         a) This is the circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

      6) If there is not 115 VAC at circuit breaker, C1449, then do these steps:
         a) Replace the circuit breaker, C1449.
         b) Do the Repair Confirmation at the end of this task.
   (d) If there is not 28 VDC at pin 25 of connector D11448, then continue.

(3) Do this wiring check:
   (a) Remove the flight controls panel, P5-3. To remove it, do this task: Flight Controls Panel Removal, AMM TASK 27-09-45-000-801
   (b) Disconnect connector D624 from the flight controls panel, P5-3.
   (c) Do a wiring check between these pins of connector D624 at the P5-3 panel and D11448 at the panel P92 (WDM 29-22-11):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D624	D11448
pin 21	pin 25

(d) If there is a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect connector D11448.
3) Re-connect connector D624.
4) Re-install the flight controls panel, P5-3. To install it, do this task: Flight Controls Panel Installation, AMM TASK 27-09-45-400-801.
5) Do the Repair Confirmation at the end of this task.

(e) If you do not find a problem with the wiring, then continue.

1) Re-connect connector D11448.

(4) Do this check of the flight controls panel, P5-3:

(a) Do a check for 28 VDC at pin 11 of connector D624 to structure ground (WDM 29-22-11).

(b) If there is 28 VDC at pin 11 of connector D624, then do these steps:

1) Install a new flight controls panel, P5-3. To install it, do this task: Flight Controls Panel Installation, AMM TASK 27-09-45-400-801.
2) Do the Repair Confirmation at the end of this task.

(c) If there is not 28 VDC at pin 11 of connector D624, then continue.

(5) Do this check of the circuit breaker, C362:

(a) Do a check for 28 VDC from the load terminal of this circuit breaker, C362, to structure ground (WDM 29-22-11):

1) This is the circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD

(b) If there is not 28 VDC at circuit breaker, C362, then do these steps:

1) Replace the circuit breaker, C362.
2) Re-connect connector D624.
3) Re-install the flight controls panel, P5-3. To install it, do this task: Flight Controls Panel Installation, AMM TASK 27-09-45-400-801.
4) Do the Repair Confirmation at the end of this task.

(c) If there is 28 VDC at circuit breaker, C362, then do these steps:

1) Repair the wiring between pin 11 of connector D624 at the panel P5-3 to the load terminal of circuit breaker C362 at panel P6 (WDM 29-22-11).
2) Re-connect connector D624.
3) Re-install the flight controls panel, P5-3. To install it, do this task: Flight Controls Panel Installation, AMM TASK 27-09-45-400-801.
4) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Automatic Mode Fault Isolation Procedure

(1) Do these steps to prepare the airplane for the following check:

(a) If the flaps are not extended, then do this task to extend the Trailing Edge Flaps: Extend the Trailing Edge Flaps, AMM TASK 27-51-00-860-803

NOTE: The position of the flaps is not important as long as they are extended.

(b) Do this task: Standby Hydraulic System Power Removal, AMM TASK 29-21-00-000-802.

(c) Do this task to put the airplane in air mode:

Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801

(d) Put the System A Flight Control Switch on the Flight Controls Panel, P5-3, to the ON position.

(e) Put the System B Flight Control Switch on the Flight Controls Panel, P5-3, to the ON position.

(f) Put the Alternate Flaps Switch on the Flight Controls Panel, P5-3, to the OFF position.

(2) With the airplane configured as shown above, do this wiring check of the Auto Standby Relay, K1, on the P5-3 Flight Control Panel:

(a) Check the X2 pin of the Auto Standby Relay, K1, for continuity to ground.

(b) If the X2 pin does have continuity to ground, then replace the Auto Standby Relay, K1.

1) Do the Repair Confirmation at the end of this task.

(c) If the X2 pin does not have continuity to ground, then repair the wiring as necessary.

1) Do the Repair Confirmation at the end of this task.

H. Repair Confirmation

(1) Do this check of the standby hydraulic system EMDP, M242:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the standby hydraulic system. To pressurize it, do this task: Standby Hydraulic System Pressurization, AMM TASK 29-21-00-000-801.

(2) Do this check of the automatic operation of the standby hydraulic system EMDP:

(a) If the flaps are not extended, then do this task to extend the Trailing Edge Flaps: Extend the Trailing Edge Flaps, AMM TASK 27-51-00-860-803

NOTE: The position of the flaps is not important as long as they are extended.

(b) Do this task: Standby Hydraulic System Power Removal, AMM TASK 29-21-00-000-802.

(c) Do this task to put the airplane in air mode:

Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801

(d) Put the System A Flight Control Switch on the Flight Controls Panel, P5-3, to the ON position.

(e) Put the System B Flight Control Switch on the Flight Controls Panel, P5-3, to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Put the Alternate Flaps Switch on the Flight Controls Panel, P5-3, to the OFF position.

(3) If the standby hydraulic system EMDP operates, then you corrected the fault.

I. Put the Airplane Back to Its Usual Condition

(1) Do this task to put the airplane in ground mode: Return the Airplane to the Ground Mode, AMM TASK 32-09-00-860-802.

(2) Do this task to raise the trailing edge flaps: Retract the Trailing Edge Flaps, AMM TASK 27-51-00-860-804.

(3) Do this task: Standby Hydraulic System Power Removal, AMM TASK 29-21-00-000-802.

——— END OF TASK ———

804. Power Transfer Unit (PTU) Does Not Operate - Fault Isolation

A. Description

(1) The power transfer unit (PTU) supplies alternative hydraulic pressure to the leading edge flaps and slats.

(2) The PTU has a hydraulic motor and a hydraulic pump on a common shaft. The PTU motor receives pressure from the hydraulic system A through the PTU control valve to turn the motor. The PTU pump receives fluid supply from the hydraulic system B reservoir. The PTU pump supplies alternative pressure to the leading edge flaps and slats.

B. Possible Causes

(1) Flaps up switch, S1051

(2) Flap landing warning switch, S138

(3) Standby shutoff valve indication relay, R625

(4) PTU control valve, V133

(5) Hydraulic system B EDP pressure switch, S855

(6) Wiring problem

(7) Power transfer unit (PTU)

(8) System B EDP relay, R353

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	15	C01081	HYDRAULIC SYSTEM PTU VALVE CONT 1
A	16	C01085	HYDRAULIC SYSTEM PTU VALVE CONT 2

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	16	C01356	LANDING GEAR AIR/GND SYS 1

D. Related Data

(1) (SSM 29-25-11)

(2) (WDM 29-25-11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Initial Evaluation

(1) Do this check of the power transfer unit (PTU):

(a) Make sure the ALTERNATE FLAPS ARM switch, on the flight controls panel, P5-3, is not in the ARM position.

(b) Make sure the ALTERNATE FLAPS switch, on the flight controls panel, P5-3, is not in the DOWN position.

(c) Do this task: Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-801 or Hydraulic Reservoirs Pressurization, AMM TASK 29-09-00-860-807.

(d) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(e) Pressurize hydraulic system B. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(f) Move the flap control lever to the 5-unit detent.

(g) Remove hydraulic power from the hydraulic system B. To remove it, do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(h) Pressurize hydraulic system A. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE FLAPS AND SLATS. THE LEADING EDGE FLAPS AND SLATS CAN MOVE AUTOMATICALLY. THIS CAN OCCUR WHEN THE HYDRAULIC SYSTEM B EDP PRESSURE IS LESS THAN 2400 PSI, THE TRAILING EDGE FLAPS ARE NOT IN THE UP POSITION, AND THE MAIN LANDING GEAR AIR/GROUND RELAYS IS IN THE AIR MODE. THIS CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(i) Do this task: Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801.

(j) If the PTU control valve does not open, then do the Fault Isolation Procedure - Valve Not Open below.

(k) If the PTU control valve opens, then do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If the PTU does not operate, then do the Fault Isolation Procedure - Valve Open below.

2) If the PTU operates, then there was an intermittent fault. Do these steps to complete the task:

a) Do this task: Return the Airplane to the Ground Mode, AMM TASK 32-09-00-860-802.

b) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

c) Remove hydraulic power from the hydraulic system A. To remove it, do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

F. Fault Isolation Procedure - Valve Not Open

(1) Do this check of the power transfer unit (PTU) control valve, V133:

(a) Disconnect connector D3244 from the PTU control valve, V133.

(b) Do a check for 28 VDC from pin 3 of connector D3244 to structure ground (WDM 29-25-11).

(c) If there is 28 VDC at pin 3 of connector D3244, then do these steps:

1) Replace the PTU control valve, V133. These are the tasks:

- Power Transfer Unit (PTU) Control Valve Removal, AMM TASK 29-22-31-000-801
- Power Transfer Unit (PTU) Control Valve Installation, AMM TASK 29-22-31-400-801

2) Do the operational test of the PTU. Do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

3) If the operational test is satisfactory, then you corrected the fault.

(d) If there is not 28 VDC at pin 3 of connector D3244, then continue.

1) Re-connect connector D3244.

(2) Do this power wiring check:

(a) Disconnect connector D3254 from the hydraulic system B EDP relay, R353, in the junction box J11.

NOTE: The junction box J11 is in the aft cargo bay near aft bulkhead.

(b) Do a check of 28 VDC from pins B1 and X1 of connector D3254 to structure ground (WDM 29-25-11).

(c) If there is 28 VDC at pins B1 and X1 of connector D3254, then do these steps:

1) Do a check for 28 VDC from pin X1 to X2 (ground) of connector D3254 (WDM 29-25-11).

2) If there is 28 VDC at pin X1 of connector D3254, then do these steps:

a) Install a new hydraulic system B EDP relay, R353.

b) Do the operational test of the PTU. To do it, do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

c) If the operational test is satisfactory, then you corrected the fault.

d) If the operational test is not satisfactory, then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

e) Disconnect connector D3244 from the PTU control valve, V133.

f) Remove the hydraulic system B EDP relay, R353.

g) Repair the wiring between these pins of connector D3244 at the PTU control valve, V133 and connector D3254 at the junction box J11 (WDM 29-25-11):

D3244		D3254
pin 3	. . . . .	pin B2

h) Re-connector connector D3244.

i) Do the operational test of the PTU. Do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

j) If the operational test is satisfactory, then you corrected the fault.

3) If there is not 28 VDC at pin X1 or ground at pin X2 of connector D3254, then do these steps:

a) Replace the hydraulic system B EDP pressure switch, S855. These are the tasks:

- Engine Driven Pump (EDP) Pressure Switch Auto Slat System Removal, AMM TASK 29-22-41-000-801
- Engine Driven Pump (EDP) Pressure Switch Auto Slat System Installation, AMM TASK 29-22-41-400-801

b) Re-connect connector D3254.

c) Do the operational test of the PTU. Do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

d) If the operational test is satisfactory, then you corrected the fault.

e) If the operational test is not satisfactory, then continue.

f) Disconnect connector D3242 from the hydraulic system B EDP pressure switch, S855.

g) Remove the hydraulic system B EDP relay, R353.

h) Repair wiring between these pins of connector D3242 at the hydraulic system B EDP pressure switch, S855 and connector D3254 at the junction box J11 (WDM 29-25-11):

D3242		D3254
pin 3	. . . . .	pin X2

i) Repair the wiring between pin 2 of connector D3242 to DC ground (WDM 29-25-11).

j) Re-connect connector D3242.

k) Re-connect connector D3254.

l) Do the operational test of the PTU. Do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

m) If the operational test is satisfactory, then you corrected the fault.

(d) If there is not 28 VDC at pin B1 and/or X1 of connector D3254, then continue.

(3) Do this wiring check:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the standby shutoff valve indication relay, R625 from the junction box J11.

NOTE: The junction box J11 is in the aft cargo bay near aft bulkhead.

(b) Do a wiring check between these pins of connector D11370 at standby shutoff valve indication relay, R625 and connector D3254 at the hydraulic system B EDP relay, R353 (WDM 29-25-11):

D11370		D3254
pin A3	. . . . .	pin B1
pin A3	. . . . .	pin X1

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-install the standby shutoff valve indication relay, R625.
3) Re-connect connector D3254.
4) Do the operational test of the PTU. Do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.
5) If the operational test is satisfactory, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

1) Re-install the hydraulic system B EDP relay, R353.

OMR 701-799, 802, 805, 807, 809, 810 PRE SB 737-27-1290 AND PRE SB 737-27-1304

(4) Do this check of the two flap position switches.

(a) Disconnect connector D11370 from the standby shutoff valve indicator relay, R625.

(b) Do a check of 28 VDC at pin A2 of connector D11370 (WDM 29-25-11).

(c) If there is not 28 VDC at pin A2 of connector D11370, then do these steps:

1) Disconnect connector D46040P from the flap landing warning switch, S138.
2) Do a check for 28 VDC at pin 19 of connector D46040P (WDM 29-25-11).
3) If there is 28 VDC at pin 19 of connector D46040P, then do these steps:

a) Replace the flap landing warning switch, S138.
b) Do the operational test of the PTU. To do it, do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.
c) If the operational test is satisfactory, then you corrected the fault.
d) If the operational test is not satisfactory, then continue.
e) Disconnect connector D11370 from the standby shutoff valve indication relay, R625.
f) Disconnect connector D46040P from the flap landing warning switch, S138.
g) Repair wiring between these pins of connector D11370 at the standby shutoff valve indication relay, R625 and connector D46040P at the flap landing warning switch, S138 (WDM 29-25-11):

D11370		D46040P
pin A2	. . . . .	pin 9

h) Re-connect connector D11370.
i) Re-connect connector D46040P.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

j) Do the operational test of the PTU. Do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

k) If the operational test is satisfactory, then you corrected the fault.

l) If the operational test is not satisfactory, then continue.

4) If there is not 28 VDC at pin 19 of connector D46040P, then do these steps:

a) Replace the trailing edge flaps up switch, S1051.

b) Do the operational test of the PTU. To do it, do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

c) If the operational test is satisfactory, then you corrected the fault.

d) If the operational test is not satisfactory, then continue.

e) Disconnect connector D46040P from the flap landing warning switch, S138 and the trailing edge flaps up switch, S1051.

f) Repair wiring between these pins of connector D46040P at the flap landing warning switch, S138 and the trailing edge flaps up switch, S1051 (WDM 29-25-11):

D46040P	D46040P
pin 19	pin 18

g) Re-connect connector D46040P.

h) Do the operational test of the PTU. Do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

i) If the operational test is satisfactory, then you corrected the fault.

(d) If there is 28VDC at pin A2 of connector D11370, then continue.

OMR 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802, 805, 807, 809, 810 POST SB 737-27-1290 OR POST SB 737-27-1304

(5) Do this check of the two flap position switches.

(a) Disconnect connector D11370 from the standby shutoff valve indicator relay, R625.

(b) Do a check of 28 VDC at pin A2 of connector D11370 (WDM 29-25-11).

(c) If there is not 28 VDC at pin A2 of connector D11370, then do these steps:

1) Disconnect connector D14730 from the flap landing warning switch, S138.

2) Do a check for 28 VDC at pin G of connector D14730 (WDM 29-25-11).

3) If there is 28 VDC at pin G of connector D14730, then do these steps:

a) Replace the flap landing warning switch, S138.

b) Do the operational test of the PTU. To do it, do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

c) If the operational test is satisfactory, then you corrected the fault.

d) If the operational test is not satisfactory, then continue.

e) Disconnect connector D11370 from the standby shutoff valve indication relay, R625.

f) Disconnect connector D14730 from the flap landing warning switch, S138.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

g) Repair wiring between these pins of connector D11370 at the standby shutoff valve indication relay, R625 and connector D14730 at the flap landing warning switch, S138 (WDM 29-25-11):

D11370		D14730
pin A2	. . . . .	pin H

h) Re-connect connector D14730.

i) Re-connect connector D11370.

j) Do the operational test of the PTU. Do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

k) If the operational test is satisfactory, then you corrected the fault.

l) If the operational test is not satisfactory, then continue.

4) If there is not 28 VDC at pin G of connector D14730, then do these steps:

a) Replace the trailing edge flaps up switch, S1051.

b) Do the operational test of the PTU. To do it, do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

c) If the operational test is satisfactory, then you corrected the fault.

d) If the operational test is not satisfactory, then continue.

e) Disconnect connector D14730 from the flap landing warning switch, S138.

f) Dicsonnect connector D14360 from the trailing edge flaps up switch, S1051.

g) Repair wiring between these pins of connector D14730 at the flap landing warning switch, S138 and connector D14360 at the trailing edge flaps up switch, S1051 (WDM 29-25-11):

D14730		D14360
pin G	. . . . .	pin U

h) Re-connect connector D14360.

i) Re-connect connector D14730.

j) Do the operational test of the PTU. Do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

k) If the operational test is satisfactory, then you corrected the fault.

(d) If there is 28VDC at pin A2 of connector D11370, then continue.

OMR ALL

(6) Install a new standby shutoff valve indication relay, R625.

(a) Do the operational test of the PTU. Do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

1) If the operational test is satisfactory, then you corrected the fault.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Fault Isolation Procedure - Valve Open

(1) Replace the power transfer unit (PTU). These are the tasks:

- Power Transfer Unit (PTU) Removal, AMM TASK 29-22-11-000-801
- Power Transfer Unit (PTU) Installation, AMM TASK 29-22-11-400-801

(a) Do the operational test of the PTU. Do this task: Power Transfer Unit Operational Test, AMM TASK 29-22-00-710-801.

1) If the operational test is satisfactory, then you corrected the fault.

——— END OF TASK ———

805. Primary Landing Gear Transfer Valve Signal to PSEU Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) 29-24001 PRI LGTV FAIL

(2) The proximity switch electronics unit (PSEU) has detected a fault from the landing gear transfer valve.

B. Possible Causes

(1) Landing gear transfer valve, V123

(2) Proximity switch electronics unit (PSEU), M2061

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

D. Related Data

(1) (SSM 29-23-11)

(2) (WDM 29-23-11)

E. Initial Evaluation

(1) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of landing gear transfer valve, V123:

(a) Disconnect connector D2860 from the landing gear transfer valve, V123.

(b) Measure the resistance between pin 2 and pin 1 of the landing gear transfer valve, V123.

(c) If the resistance is not between 60-80 ohms, then do these steps:

1) Replace the landing gear transfer valve, V123. These are the tasks:

- Landing Gear Transfer Valve Removal, AMM TASK 32-31-71-020-801
- Landing Gear Transfer Valve Installation, AMM TASK 32-31-71-400-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(d) If the resistance is between 60-80 ohms, then continue.

(2) Do this wiring check:

(a) Disconnect connector D11138 from the proximity switch electronics unit (PSEU).

(b) Do a wiring check between these pins of connector D11138 at the E11 shelf and connector D2860 at the landing gear transfer valve, V123:

D11138		D2860
pin 2	. . . . .	pin 1

1) Do a check for the ground at pin 2 of connector D2860.

2) If there is a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D2860.

c) Re-connect connector D11138 to the PSEU.

d) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

e) If the maintenance message does not show, then you corrected the fault.

(c) If you do not find a problem with the wiring, then continue.

1) Re-connect connector D2860.

(3) Replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

(a) If the maintenance message does not show after the PSEU installation test, then you corrected the fault.

——— END OF TASK ———

806. Secondary Landing Gear Transfer Valve Signal to PSEU Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) 29-24002 SEC LGTV FAIL

(2) The proximity switch electronics unit (PSEU) has detected a fault from the landing gear transfer valve.

B. Possible Causes

(1) Landing gear transfer valve, V123

(2) Proximity switch electronics unit (PSEU), M2061

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

D. Related Data

(1) (SSM 29-23-11)

(2) (WDM 29-23-11)

E. Initial Evaluation

(1) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of landing gear transfer valve, V123:

(a) Disconnect connector D2860 from the landing gear transfer valve, V123.

(b) Measure the resistance between pin 3 and pin 4 of the landing gear transfer valve, V123.

(c) If the resistance is not between 60-80 ohms, then do these steps:

1) Replace the landing gear transfer valve, V123. These are the tasks:

- Landing Gear Transfer Valve Removal, AMM TASK 32-31-71-020-801
- Landing Gear Transfer Valve Installation, AMM TASK 32-31-71-400-801

2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

3) If the maintenance message does not show, then you corrected the fault.

(d) If the resistance is between 60-80 ohms, then continue.

(2) Do this wiring check:

(a) Disconnect connector D11140 from the proximity switch electronics unit (PSEU).

(b) Do a wiring check between these pins of connector D11140 at the E11 shelf and connector D2860 at the landing gear transfer valve, V123:

D11140		D2860
pin 2	. . . . .	pin 3

1) Do a check for the ground at pin 4 of connector D2860.

2) If there is a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D2860.

c) Re-connect connector D11140 to the PSEU.

d) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

e) If the maintenance message does not show, then you corrected the fault.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) If you do not find a problem with the wiring, then continue.

1) Re-connect connector D2860.

(3) Replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

(a) If the maintenance message does not show after the PSEU installation test, then you corrected the fault.

——— END OF TASK ———

807. Landing Gear Primary Power Signal to PSEU Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) 29-20001 NO LGTV PRI PWR

(2) The proximity switch electronics unit (PSEU) has detected a fault from the primary landing gear circuit breaker.

NOTE: Power is only provided through the circuit breaker when the 115V AC transfer busses are powered.

B. Possible Causes

(1) Wiring problem

(2) Proximity switch electronics unit (PSEU), M2061

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

D. Related Data

(1) (SSM 29-23-11))

(2) (WDM 29-23-11))

E. Initial Evaluation

(1) Supply Electrical Power to the 115VAC Transfer Buses: (Supply Electrical Power, AMM TASK 24-22-00-860-811).

(2) Make sure these lights on the P5-4 are OFF:

(a) 1 TRANSFER BUS OFF

(b) 2 TRANSFER BUS OFF

(3) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

(4) Do this task: (Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

(5) Do this check of the circuit breaker, C781:

(a) Do a check for 28 VDC from the load terminal of this circuit breaker, C781, to structure ground.

1) This is the circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

(b) If there is not 28 VDC at circuit breaker C781, then do these steps:

1) Replace the circuit breaker, C781.

2) Do this task: (Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801).

3) If the maintenance message does not show, then you corrected the fault.

(c) If there is 28 VDC at circuit breaker C781, then continue.

F. Fault Isolation Procedure

(1) Do this check of the landing gear primary power signal to the PSEU:

(a) Push the ON/OFF button on the front panel of the PSEU to turn on the BITE display.

NOTE: The display will show EXISTING FAULTS.

(b) Select OTHER FUNCTNS from the MAIN MENU on the PSEU.

(c) Select I/O MONITOR from the OTHER FUNCTNS menu

(d) Select INPUTS from the I/O MONITOR menu.

(e) Select Connector D11138 PIN 7 from the list of inputs.

(f) (f) If the PSEU BITE display shows VLTGE, then do these steps:

1) Remove the proximity switch electronics unit (PSEU), M2061. To remove it, do this task: (Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801).

2) Install a new PSEU, M2061. To install it, do this task: (Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801).

3) Do this task: (Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801).

4) If the maintenance message does not show, then you corrected the fault.

(g) If the PSEU BITE display shows NO VLTGE, then do these steps:

1) Repair the wiring between pin 7 of connector D11138 at the PSEU to the load terminal of circuit breaker C781 at the panel P6.

2) Do this task: (Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801).

3) If the maintenance message does not show, then you corrected the fault.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

808. Landing Gear Secondary Power Signal to PSEU Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) 29-20002 NO LGTV SEC PWR

(2) The proximity switch electronics unit (PSEU) has detected a fault from the secondary landing gear circuit breaker.

NOTE: Power is only provided through the circuit breaker when the 115V AC transfer busses are powered.

B. Possible Causes

(1) Wiring problem

(2) Proximity switch electronics unit (PSEU), M2061

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

D. Related Data

(1) (SSM 29-23-11)

(2) (WDM 29-23-11)

E. Initial Evaluation

(1) Supply Electrical Power to the 115VAC Transfer Buses. Do this task: Supply Electrical Power, AMM TASK 24-22-00-860-811.

(2) Make sure these lights on the P5-4 are OFF:

(a) 1 TRANSFER BUS OFF

(b) 2 TRANSFER BUS OFF

(3) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC

(4) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the circuit breaker, C799:

(a) Do a check for 28 VDC from the load terminal of this circuit breaker, C799, to structure ground.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) This is the circuit breaker

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC

(b) If there is not 28 VDC at pin 7 of connector D11140, then continue.

1) Replace the circuit breaker, C799.

2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure: (Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801).

3) If the maintenance message does not show, then you corrected the fault.

(c) If there is 28 VDC at circuit breaker C799, then continue.

(2) Do this check of the landing gear secondary power signal to the PSEU:

(a) Push the ON/OFF button on the front panel of the PSEU to turn on the BITE display.

NOTE: The display will show EXISTING FAULTS.

(b) Select OTHER FUNCTNS from the MAIN MENU on the PSEU.

(c) Select I/O MONITOR from the OTHER FUNCTNS menu

(d) Select INPUTS from the I/O MONITOR menu

(e) Select Connector D11140 PIN 7 from the list of inputs

(f) If the PSEU BITE display shows VLTGE, then do these steps:

1) Remove the proximity switch electronics unit (PSEU), M2061. To remove it, do this task: (Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801).

2) Install a new PSEU, M2061. To install it, do this task: (Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801).

3) Do this task: (Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801).

4) If the maintenance message does not show, then you corrected the fault.

(g) If the PSEU BITE display shows NO VLTGE, then do these steps:

1) Repair the wiring between pin 7 of connector D11140 at the PSEU to the load terminal of circuit breaker C799 at the panel P6

2) Do this task: (Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801).

3) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

809. Left Engine Running Signal to PSEU Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 29-22001 ENG RUN L FAULT

(b) 29-23001 ENG NOT RUNNING

(2) The maintenance messages show when these conditions occur:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) The left engine is on, but the left engine-running relay is not providing a ground signal to the PSEU input.

(b) The left engine thrust lever is advanced more than 64 degrees.

B. Possible Causes

(1) Left engine running relay, R564

(2) Proximity switch electronics unit (PSEU), M2061

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

D. Related Data

(1) (SSM 29-23-11)

(2) (WDM 29-23-11)

(3) (SSM 73-22-31)

(4) (WDM 73-22-31)

E. Initial Evaluation

(1) Do these steps to prepare for the initial evaluation:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that a minimum of 3 out of 4 entry/service doors are closed.

(c) Make sure that the airplane is in ground mode.

(d) Make sure that the engine start switches on the P5-20 panel are in the OFF position.

(e) Move the engine start levers on the control stand to IDLE.

(f) Wait for 5 minutes, then move the engine thrust levers on the control stand to the fully advanced position.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Replace the left engine running relay, R564.

NOTE: The left engine running relay, R564 is installed in the junction box, J22.

(a) Do the Repair Confirmation at the end of this task:

1) If the maintenance message does not show, then you corrected the fault.

2) If the maintenance message shows, then continue.

(2) Do this wiring check:

(a) Remove the proximity switch electronics unit (PSEU). To remove it, do this task: Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801.

(b) Remove the left engine running relay, R564, from the J22 junction box.

(c) Do a wiring check between these pins of connector D10986 at the E11 and connector D10918 at the J22 junction box:

D10986	D10918
pin 39	pin A1

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the left engine running relay, R564.

c) Re-install the PSEU, M2061. To install it, do this task: Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

d) Do the Repair Confirmation at the end of this task:

<1> If the maintenance message does not show, then you corrected the fault.

<2> If you do not find a problem with the wiring, then continue.

(3) Install a new PSEU, M2061. To install it, do this task: Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

(a) Do the Repair Confirmation at the end of this task:

G. Repair Confirmation

(1) Do these steps to prepare for the Repair Confirmation:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

(b) Make sure that the right-engine running relay, R563 is installed.

(c) Make sure that a minimum of 3 out of 4 entry/service doors are closed.

(d) Make sure that the airplane is in ground mode.

(e) Make sure that the engine start switches on the P5-20 panel are in the OFF position.

(f) Move the engine start levers on the control stand to IDLE.

(g) Wait for 5 minutes, then move the engine thrust levers on the control stand to the fully advanced position.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message shows, return to the Fault Isolation Procedure and continue.

(b) If the maintenance message does not show, then you corrected the fault, continue the procedure below.

H. Put the Airplane in its Usual Condition

(1) Do these steps to put the airplane back to its usual condition.

(a) Move the engine start levers to CUTOFF.

(b) Move the engine thrust levers to idle.

(c) Make sure the engine start switches are OFF.

(d) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

END OF TASK

810. Hydraulic Quantity B Normal Signal to PSEU Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) 29-22002 HYD QTY B FAULT

(2) The proximity switch electronics unit (PSEU) has detected a fault from the hydraulic system B quantity/transmitter.

B. Possible Causes

(1) Hydraulic system B quantity transmitter, T184

(2) Proximity switch electronics unit (PSEU), M2061

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

D. Related Data

(1) (SSM 29-23-11)

(2) (WDM 29-23-11)

E. Initial Evaluation

(1) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Look at the QTY indication for the hydraulic system B on the pilots center panel, P2.

(a) If the QTY indication for hydraulic system B is less than 76%, then, do this task: Hydraulic Fluid Quantity Indication Problem - Fault Isolation, 29-30 TASK 807.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) if the QTY indication for the hydraulic system B is greater than 76%, then continue.

(2) Do this wiring check:

(a) Remove the proximity switch electronics unit (PSEU). To remove it, do this task: Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801.

(b) Disconnect the connector D2674 from the hydraulic system B quantity/transmitter, T184.

(c) Do a wiring check between these pins of connector D10986 at the E11 and connector D2674 at the hydraulic system B quantity/transmitter, T184:

D10986		D2674
pin 9	. . . . .	pin 5

1) Do a check for ground at pin 4 of connector D2674.

2) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D2674.

c) Re-install the PSEU, M2061. To install it, do this task: Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

d) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

e) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(3) Replace the hydraulic system B quantity/transmitter, T184. These are the tasks:

- Hydraulic Fluid Quantity Transmitter/Indicator Removal, AMM TASK 29-33-12-000-801
- Hydraulic Fluid Quantity Transmitter/Indicator Installation, AMM TASK 29-33-12-400-801

(a) Re-install the PSEU, M2061. To install it, do this task: Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

(b) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(c) If the maintenance message does not show, then you corrected the fault.

(d) If the maintenance message shows, then continue.

(4) Replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

(a) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

811. Hydraulic Quantity B Normal BITE Signal to PSEU Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) 29-22003 HYD QTY B NORM BITE

(2) The proximity switch electronics unit (PSEU) has detected a fault from the hydraulic system B quantity/transmitter.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Possible Causes

(1) Hydraulic system B quantity transmitter, T184

(2) Proximity switch electronics unit (PSEU), M2061

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

D. Related Data

(1) (SSM 29-23-11)

(2) (WDM 29-23-11)

E. Initial Evaluation

(1) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this wiring check:

(a) Remove the proximity switch electronics unit (PSEU). To remove it, do this task: Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801.

(b) Disconnect the connector D2674 from the hydraulic system B quantity/transmitter, T184.

(c) Do a wiring check between these pins of connector D10986 at the E11 and connector D1674 at the hydraulic system B quantity/transmitter, T184:

D10986	D2674
pin 21	pin 5

1) Do a check for ground at pin 4 of connector D2674.

2) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D2674.

c) Re-install the PSEU, M2061. To install it, do this task: Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

d) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

e) If the maintenance message does not show, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(2) Replace the hydraulic system B quantity/transmitter, T184. These are the tasks:

- Hydraulic Fluid Quantity Transmitter/Indicator Removal, AMM TASK 29-33-12-000-801
- Hydraulic Fluid Quantity Transmitter/Indicator Installation, AMM TASK 29-33-12-400-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Re-install the PSEU, M2061. To install it, do this task: Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

(b) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(c) If the maintenance message does not show, then you corrected the fault.

(d) If the maintenance message shows, then continue.

(3) Replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

(a) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

812. Alternate Nose Wheel Steering Signal to PSEU Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 29-22004 ALT NOSE STRG FLT

(b) 29-23002 ALT NOSE STRG SEL

(2) The proximity switch electronics unit (PSEU) has detected a fault from the alternate nose wheel steering switch, S1.

B. Possible Causes

(1) Alternate nose wheel steering switch, S1

(2) Proximity switch electronics unit (PSEU), M2061

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

D. Related Data

(1) (SSM 29-23-11)

(2) (WDM 29-23-11)

E. Initial Evaluation

(1) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this wiring check:

(a) Remove the proximity switch electronics unit (PSEU). To remove it, do this task: Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801.

(b) Remove the captain's instrument panel, P1.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Disconnect connector D903 from the alternate nose wheel steering switch, S1.

(d) Do a wiring check between these pins of connector D10982 at the E11 and connector D903 at the alternate nose wheel steering switch, S1:

D10982	D903
pin 39	pin 5

1) Do a check for ground at pin 6 of connector D903.

2) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-connect connector D903.

c) Re-install the captain's instrument panel, P1.

d) Re-install the PSEU, M2061. To install it, do this task: Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

e) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

f) If the maintenance message does not show, then you corrected the fault.

(e) If you do not find a problem with the wiring, then continue.

(2) Replace the alternate wheel steering switch, S1.

(a) Re-install the captain's instrument panel, P1.

(b) Re-install the PSEU, M2061. To install it, do this task: Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

(c) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(d) If the maintenance message does not show, then you corrected the fault.

(e) If the maintenance message shows, then continue.

(3) Replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

(a) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(b) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

813. Landing Gear System Transfer Valve Set Signal to PSEU Problem - Fault Isolation

A. Description

(1) This task is for these Maintenance Messages:

(a) 29-22005 LGTV SET FAULT

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SBC 285A1600-32-04; AIRPLANES WITH PSEU -5 OR -6

(b) 29-22105 LGTV SET FAULT

OMR ALL

(c) 29-24003 LGTV RESET

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) The Proximity Switch Electronic Unit (PSEU), M02061 has detected a fault from the Landing Gear Transfer Valve (LGTV) set signal.

B. Possible Causes

(1) Wiring

(2) LGTV, V123

(3) PSEU, M02061

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

D. Related Data

(1) SSM 29-23-11

(2) WDM 29-23-11

E. Initial Evaluation

(1) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the Maintenance Message does not show, then there was an intermittent problem.

(b) If the Maintenance Message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Disconnect Connector D2860 from the LGTV, V123 (WDM 29-23-11).

(a) Clean the electrical Connector (SWPM 20-60-01).

(2) Do this Wiring Check (WDM 29-23-11):

(a) Disconnect Connector D10984 from the PSEU, M02061.

(b) Disconnect Connector D2860 from the LGTV, V123.

(c) Examine the wiring between the PSEU, M02061 and LGTV, V123 as follows:

PSEU D10984	**LGTV D2860**
pin 57	pin 11

LGTV D2860	
pin 9	GND

1) Make sure that the connectors do not have bent, broken, damaged, or corroded pins.

2) If you find a problem with the Wiring, then do these steps:

a) Repair the Wiring as necessary (SWPM 20-10-13).

b) Connect Connector D2860 to the LGTV, V123.

c) Connect Connector D10984 to the PSEU, M02061.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

d) Do the Repair Confirmation at the end of this task.

3) If you do not find a problem with the Wiring, then continue.

(3) Replace the LGTV, V123. These are the tasks:

- Landing Gear Transfer Valve Removal, AMM TASK 32-31-71-020-801
- Landing Gear Transfer Valve Installation, AMM TASK 32-31-71-400-801

(a) Connect Connector D2860 to the LGTV, V123.

(b) Connect Connector D10984 to the PSEU, M02061.

(c) Do the Repair Confirmation at the end of this task.

(4) Replace the PSEU, M02061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do the Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the Maintenance Message does not show, then you corrected the problem.

(b) If the Maintenance Message still shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

803. System A Hydraulic Pressure Reads Zero - Fault Isolation

A. Description

(1) The system pressure transmitter monitors the system pressure and sends the information to the flight compartment. The system pressure transmitter sends pressure signals between 0 and 4000 psi.

(2) The hydraulic system A pressure transmitter is on the hydraulic system A pressure module.

B. Possible Causes

(1) Hydraulic leak

(2) Wiring problem

(3) System A pressure transmitter, T424

(4) Display electronic unit 1 (DEU-1), M1808

C. Related Data

(1) (SSM 29-32-11)

(2) (WDM 29-32-11)

D. Initial Evaluation

(1) Do this check of the system A hydraulic pressure:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system A. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) If the system A hydraulic quantity is low and the pressure indication on the center instrument panel, P2, is zero psi, then do the Fault Isolation Procedure - Hydraulic Fluid Leakage.

(c) If the system A hydraulic quantity is not low and the pressure indication on the center instrument panel, P2, is zero psi, then do the Fault Isolation Procedure - Indication Problem.

(d) If the system A hydraulic pressure indication on the center instrument panel, P2, is between 2900 and 3200 psi, then there was an intermittent fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

E. Fault Isolation Procedure - Hydraulic Fluid Leakage

(1) Do a check of the hydraulic system A for external leakage.

(a) Do this task: Hydraulic System External Leakage Check, AMM TASK 29-00-00-790-801.

1) If you find external leakage of hydraulic fluid, then repair the cause of the leak and continue.

2) If you do not find external leakage of hydraulic fluid, then do continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Fill the system A hydraulic reservoir with hydraulic fluid. To fill it, do this task: Hydraulic Reservoir Servicing, AMM TASK 12-12-00-610-801.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(c) Pressurize the hydraulic system A. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(d) If the system A hydraulic pressure indication on the center instrument panel, P2, is between 2900 and 3200 psi, then you corrected the fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(e) If the system A hydraulic pressure indication on the center instrument panel, P2, is zero then do the Fault Isolation Procedure - Indication Problem

F. Fault Isolation Procedure - Indication Problem

(1) Do this check of the wiring:

(a) Remove the display electronic unit 1 (DEU-1), M1808. To remove it, do this task: Display Electronic Unit Removal, AMM TASK 31-62-21-000-801.

(b) Disconnect connector D662 from the system A pressure transmitter, T424.

(c) Do a wiring check between these pins of connector D3973A at E3-1 shelf and connector D662 at system A pressure transmitter, T424 (WDM 29-32-11):

D3973A		D662
pin A9	. . . . .	pin 3
pin B9	. . . . .	pin 2

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the DEU-1, M1808. To install it, do this task: Display Electronic Unit Installation, AMM TASK 31-62-21-400-801.

3) Re-connect connector D662.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

4) Pressurize the hydraulic system A. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

5) If the system A hydraulic pressure indication on the center instrument panel, P2, is between 2900 and 3200 psi, then you corrected the fault. Do this step to complete the task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(e) If you do not find a problem with the wiring, then do these steps and continue.

1) Re-install the DEU-1, M1808. To install it, do this task: Display Electronic Unit Installation, AMM TASK 31-62-21-400-801.

2) Re-connect connector D662.

(2) Replace the system A pressure transmitter, T424. These are the tasks:

- System Pressure Transmitter Removal, AMM TASK 29-31-12-000-801
- System Pressure Transmitter Installation, AMM TASK 29-31-12-400-801

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system A. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) If the system A hydraulic pressure indication on the center instrument panel, P2, is between 2900 and 3200 psi, then you corrected the fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(c) If the system A hydraulic pressure indication on the center instrument panel, P2, is zero psi, then continue.

(3) Replace the display electronic unit 1 (DEU-1), M1808. These are the tasks:

- Display Electronic Unit Removal, AMM TASK 31-62-21-000-801
- Display Electronic Unit Installation, AMM TASK 31-62-21-400-801

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(a) Pressurize the hydraulic system A. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) If the system A hydraulic pressure indication on the center instrument panel, P2, is between 2900 and 3200 psi, then you corrected the fault. Do this step to complete the task:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

——— END OF TASK ———

804. System B Hydraulic Pressure Reads Zero - Fault Isolation

A. Description

(1) The system pressure transmitter monitors the system pressure and sends the information to the flight compartment. The system pressure transmitter sends pressure signals between 0 and 4000 psi.

(2) The hydraulic system B pressure transmitter is on the hydraulic system B pressure module.

B. Possible Causes

(1) Hydraulic leak

(2) Wiring problem

(3) System B pressure transmitter, T423

(4) Display electronic unit 2 (DEU-2), M1809

C. Related Data

(1) (SSM 29-32-11)

(2) (WDM 29-32-11)

D. Initial Evaluation

(1) Do this check of the system B hydraulic pressure:

WARNING

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(a) Pressurize the hydraulic system B. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) If the system B hydraulic quantity is low and the pressure indication on the center instrument panel, P2, is zero psi, then do the Fault Isolation Procedure - Hydraulic Fluid Leakage.

(c) If the system B hydraulic quantity is not low and the pressure indication on the center instrument panel, P2, is zero psi, then do the Fault Isolation Procedure - Indication Problem.

(d) If the system B hydraulic pressure indication on the center instrument panel, P2, is between 2900 and 3200 psi, then there was an intermittent fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

E. Fault Isolation Procedure - Hydraulic Fluid Leakage

(1) Do a check of the hydraulic system B for external leakage.

(a) Do this task: Hydraulic System External Leakage Check, AMM TASK 29-00-00-790-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If you find external leakage of hydraulic fluid, then repair the cause of the leak and continue.

2) If you do not find external leakage of hydraulic fluid, then do continue.

(b) Fill the system B hydraulic reservoir with hydraulic fluid. To fill it, do this task: Hydraulic Reservoir Servicing, AMM TASK 12-12-00-610-801.

WARNING

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(c) Pressurize the hydraulic system B. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(d) If the system B hydraulic pressure indication on the center instrument panel, P2, is between 2900 and 3200 psi, then you corrected the fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(e) If the system B hydraulic pressure indication on the center instrument panel, P2, is zero then do the Fault Isolation Procedure - Indication Problem

F. Fault Isolation Procedure - Indication Problem

(1) Do this wiring check:

(a) Remove the display electronic unit 2 (DEU-2), M1809. To remove it, do this task: Display Electronic Unit Removal, AMM TASK 31-62-21-000-801.

(b) Disconnect connector D660 from the system B pressure transmitter, T423.

(c) Do a wiring check between these pins of connector D3975A at E3-1 shelf and connector D660 at system B pressure transmitter, T423 (WDM 29-32-11):

D3975A		D660
pin A9	. . . . .	pin 3
pin B9	. . . . .	pin 2

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the DEU-2, M1809. To install it, do this task: Display Electronic Unit Installation, AMM TASK 31-62-21-400-801.

3) Re-connect connector D660.

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4) Pressurize the hydraulic system B. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

5) If the system B hydraulic pressure indication on the center instrument panel, P2, is between 2900 and 3200 psi, then you corrected the fault. Do this step to complete the task:

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(e) If you do not find a problem with the wiring, then continue.

1) Re-install the DEU-2, M1809. To install it, do this task: Display Electronic Unit Installation, AMM TASK 31-62-21-400-801.

2) Re-connect connector D660.

(2) Replace the system B pressure transmitter, T423. These are the tasks:

- System Pressure Transmitter Removal, AMM TASK 29-31-12-000-801
- System Pressure Transmitter Installation, AMM TASK 29-31-12-400-801

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(a) Pressurize the hydraulic system B. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) If the system B hydraulic pressure indication on the center instrument panel, P2, is between 2900 and 3200 psi, then you corrected the fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(c) If the system B hydraulic pressure indication on the center instrument panel, P2, is zero psi, then continue.

(3) Replace the display electronic unit 2 (DEU-2), M1809. These are the tasks:

- Display Electronic Unit Removal, AMM TASK 31-62-21-000-801
- Display Electronic Unit Installation, AMM TASK 31-62-21-400-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

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(a) Pressurize the hydraulic system B. To pressurize it, do this task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) If the system B hydraulic pressure indication on the center instrument panel, P2, is between 2900 and 3200 psi, then you corrected the fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

——— END OF TASK ———

805. ELEC 1 OVERHEAT Light On - Fault Isolation

A. Description

(1) The ELEC 1 OVERHEAT light comes ON when the hydraulic fluid temperature in the case drain line below the Electric Motor-Driven Pump (EMDP) increases to more than normal.

(2) The hydraulic fluid Overheat Warning Switch in the case drain line monitors the temperature of the EMDP. This Overheat Warning Switch supplies a ground for the amber OVERHEAT light when the case drain fluid temperature is more than 225°F (107°C).

(3) The hydraulic fluid Overheat Warning Switch for the hydraulic system B is installed directly below the hydraulic system B EMDP.

(4) The EMDP Overheat Warning Switch monitors the temperature of the EMDP electric motor. It is internal to the EMDP and it closes at 235°F (113°C). The EMDP Overheat Warning Switch supplies a ground for the amber OVERHEAT light separate from the hydraulic fluid Overheat Warning Switch.

(5) The hydraulic fluid is cooled through the heat exchanger in the main fuel tank.

(6) This is normal operation of the ELEC 1 OVERHEAT light:

(a) When the system B EMDP operates, the ELEC 1 OVERHEAT light is off.

(b) When the temperature of the system B EMDP case drain fluid is above 225°F (107°C), the ELEC 1 OVERHEAT light will come ON.

(c) When the temperature of the system B EMDP electric motor is above 235°F (113°C), the ELEC 1 OVERHEAT light will come ON even if the case drain fluid is below 225°F (107°C).

B. Possible Causes

(1) Wiring problem

(2) Hydraulic system B hydraulic fluid Overheat Warning Switch, S799

(3) Supply line blockage

(4) Hydraulic system B EMDP Case Drain Filter

(5) Hydraulic system B EMDP, M1104

(6) Hydraulic system B Engine Driven Pump (EDP)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(7) System B Pressure Module Relief Valve is open

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	11	C00317	INDICATOR MASTER DIM SECT 5

D. Related Data

(1) SSM 29-34-11

(2) SSM 33-18-35

(3) WDM 29-34-11

(4) WDM 33-18-35

E. Initial Evaluation

(1) Do this check of the ELEC 1 OVERHEAT light, L4:

(a) Make sure that the system B EMDP temperature is less than 185°F (85°C).

NOTE: If an overheat condition has occurred, then the EMDP temperature must decrease to less than 185°F (85°C) for the temperature switch to reset to the open position.

(b) Make sure that the fuel tank has a minimum of 250 gal (946 l) (1675 lb (760 kg)) of fuel in it.

WARNING

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(c) Pressurize the hydraulic system B with the EMDP. This is the task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(d) If the ELEC 1 OVERHEAT light is ON after the EMDP has operated long enough for its temperature to become stable, then do the Fault Isolation Procedure below.

(e) If the ELEC 1 OVERHEAT light is not ON, then there was an intermittent fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

F. Fault Isolation Procedure

(1) Do this check of the wire bundle W5230:

NOTE: Damage to the wire bundle W5230 is possible because of contact friction with the NGS SOV.

(a) Get access to the Nitrogen Generation System (NGS) Shutoff Valve (SOV). This is the task: NGS Shutoff Valve Removal, AMM TASK 47-32-01-000-801.

(b) Do a visual inspection of the wire bundle for chafing and damage.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If you find a problem with the wire bundle W5230, then do these steps:
   a) Repair the wiring.
      <1> Make sure that the protective sleeve of the wire bundle is placed correctly.
      <2> If it is necessary, adjust the wire bundle separation distance between the NGS SOV, adjacent equipment, and structure (SWPM 20-10-11).
   b) Close access to the NGS SOV. This is the task: NGS Shutoff Valve Installation, AMM TASK 47-32-01-400-801.
   c) Do the Repair Confirmation at the end of this task.
2) If you did not find a problem with the wire bundle W5230, then do these steps and continue:
   a) Make sure that the protective sleeve of the wire bundle is placed correctly.
   b) If it is necessary, adjust the wire bundle separation distance between the NGS SOV, adjacent equipment, and structure (SWPM 20-10-11).
   c) Close access to the NGS SOV. This is the task: NGS Shutoff Valve Installation, AMM TASK 47-32-01-400-801.

(2) Do this wiring check:
   (a) Disconnect connector D2690 from the system B hydraulic fluid Overheat Warning Switch, S799.
   (b) Disconnect connector D2666 from the system B EMDP, M1104.
   (c) Do a check for 28v DC from pin B to pin A (ground) of connector D2690 (WDM 29-34-11).
   (d) Do a check for 28v DC from pin 7 of connector D2666 to ground (WDM 29-34-11).
   (e) If you do not find 28v DC or if you do not find ground at pin A of connector D2690, then do these steps:
      1) Repair the wiring (WDM 29-34-11).
      2) Re-connect connector D2690.
      3) Do a check for ground at pin 7 of the system B EMDP connector.
      4) If you find ground at pin 7 of the EMDP connector and the EMDP temperature is less than 185°F (85°C), then replace the system B EMDP, M1104. These are the tasks:
         • Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
         • Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001
      5) Re-connect connector D2666.
      6) Do the Repair Confirmation at the end of this task.
   (f) If you find 28v DC and there is ground at pin A of connector D2690, then continue.

(3) Replace the hydraulic fluid Overheat Warning Switch, S799. These are the tasks:
   • Hydraulic Fluid Overheat Warning Switch Removal, AMM TASK 29-32-12-000-801
   • Hydraulic Fluid Overheat Warning Switch Installation, AMM TASK 29-32-12-400-801
   (a) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(4) Do a check for correct connection and condition of the quick-disconnect for hydraulic system B.

NOTE: The quick-disconnect for the hydraulic system B is on the inboard side of the hydraulic system B reservoir.

(a) Make sure that there are no missing or damaged parts in the quick-disconnect.

NOTE: Missing or damaged parts in the quick-disconnect can block the hydraulic line or cause damage to other parts in the system.

(b) If the quick-disconnect is not installed correctly, then do these steps:

1) Re-install the quick-disconnect correctly.

2) Do the Repair Confirmation at the end of this task.

(c) If the quick-disconnect is damaged, then do these steps:

1) Replace the quick-disconnect.

2) Repair any damage cause by missing parts from the quick-disconnect.

3) Do the Repair Confirmation at the end of this task.

(d) If the quick-disconnect is installed correctly and in good condition, then continue.

(5) Do a check for contamination of the Case Drain Filter:

(a) Remove the system B EMDP Case Drain Filter element. This is the task: EMDP Case Drain Filter Element Removal, AMM TASK 29-11-41-000-801.

(b) Look for metal contamination on the EMDP Case Drain Filter.

(c) If you find metal contamination, then do these steps:

1) Repair or replace those component that have metal contamination. This is the task: EMDP Case Drain Filter - Metal Contamination Check, AMM TASK 29-11-41-700-801.

2) Re-install the system B Case Drain Filter element. This is the task: EMDP Case Drain Filter Element Installation, AMM TASK 29-11-41-400-801.

3) If the system B EMDP is replaced because metal contamination is found in the filter, then flush the hydraulic lines. This is the task: Hydraulic System A or B Flushing, AMM TASK 29-11-00-170-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find metal contamination, then continue.

1) Re-install the system B Case Drain Filter element. This is the task: EMDP Case Drain Filter Element Installation, AMM TASK 29-11-41-400-801.

(6) Replace the system B EMDP, M1104. These are the tasks:

- Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
- Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001

(a) Do the Repair Confirmation at the end of this task.

(b) If the ELEC 1 OVERHEAT light is ON, then continue.

(7) Do a check of the EDP case drain flow:

(a) Run the system B EDP while maintaining hydraulic pressure with the associated EMDP.

(b) Remove the pressure from the system B EDP while maintaining hydraulic pressure with the associated EMDP.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) If the light extinguishes, the system B EDP is not the cause of the ELEC 1 OVERHEAT light.

(d) If the light persists, then continue.

NOTE: This test may take some time to complete as the EDP case drain line does not have a temperature switch. This means that the fluid will have to be cooled and returned to the EMDP case drain line prior to the light extinguishing.

(8) Replace the system B EDP. These are the tasks:

- Hydraulic Systems A and B Engine-Driven Pump (EDP) Removal, AMM TASK 29-11-11-000-801-001
- Hydraulic Systems A and B Engine-Driven Pump (EDP) Installation, AMM TASK 29-11-11-400-801-001

(a) Do the Repair Confirmation at the end of this task.

(9) Do this check of the System B Pressure Module Relief Valve:

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(a) Pressurize the hydraulic system B. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) Read the temperature at the System B Pressure Module Relief Valve. This is the task: Thermal Imaging Checks, AMM TASK 20-10-75-700-801.

1) If there is a significant temperature on the System B Pressure Module Relief Valve and it is similar to that of the connected return line, then replace the System B Pressure Module Relief Valve. This is the task:

NOTE: The relief valve is open and porting fluid back to the reservoir.

- Hydraulic Systems A and B Pressure Module Relief Valve Removal, AMM TASK 29-11-71-000-804
- Hydraulic Systems A and B Pressure Module Relief Valve - Installation, AMM TASK 29-11-71-400-804

a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the ELEC 1 OVERHEAT light, L4:

(a) Make sure that the fuel tank has a minimum of 250 gal (946 l) (1675 lb (760 kg)) of fuel in it.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

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(b) Pressurize the hydraulic system B with the EMDP. This is the task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(c) If the ELEC 1 OVERHEAT light is not ON, then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(d) If the ELEC 1 OVERHEAT light is ON, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

806. ELEC 2 OVERHEAT Light On - Fault Isolation

A. Description

(1) The ELEC 2 OVERHEAT light comes ON when the hydraulic fluid temperature in the case drain line below the EMDP increases to more than normal.

(2) The hydraulic fluid Overheat Warning Switch in the case drain line monitors the temperature of the EMDP. This Overheat Warning Switch supplies a ground for the amber OVERHEAT light when the case drain fluid temperature is more than 225°F (107°C).

(3) The hydraulic fluid Overheat Warning Switch for the hydraulic system A is installed directly below the hydraulic system A EMDP.

(4) The EMDP Overheat Warning Switch monitors the temperature of the EMDP electric motor. It is internal to the EMDP and it closes at 235°F (113°C). The EMDP Overheat Warning Switch supplies a ground for the amber OVERHEAT light separate from the hydraulic fluid Overheat Warning Switch.

(5) The hydraulic fluid is cooled through the heat exchanger in the main fuel tank.

(6) This is normal operation of the ELEC 2 OVERHEAT light:

(a) When the system A EMDP operates, the ELEC 2 OVERHEAT light is OFF.

(b) When the temperature of the system A EMDP case drain fluid is above 225°F (107°C), the ELEC 2 OVERHEAT light will come ON.

(c) When the temperature of the system A EMDP is above 235°F (113°C), the ELEC 2 OVERHEAT light will come ON even if the case drain fluid is below 225°F (107°C).

B. Possible Causes

(1) Wiring problem

(2) Hydraulic System A hydraulic fluid Overheat Warning Switch, S798

(3) Supply line blockage

(4) Hydraulic system A EMDP Case Drain Filter

(5) Hydraulic system A EMDP, M1103

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(6) Hydraulic system A EDP

(7) System A Pressure Module Relief Valve is open

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	12	C00318	INDICATOR MASTER DIM SECT 6

D. Related Data

(1) (SSM 29-34-11)

(2) (SSM 33-18-35)

(3) (WDM 29-34-11)

(4) (WDM 33-18-35)

E. Initial Evaluation

(1) Do this check of the ELEC 2 OVERHEAT light, L3:

(a) Make sure that the system A EMDP temperature is less than 185°F (85°C).

NOTE: If an overheat condition has occurred, then the EMDP temperature must decrease to less than 185 degrees F (85 degrees C) for the temperature switch to reset to the open position.

(b) Make sure that the fuel tank has a minimum of 250 gallons (1675 lbs, 760 kg) of fuel in it.



WARNING

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(c) Pressurize the hydraulic system A with the EMDP. This is the task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(d) If the ELEC 2 OVERHEAT light is ON after the EMDP has operated long enough for its temperature to become stable, then do the Fault Isolation Procedure below.

(e) If the ELEC 2 OVERHEAT light is not ON, then there was an intermittent fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

F. Fault Isolation Procedure

(1) Do this wiring check:

(a) Disconnect connector D2688 from the system A hydraulic fluid Overheat Warning Switch, S798.

(b) Disconnect connector D2664 from the system A EMDP, M1103.

(c) Do a check for 28v DC from pin B to pin A (ground) of connector D2688 (WDM 29-34-11).

(d) Do a check for 28v DC from pin 7 of connector D2664 to ground (WDM 29-34-11).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) If you do not find 28v DC or if you do not find ground at pin A of connector D2688, then do these steps:

1) Repair the wiring (WDM 29-34-11).

2) Re-connect connector D2688.

3) Do a check for ground at pin 7 of the system A EMDP connector.

4) If you find ground at pin 7 of the EMDP connector and the EMDP temperature is less than 185°F (85°C), then replace the system A EMDP, M1103. These are the tasks:

- Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
- Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001

5) Re-connect connector D2664.

6) Do the Repair Confirmation at the end of this task.

(f) If you find 28v DC and there is ground at pin A of connector D2688, then continue.

(2) Replace the hydraulic fluid Overheat Warning Switch, S798. These are the tasks:

- Hydraulic Fluid Overheat Warning Switch Removal, AMM TASK 29-32-12-000-801
- Hydraulic Fluid Overheat Warning Switch Installation, AMM TASK 29-32-12-400-801

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

(3) Do a check for correct connection and condition of the quick-disconnect for hydraulic system A.

NOTE: The quick-disconnect for the hydraulic system A is at the bottom the hydraulic system A reservoir.

(a) Make sure that there are no missing or damaged parts in the quick-disconnect.

NOTE: Missing or damaged parts in the quick-disconnect can block the hydraulic line or cause damage to other parts in the system.

(b) If the quick-disconnect is not installed correctly, then do these steps:

1) Re-install the quick-disconnect correctly.

2) Do the Repair Confirmation at the end of this task.

(c) If the quick-disconnect is damaged, then do these steps:

1) Replace the quick-disconnect.

2) Repair any damage cause by missing parts from the quick-disconnect.

3) Do the Repair Confirmation at the end of this task.

(d) If the quick-disconnect is installed correctly and in good condition, then continue.

(4) Do a check for contamination of the EMDP Case Drain Filter:

(a) Remove the system A EMDP Case Drain Filter element. This is the task: EMDP Case Drain Filter Element Removal, AMM TASK 29-11-41-000-801.

(b) Look for metal contamination on the EMDP Case Drain Filter.

(c) If you find metal contamination, then do these steps:

1) Repair or replace those component that have metal contamination. This is the task: EMDP Case Drain Filter - Metal Contamination Check, AMM TASK 29-11-41-700-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Re-install the system A Case Drain Filter element. This is the task: EMDP Case Drain Filter Element Installation, AMM TASK 29-11-41-400-801.

3) If the system A EMDP is replaced because metal contamination is found in the filter, then flush the hydraulic lines. This is the task: Hydraulic System A or B Flushing, AMM TASK 29-11-00-170-801.

4) Do the Repair Confirmation at the end of this task.

(d) If you do not find metal contamination, then continue.

1) Re-install the system A Case Drain Filter element. This is the task: EMDP Case Drain Filter Element Installation, AMM TASK 29-11-41-400-801.

(5) Replace the system A EMDP, M1103. These are the tasks:

- Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-11-21-000-801-001
- Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-11-21-400-801-001

(a) Do the Repair Confirmation at the end of this task.

(b) If the ELEC 2 OVERHEAT light is ON, then continue.

(6) Do a check of the EDP case drain flow:

(a) Run the system A EDP while maintaining hydraulic pressure with the associated EMDP.

(b) Remove the pressure from the system A EDP while maintaining hydraulic pressure with the associated EMDP.

(c) If the light extinguishes, the system A EDP is not the cause of the ELEC 2 OVERHEAT light.

(d) If the light persists, then continue.

NOTE: This test may take some time to complete as the EDP case drain line does not have a temperature switch. This means that the fluid will have to be cooled and returned to the EMDP case drain line prior to the light extinguishing.

(7) Replace the system A EDP. These are the tasks:

- Hydraulic Systems A and B Engine-Driven Pump (EDP) Removal, AMM TASK 29-11-11-000-801-001
- Hydraulic Systems A and B Engine-Driven Pump (EDP) Installation, AMM TASK 29-11-11-400-801-001

(a) Do the Repair Confirmation at the end of this task.

(8) Do this check of the System A Pressure Module Relief Valve:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system A. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) Read the temperature at the System A Pressure Module Relief Valve. This is the task: Thermal Imaging Checks, AMM TASK 20-10-75-700-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If there is a significant temperature on the System A Pressure Module Relief Valve and it is similar to that of the connected return line, then replace the System A Pressure Module Relief Valve. These are the tasks:

NOTE: The relief valve is open and porting fluid back to the reservoir.

- Hydraulic Systems A and B Pressure Module Relief Valve Removal, AMM TASK 29-11-71-000-804
- Hydraulic Systems A and B Pressure Module Relief Valve - Installation, AMM TASK 29-11-71-400-804

a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the ELEC 2 OVERHEAT light, L3:

(a) Make sure that the fuel tank has a minimum of 250 gallons (1675 lbs, 760 kg) of fuel in it.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Pressurize the hydraulic system A with the EMDP. This is the task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(c) If the ELEC 2 OVERHEAT light is not ON, then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

——— END OF TASK ———

807. Hydraulic Fluid Quantity Indication Problem - Fault Isolation

A. Description

(1) The Hydraulic Fluid Quantity Transmitter/Indicators show the hydraulic fluid level in the Hydraulic System A and B Reservoirs.

(2) The indicator shows the fluid quantity on the mechanical gage on the reservoir in the Main Landing Gear (MLG) Wheel Well (W/W).

(3) The transmitter sends signal to the Display Electronic Unit (DEU)s. The fluid quantity shows on the engine display.

(4) This fault can cause the 31-63210 HYDRAULIC OIL QUANTITY SYS A INVALID or 31-63220 HYDRAULIC OIL QUANTITY SYS B INVALID maintenance message to show on the Control Display Unit (CDU).

B. Possible Causes

(1) Hydraulic Fluid Quantity Transmitter/Indicator

(2) Wiring

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Related Data

(1) SSM 29-31-11

(2) WDM 29-31-11

D. Initial Evaluation

(1) If the hydraulic fluid quantity shows on the engine display, then there was an intermittent fault.

(2) If the hydraulic fluid quantity does not show on the engine display, then do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Replace the applicable Hydraulic Fluid Quantity Transmitter/Indicator. These are the tasks:

- Hydraulic Fluid Quantity Transmitter/Indicator Removal, AMM TASK 29-33-12-000-801
- Hydraulic Fluid Quantity Transmitter/Indicator Installation, AMM TASK 29-33-12-400-801

(a) If the test in the installation task for the Hydraulic Fluid Quantity Transmitter/Indicator is satisfactory, then you corrected the problem.

(b) If the test in the installation task for the Hydraulic Fluid Quantity Transmitter/Indicator is not satisfactory, then continue.

(2) For the Hydraulic System A Fluid Quantity Indication, do this check of the wiring (WDM 29-31-11):

(a) Remove the DEU-1, M1808 and DEU-2, M1809. This is the task: Display Electronic Unit Removal, AMM TASK 31-62-21-000-801.

(b) Disconnect connector D674 from the Hydraulic Oil Quantity System A Transmitter, T436 in the MLG W/W.

(c) Do a wiring check as follows:

	SYS A HYD QTY
DEU-1, M1808	**XMTR, T436**
D3973A	**D674**
pin J4	pin 2
pin K4	pin 1
D3973B	**D674**
pin E14	pin 3

	SYS A HYD QTY
DEU-2, M1809	**XMTR, T436**
D3975A	**D674**
pin J4	pin 2
pin K4	pin 1
D3975B	**D674**
pin E14	pin 3

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the DEU-1, M1808 and DEU-2, M1809. This is the task: Display Electronic Unit Installation, AMM TASK 31-62-21-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Re-connect connector D674.

d) If the Hydraulic System A fluid quantity shows on the engine display, then you corrected the problem.

(3) For the Hydraulic System B Fluid Quantity Indication, do this check of the wiring (WDM 29-31-11):

(a) Remove the DEU-1, M1808 and DEU-2, M1809. This is the task: Display Electronic Unit Removal, AMM TASK 31-62-21-000-801.

(b) Disconnect connector D2674 from the Hydraulic Oil Quantity System B Transmitter, T184 in the MLG W/W.

(c) Do a wiring check as follows:

	SYS B HYD QTY
DEU-1, M1808	**XMTR, T184**
D3973D	**D2674**
pin J4	pin 2
pin K4	pin 1
D3973E	**D2674**
pin E14	pin 3

	SYS B HYD QTY
DEU-2, M1809	**XMTR, T184**
D3975D	**D2674**
pin J4	pin 2
pin K4	pin 1
D3975E	**D2674**
pin E14	pin 3

1) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the DEU-1, M1808 and DEU-2, M1809. This is the task: Display Electronic Unit Installation, AMM TASK 31-62-21-400-801.

c) Re-connect connector D2674.

d) If the Hydraulic System B fluid quantity shows on the engine display, then you corrected the problem.

——— END OF TASK ———

808. Hydraulic Reservoir Gage Problem - Fault Isolation

A. Description

(1) The hydraulic fluid quantity transmitter/indicators show the hydraulic fluid level in the hydraulic system A and B reservoirs.

(2) The indicator shows the fluid quantity on the mechanical gage on the reservoir in the main landing gear wheel well.

(3) The transmitter sends signal to the display electronic units (DEUs). The fluid quantity shows on the engine display.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(4) Reservoir quantity may be reduced in flight by thermal contraction of the hydraulic fluid. Reservoir quantity should be checked on the ground when the hydraulic fluid has reached ambient temperature.

B. Possible Causes

(1) Hydraulic fluid quantity transmitter/indicator

C. Related Data

(1) (SSM 29-31-11)

(2) (WDM 29-31-11)

D. Initial Evaluation

(1) Do this check of the hydraulic reservoir gage:

(a) Fully lower the hydraulic fluid to the applicable hydraulic system reservoir. To do this, do this task: Hydraulic Reservoir Servicing, AMM TASK 12-12-00-610-801.

(b) Slowly fill the applicable hydraulic reservoir. To do this, do this task: Hydraulic Reservoir Servicing, AMM TASK 12-12-00-610-801.

NOTE: Make sure the hydraulic reservoir gage and the HYD Q% indication on the engine display show increasing fluid level as you fill the reservoir.

(c) If the hydraulic reservoir gage and the HYD Q% indication on the engine display does not show increasing fluid level, then do the Fault Isolation Procedure below.

(d) If the hydraulic reservoir gage and the HYD Q% indication on the engine display show increasing fluid level, then there was an intermittent fault.

E. Fault Isolation Procedure

(1) Replace the applicable hydraulic fluid quantity transmitter/indicator. These are the tasks:

- Hydraulic Fluid Quantity Transmitter/Indicator Removal, AMM TASK 29-33-12-000-801
- Hydraulic Fluid Quantity Transmitter/Indicator Installation, AMM TASK 29-33-12-400-801

(a) If the test in the installation task for the hydraulic fluid quantity transmitter/indicator is satisfactory, then you corrected the fault.

END OF TASK

809. STANDBY HYD LOW QUANTITY Light On - Fault Isolation

A. Description

(1) The standby hydraulic system reservoir low quantity switch sends a signal when the fluid level is less than 50 percent. The signal goes to the STANDBY HYD LOW QUANTITY light amber light on the flight control panel.

(2) This is normal operation of the STANDBY HYD LOW QUANTITY light:

(a) When the standby EMDP operates with less than 50 percent of hydraulic fluid, the STANDBY HYD LOW QUANTITY light will come on.

B. Possible Causes

(1) Hydraulic leakage in the standby hydraulic system

(2) STANDBY HYD LOW QUANTITY light, L4

(3) Standby hydraulic system reservoir low quantity switch, S49

(4) Wiring problem

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	14	C00316	INDICATOR MASTER DIM SECT 4

D. Related Data

(1) (SSM 29-31-11)

(2) (SSM 33-18-31)

(3) (WDM 29-31-11)

(4) (WDM 33-18-31)

E. Initial Evaluation

(1) Do this check of the STANDBY HYD LOW QUANTITY light, L4:

(a) Make sure the standby hydraulic reservoir is full.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Pressurize the standby hydraulic system. To pressurize it, do this task: Standby Hydraulic System Pressurization, AMM TASK 29-21-00-000-801.

(c) If the STANDBY HYD LOW QUANTITY light comes on, then do the Fault Isolation Procedure below.

(d) If the STANDBY HYD LOW QUANTITY light does not come on, then there was an intermittent fault. Do this step to complete the task:

1) Do this task: Standby Hydraulic System Power Removal, AMM TASK 29-21-00-000-802.

F. Fault Isolation Procedure

(1) Do an external leakage check of the standby hydraulic system. To do it, do this task: Hydraulic System External Leakage Check, AMM TASK 29-00-00-790-801.

(a) If there is external hydraulic leakage, then do these steps:

1) Repair or replace the component that causes the leakage (AMM TASK 29-00-00-790-801).

2) Do the Repair Confirmation at the end of this task.

(b) If there is no external hydraulic leakage, then continue.

(2) Do an internal leakage check of the standby hydraulic system (AMM 29-00-00-6).

(a) If there is internal hydraulic leakage, then do these steps:

1) Repair or replace the component that causes the leakage.

2) Do the Repair Confirmation at the end of this task.

(b) If there is no internal hydraulic leakage, then continue.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(3) Replace the STANDBY HYD LOW QUANTITY light, L4 on the flight controls panel, P5-3 (WDM 29-31-11).

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

(4) Do this wiring check:

(a) Disconnect connector D678 from the standby hydraulic system reservoir low quantity switch, S49.

NOTE: You can find this low quantity switch on the standby hydraulic reservoir.

(b) Do a check for 28v DC from pin 2 to pin 1 (ground) of connector D678 (WDM 29-31-11).

(c) If there is not 28v DC at pin 2 or ground at pin 1 of connector D678, then do these steps:

1) Repair the wiring (WDM 29-33-11).

2) Re-connect connector D678.

3) Do the Repair Confirmation at the end of this task.

(d) If there is 28v DC at pin 2 of connector D678, then continue.

(5) Replace the standby hydraulic system reservoir low quantity switch, S49. These are the tasks:

- Standby Reservoir Low Quantity Switch Removal, AMM TASK 29-33-21-000-801
- Standby Reservoir Low Quantity Switch Installation, AMM TASK 29-33-21-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the STANDBY HYD LOW QUANTITY light, L4:

(a) Make sure the standby hydraulic reservoir is full.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Pressurize the standby hydraulic system. To pressurize it, do this task: Standby Hydraulic System Pressurization, AMM TASK 29-21-00-000-801.

(c) If the STANDBY HYD LOW QUANTITY light does not come on, then you corrected the fault. Do this step to complete the task:

1) Do this task: Standby Hydraulic System Power Removal, AMM TASK 29-21-00-000-802.

END OF TASK

810. ELEC 1 LOW PRESSURE Light On - Fault Isolation

A. Description

(1) The ELEC 1 LOW PRESSURE light comes ON when the EMDP for the hydraulic system B is below 1200 psi (8274 kPa).

(2) This is normal operation of the ELEC 1 LOW PRESSURE light:

(a) When the system B EMDP operates, the ELEC 1 LOW PRESSURE light is OFF.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) When the pressure of the system B EMDP is below normal, the ELEC 1 LOW PRESSURE light will come ON.

(c) When the system B EMDP does not operate, the ELEC 1 LOW PRESSURE light will come ON.

B. Possible Causes

(1) Hydraulic system B EMDP, M1104

(2) ELEC 1 LOW PRESSURE light, L2

(3) Hydraulic system B EMDP Low Pressure Switch, S795

(4) Wiring problem

(5) System B Pressure Module Relief Valve is open

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	11	C00317	INDICATOR MASTER DIM SECT 5

D. Related Data

(1) SSM 29-33-11

(2) SSM 33-18-35

(3) WDM 29-33-11

(4) WDM 33-18-35

E. Initial Evaluation

(1) Do this check of the ELEC 1 LOW PRESSURE light, L2:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS AND THE THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system B with the EMDP. This is the task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(b) If the system B EMDP does not operate, or operates intermittently, then do this task:

Electrical Power to the System B EMDP Problem - Fault Isolation, 29-10 TASK 804.

(c) If the ELEC 1 LOW PRESSURE light goes OFF, then there was an intermittent fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(d) If the ELEC 1 LOW PRESSURE light is ON, then do the Fault Isolation Procedure below.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Fault Isolation Procedure

(1) Do this check of the hydraulic system B EMDP:

(a) Look at the hydraulic pressure indication on the engine display.

(b) If the hydraulic system B pressure is less than 2800 psi (19,305 kPa), then do this task: System B Hydraulic Pressure Below 2800 psi Problem - Fault Isolation, 29-10 TASK 811.

(c) If the hydraulic system B pressure is greater than 2800 psi (19,305 kPa), then continue.

(2) Replace the ELEC 1 LOW PRESSURE light, L2 on the Hydraulic Control Module, P5-8 (WDM 29-33-11).

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

(3) Do this wiring check:

(a) Disconnect connector D2682 from the hydraulic system B EMDP Low Pressure Switch, S795.

(b) Do a check for 28v DC from pin 2 to pin 3 (ground) of connector D2682 (WDM 29-33-11).

(c) If there is not 28v DC at pin 2 or ground at pin 3 of connector D2682, then do these steps:

1) Repair the wiring (WDM 29-33-11).

2) Re-connect connector D2682.

3) Do the Repair Confirmation at the end of this task.

(d) If there is 28v DC at pin 2 of connector D2682, then continue.

(4) Replace the hydraulic system B EMDP Low Pressure Switch, S795. These are the tasks:

- Hydraulic System A or B EMDP Low Pressure Switch Removal, AMM TASK 29-34-21-000-801
- Hydraulic System A or B EMDP Low Pressure Switch Installation, AMM TASK 29-34-21-400-801

(a) Do the Repair Confirmation at the end of this task.

(5) Do this check of the System B Pressure Module Relief Valve:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system B. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) Read the temperature at the System B Pressure Module Relief Valve. This is the task: Thermal Imaging Checks, AMM TASK 20-10-75-700-801.

1) If there is a significant temperature on the System B Pressure Module Relief Valve and it is similar to that of the connected return line, then replace the System B Pressure Module Relief Valve. These are the tasks:

NOTE: The relief valve is open and porting fluid back to the reservoir.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Hydraulic Systems A and B Pressure Module Relief Valve Removal, AMM TASK 29-11-71-000-804
- Hydraulic Systems A and B Pressure Module Relief Valve - Installation, AMM TASK 29-11-71-400-804

a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the ELEC 1 LOW PRESSURE light, L2:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS AND THE THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system B with the EMDP. This is the task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(b) If the ELEC 1 LOW PRESSURE light goes OFF, then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(c) If the ELEC 1 LOW PRESSURE light does not go OFF, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

811. ELEC 2 LOW PRESSURE Light On - Fault Isolation

A. Description

(1) The ELEC 2 LOW PRESSURE light comes ON when the EMDP for the hydraulic system A is below 1200 psi (8274 kPa).

(2) This is normal operation of the ELEC 2 LOW PRESSURE light:

(a) When the system A EMDP operates, the ELEC 2 LOW PRESSURE light is OFF.

(b) When the pressure of the system A EMDP is below normal, the ELEC 2 LOW PRESSURE light will come ON.

(c) When the system A EMDP does not operate, the ELEC 2 LOW PRESSURE light will come ON.

B. Possible Causes

(1) Hydraulic system A EMDP, M1103

(2) ELEC 2 LOW PRESSURE light, L1

(3) Hydraulic system A EMDP Low Pressure Switch, S794

(4) Wiring problem

(5) System A Pressure Module Relief Valve is open

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	12	C00318	INDICATOR MASTER DIM SECT 6

D. Related Data

(1) (SSM 29-33-11)

(2) (SSM 33-18-35)

(3) (WDM 29-33-11)

(4) (WDM 33-18-35)

E. Initial Evaluation

(1) Do this check of the ELEC 2 LOW PRESSURE light, L1:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS AND THE THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system A with the EMDP. This is the task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(b) If the system A EMDP does not operate, then do this task: Electrical Power to the System A EMDP Problem - Fault Isolation, 29-10 TASK 803.

(c) If the ELEC 2 LOW PRESSURE light goes OFF, then there was an intermittent fault. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(d) If the ELEC 2 LOW PRESSURE light is ON, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do this check of the hydraulic system A EMDP:

(a) Look at the hydraulic pressure indication on the engine display.

(b) If the hydraulic system A pressure is less than 2800 psi (19,305 kPa), then, do this task: System A Hydraulic Pressure Below 2800 psi Problem - Fault Isolation, 29-10 TASK 810.

(c) If the hydraulic system A pressure is greater than 2800 psi (19,305 kPa), then continue.

(2) Replace the ELEC 2 LOW PRESSURE light, L1 on the hydraulic control module, P5-8 (WDM 29-33-11).

(a) Do the Repair Confirmation at the end of this task.

(3) Do this wiring check:

(a) Disconnect connector D2680 from the hydraulic system A EMDP Low Pressure Switch, S794.

(b) Do a check for 28v DC from pin 2 to pin 3 (ground) of connector D2680 (WDM 29-33-11).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) If there is not 28v DC at pin 2 or ground at pin 3 of connector D2680, then do these steps:

1) Repair the wiring.

2) Re-connect connector D2680.

3) Do the Repair Confirmation at the end of this task.

(d) If there is 28v DC at pin 2 of connector D2680, then continue.

(4) Replace the hydraulic system A Low Pressure Switch, S794. These are the tasks:

- Hydraulic System A or B EMDP Low Pressure Switch Removal, AMM TASK 29-34-21-000-801
- Hydraulic System A or B EMDP Low Pressure Switch Installation, AMM TASK 29-34-21-400-801

(a) Do the Repair Confirmation at the end of this task.

(5) Do this check of the System A Pressure Module Relief Valve:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system A. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) Read the temperature at the System A Pressure Module Relief Valve. This is the task: Thermal Imaging Checks, AMM TASK 20-10-75-700-801.

1) If there is a significant temperature on the System A Pressure Module Relief Valve and it is similar to that of the connected return line, then replace the System A Pressure Module Relief Valve. These are the tasks:

NOTE: The relief valve is open and porting fluid back to the reservoir.

- Hydraulic Systems A and B Pressure Module Relief Valve Removal, AMM TASK 29-11-71-000-804
- Hydraulic Systems A and B Pressure Module Relief Valve - Installation, AMM TASK 29-11-71-400-804

a) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Repair Confirmation

(1) Do this check of the ELEC 2 LOW PRESSURE light, L1:

> **WARNING** MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS AND THE THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system A with the EMDP. This is the task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), AMM TASK 29-11-00-860-803.

(b) If the ELEC 2 LOW PRESSURE light goes OFF, then you corrected the problem. Do this step to complete the task:

1) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

(c) If the ELEC 2 LOW PRESSURE light does not go OFF, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

812. ENG 1 LOW PRESSURE Light On - Fault Isolation

A. Description

(1) The ENG 1 LOW PRESSURE Light comes ON when the EDP for the Hydraulic System A is below normal.

(2) This is normal operation of the ENG 1 LOW PRESSURE Light:

(a) When the System A EDP operates, the ENG 1 LOW PRESSURE Light is OFF.

(b) When the pressure of the System A EDP is below 1200 psi (8274 kPa), the ENG 1 LOW PRESSURE Light will come ON.

(c) When the System A EDP does not operate, the ENG 1 LOW PRESSURE Light will come ON.

B. Possible Causes

(1) EDP Case Drain Filter Module/Check Valve is installed in the wrong direction

(2) Hydraulic System A EDP

(3) ENG 1 LOW PRESSURE Light, L5

(4) Hydraulic System A EDP Low Pressure Switch, S796

(5) Pressure and supply hose not installed correctly

(6) Case drain hose not installed correctly at the strut fitting

(7) EDP SOV

(8) Wiring

(9) System A Pressure Module Relief Valve is open

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	12	C00318	INDICATOR MASTER DIM SECT 6

D. Related Data

(1) SSM 29-33-11

(2) SSM 33-18-35

(3) WDM 29-33-11

(4) WDM 33-18-35

E. Initial Evaluation

(1) Do this check of the ENG 1 LOW PRESSURE Light, L5.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the Hydraulic System A with the EDP, do this task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

1) If the ENG 1 LOW PRESSURE Light goes OFF, then there was an intermittent fault.

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

2) If the ENG 1 LOW PRESSURE Light is ON, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Make sure that the EDP case drain filter module is installed correctly with the arrow on the filter head pointing aft.

(a) Do this task: EDP Case Drain Filter Module Installation, AMM TASK 29-11-51-400-802.

(2) Do a check of the EDP case drain quick release hose where it attaches to the pump.

(a) Make sure that the case drain quick disconnect hose is hand tight and the lock tooth is engaged.

(b) If the lock tooth is correctly engaged, then continue.

(3) Do a check of the EDP pressure and supply quick disconnect hoses where they attach to the pump.

(a) Make sure that the supply hose is connected to the pump.

(b) Make sure that the pressure quick disconnect hose is hand tight and the lock tooth is engaged.

(c) If the lock tooth is correctly engaged, then continue.

(4) Do a check of the EDP case drain quick disconnect hose where it attaches to the pylon.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the case drain quick disconnect hose is hand tight and the lock tooth is engaged.

(b) If the lock tooth is correctly engaged, then continue.

(5) Do a check of the EDP pressure and supply quick disconnect hose where they attach to the pylon.

(a) Make sure that each quick disconnect hose is hand tight and each lock tooth is engaged.

(b) If each lock tooth is correctly engaged, then continue.

(6) Do this check of the Hydraulic System A EDP:

(a) Look at the hydraulic pressure indication on the Engine Display.

1) If the Hydraulic System A pressure is less than 2800 psi (19,305 kPa), do this task: System A Hydraulic Pressure Below 2800 psi Problem - Fault Isolation, 29-10 TASK 810.

2) If the Hydraulic System A pressure is greater than 2800 psi (19,305 kPa), then continue.

(7) Do the check of the EDP SOV.

(a) Do this task: EDP Supply Shutoff Valve Operational Test, AMM TASK 29-11-81-710-802.

1) If the valve does not operate correctly, replace the EDP SOV. These are the tasks:

- EDP Supply Shutoff Valve Removal, AMM TASK 29-11-81-000-801
- EDP Supply Shutoff Valve Installation, AMM TASK 29-11-81-400-801.

2) If the valve operates correctly, then continue.

(8) Replace the ENG 1 LOW PRESSURE Light, L5 on the Hydraulics Control Module, P5-8 (WDM 29-33-11).

(a) Do the Repair Confirmation at the end of this task.

(9) Do this voltage check.

(a) Disconnect connector D2684 from the Hydraulic System A EDP Low Pressure Switch, S796.

(b) Do a voltage check as follows (WDM 29-33-11).

SYS "A" ENG 1 PUMP LOW PRESS SW, S796 D2684	**SYS "A" ENG 1 PUMP LOW PRESS SW, S796 D2684**	**VOLTAGE**
pin 3	pin 2 (GND)	28V DC

1) If there is not 28V DC at pin 3 or ground at pin 2 of connector D2684, then do these steps:

a) Repair the wiring.

b) Re-connect connector D2684.

c) Do the Repair Confirmation at the end of this task.

2) If there is 28V DC at pin 3 of connector D2684, then continue.

(10) Replace the Hydraulic System A EDP Low Pressure Switch, S796. These are the tasks:

- Engine Driven Pump (EDP) Low Pressure Switch Removal, AMM TASK 29-34-11-000-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

• Engine Driven Pump (EDP) Low Pressure Switch Installation, AMM TASK 29-34-11-400-801.

(a) Do the Repair Confirmation at the end of this task.

(11) Do this check of the System A Pressure Module Relief Valve:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system A. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) Read the temperature at the System A Pressure Module Relief Valve. This is the task: Thermal Imaging Checks, AMM TASK 20-10-75-700-801.

1) If there is a significant temperature on the System A Pressure Module Relief Valve and it is similar to that of the connected return line, then replace the System A Pressure Module Relief Valve. These are the tasks:

NOTE: The relief valve is open and porting fluid back to the reservoir.

• Hydraulic Systems A and B Pressure Module Relief Valve Removal, AMM TASK 29-11-71-000-804

• Hydraulic Systems A and B Pressure Module Relief Valve - Installation, AMM TASK 29-11-71-400-804

a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the ENG 1 LOW PRESSURE Light, L5.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the Hydraulic System A with the EDP, do this task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

1) If the ENG 1 LOW PRESSURE Light goes OFF, then you corrected the fault.

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

2) If the ENG 1 LOW PRESSURE Light is ON, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

813. ENG 2 LOW PRESSURE Light On - Fault Isolation

A. Description

(1) The ENG 2 LOW PRESSURE Light comes ON when the EDP for the Hydraulic System B is below normal.

(2) This is normal operation of the ENG 2 LOW PRESSURE Light:

(a) When the System B EDP operates, the ENG 2 LOW PRESSURE Light is OFF.

(b) When the pressure of the System B EDP is below 1200 psi (8274 kPa), the ENG 2 LOW PRESSURE Light will come ON.

(c) When the System B EDP does not operate, the ENG 2 LOW PRESSURE Light will come ON.

B. Possible Causes

(1) EDP Case Drain Filter Module/Check Valve is installed in the wrong direction

(2) Hydraulic System B EDP

(3) ENG 2 LOW PRESSURE Light, L6

(4) Hydraulic System B EDP Low Pressure Switch, S797

(5) Pressure and supply hose not installed correctly

(6) Case drain hose not installed correctly at the strut fitting

(7) EDP SOV

(8) Wiring

(9) System B Pressure Module Relief Valve is open

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	11	C00317	INDICATOR MASTER DIM SECT 5

D. Related Data

(1) SSM 29-33-11

(2) SSM 33-18-35

(3) WDM 29-33-11

(4) WDM 33-18-35

E. Initial Evaluation

(1) Do this check of the ENG 2 LOW PRESSURE Light, L6:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the Hydraulic System B with the EDP. This is the task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If the ENG 2 LOW PRESSURE Light goes OFF, then there was an intermittent fault.
   a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.
2) If the ENG 2 LOW PRESSURE Light is ON, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Make sure that the EDP case drain filter module is installed correctly with the arrow on the filter head pointing aft.
   (a) Do this task: EDP Case Drain Filter Module Installation, AMM TASK 29-11-51-400-802.

(2) Do a check of the EDP case drain quick disconnect hose where it attaches to the pump.
   (a) Make sure that the case drain quick disconnect hose is hand tight and the lock tooth is engaged.
   (b) If the lock tooth is correctly engaged, then continue.

(3) Do a check of the EDP pressure and supply quick disconnect hoses where they attach to the pump.
   (a) Make sure that the supply hose is connected to the pump.
   (b) Make sure that the pressure quick disconnect hose is hand tight and the lock tooth is engaged.
   (c) If the lock tooth is correctly engaged, then continue.

(4) Do a check of the of EDP case drain quick disconnect hose where it attaches to the pylon.
   (a) Make sure that the case drain quick disconnect hose is hand tight and the lock tooth is engaged.
   (b) If the lock tooth is correctly engaged, then continue.

(5) Do a check of the of EDP pressure and supply quick disconnect hoses where they attach to the pylon.
   (a) Make sure that each quick disconnect hose is hand tight and each lock tooth is engaged.
   (b) If each lock tooth is correctly engaged, then continue.

(6) Do this check of the Hydraulic System B EDP:
   (a) Look at the hydraulic pressure indication on the Engine Display.
      1) If the Hydraulic System B pressure is less than 2800 psi (19,305 kPa), do this task: System B Hydraulic Pressure Below 2800 psi Problem - Fault Isolation, 29-10 TASK 811.
      2) If the Hydraulic System B pressure is greater than 2800 psi (19,305 kPa), then continue.

(7) Do the check of the EDP SOV.
   (a) Do this task: EDP Supply Shutoff Valve Operational Test, AMM TASK 29-11-81-710-802.
      1) If the valve does not operate correctly, replace the EDP SOV. These are the tasks:
         • EDP Supply Shutoff Valve Removal, AMM TASK 29-11-81-000-801
         • EDP Supply Shutoff Valve Installation, AMM TASK 29-11-81-400-801.
      2) If the valve operates correctly, then continue.

(8) Replace the ENG 2 LOW PRESSURE Light, L6 on the Hydraulics Control Module, P5-8 (WDM 29-33-11).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do the Repair Confirmation at the end of this task.

(9) Do this voltage check.

(a) Disconnect connector D2686 from the Hydraulic System B EDP Low Pressure Switch, S797.

(b) Do a voltage check as follows (WDM 29-33-11).

SYS "B" ENG 2 PUMP LOW PRESS SW, S797 D2686		SYS "B" ENG 2 PUMP LOW PRESS SW, S797 D2686	VOLTAGE
pin 3	. . . . .	pin 2 (GND)	28V DC

1) If there is not 28V DC at pin 3 or ground at pin 2 of connector D2686, then do these steps:

a) Repair the wiring.

b) Re-connect connector D2686.

c) Do the Repair Confirmation at the end of this task.

2) If there is 28V DC at pin 3 of connector D2686, then continue.

(10) Replace the Hydraulic System B EDP Low Pressure Switch, S797. These are the tasks:

- Engine Driven Pump (EDP) Low Pressure Switch Removal, AMM TASK 29-34-11-000-801
- Engine Driven Pump (EDP) Low Pressure Switch Installation, AMM TASK 29-34-11-400-801.

(a) Do the Repair Confirmation at the end of this task.

(11) Do this check of the System B Pressure Module Relief Valve:

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the hydraulic system B. This is the task: Hydraulic System A or B Pressurization, AMM TASK 29-11-00-860-801.

(b) Read the temperature at the System B Pressure Module Relief Valve. This is the task: Thermal Imaging Checks, AMM TASK 20-10-75-700-801.

1) If there is a significant temperature on the System B Pressure Module Relief Valve and it is similar to that of the connected return line, then replace the System B Pressure Module Relief Valve. This is the task:

NOTE: The relief valve is open and porting fluid back to the reservoir.

- Hydraulic Systems A and B Pressure Module Relief Valve Removal, AMM TASK 29-11-71-000-804
- Hydraulic Systems A and B Pressure Module Relief Valve - Installation, AMM TASK 29-11-71-400-804

a) Do the Repair Confirmation at the end of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Repair Confirmation

(1) Do this check of the ENG 2 LOW PRESSURE Light, L6:

> **WARNING** MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the Hydraulic System B with the EDP, do this task: Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), AMM TASK 29-11-00-860-804.

1) If the ENG 2 LOW PRESSURE Light goes OFF, then you corrected the fault.

a) Do this task: Hydraulic System A or B Power Removal, AMM TASK 29-11-00-860-805.

2) If the ENG 2 LOW PRESSURE Light is ON, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

814. STANDBY HYD LOW PRESSURE Light On - Fault Isolation

A. Description

(1) The STANDBY HYD LOW PRESSURE light comes on when the standby electric motor-driven pump (EMDP) is below normal.

(2) This is normal operation of the STANDBY HYD LOW PRESSURE light:

(a) When the standby hydraulic EMDP operates, the STANDBY HYD LOW PRESSURE light is off.

(b) When the pressure of the standby hydraulic system EMDP is below 1300 psi, the STANDBY HYD LOW PRESSURE light will come on.

(c) When the standby hydraulic system EMDP not operates, the STANDBY HYD LOW PRESSURE light will come on.

B. Possible Causes

(1) Standby electric motor-driven pump (EMDP)

(2) STANDBY HYD LOW PRESSURE light, L2

(3) Standby hydraulic system low pressure switch, S143

(4) Wiring problem

(5) Reservoir tubing, and quick disconnects

NOTE: Verify the that standby hydraulic system tubing and quick disconnects are tight with no leakage. Check surfaces for cracks and bends that may prevent system operation.

C. Circuit Breaker

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	14	C00316	INDICATOR MASTER DIM SECT 4

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Related Data

(1) (SSM 29-35-11)

(2) (SSM 33-18-31)

(3) (WDM 29-35-11)

(4) (WDM 33-18-31)

E. Initial Evaluation

(1) Do this check of the STANDBY HYD LOW PRESSURE light, L2:

> **WARNING** MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the standby hydraulic system. To pressurize it, do this task: Standby Hydraulic System Pressurization, AMM TASK 29-21-00-000-801.

(b) If the STANDBY HYD LOW PRESSURE light is on, then do these steps:

1) If the STANDBY HYD LOW QUANTITY light is also on, then, do this task: STANDBY HYD LOW QUANTITY Light On - Fault Isolation, 29-30 TASK 809.

2) If the STANDBY HYD LOW QUANTITY light is not on, then do the Fault Isolation Procedure below.

(c) If the STANDBY HYD LOW PRESSURE light goes off, then there was an intermittent fault. Do this step to complete the task:

1) Do this task: Standby Hydraulic System Power Removal, AMM TASK 29-21-00-000-802.

F. Fault Isolation Procedure

(1) Do a check of the standby electric motor-driven pump (EMDP):

(a) Do this task: Operational Test of the Standby Hydraulic System, AMM TASK 29-21-00-700-801.

(b) If the test is not satisfactory, then do these steps:

1) Replace the standby EMDP. These are the tasks:

- Standby Hydraulic System Electric Motor-Driven Pump (EMDP) Removal, AMM TASK 29-21-21-000-801
- Standby Hydraulic System Electric Motor-Driven Pump (EMDP) Installation, AMM TASK 29-21-21-400-801

2) Do the Repair Confirmation at the end of this task.

(c) If the test is satisfactory, then continue.

(2) Replace the STANDBY HYD LOW PRESSURE light, L2 on the flight control panel, P5-3 (WDM 29-35-11).

(a) Do the Repair Confirmation at the end of this task.

(b) If the Repair Confirmation is not satisfactory, then continue.

(3) Do this wiring check:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Disconnect connector D824 from the standby hydraulic system low pressure switch, S143.

(b) Do a check for 28v DC from pin 3 to pin 2 (ground) of connector D824 (WDM 29-35-11).

(c) If there is not 28v DC at pin 3 or ground at pin 2 of connector D824, then do these steps:

1) Repair the wiring (WDM 29-35-11).

2) Re-connect connector D824.

3) Do the Repair Confirmation at the end of this task.

(d) If there is 28v DC at pin 3 of connector D824, then continue.

(4) Replace the standby hydraulic system low pressure switch, S143. These are the tasks:

- Standby System EMDP Low Pressure Switch Removal, AMM TASK 29-34-31-000-801
- Standby System EMDP Low Pressure Switch Installation, AMM TASK 29-34-31-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do this check of the STANDBY HYD LOW PRESSURE light, L2:

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize the standby hydraulic system. To pressurize it, do this task: Standby Hydraulic System Pressurization, AMM TASK 29-21-00-000-801.

(b) If the STANDBY HYD LOW PRESSURE light goes off, then you corrected the fault. Do this step to complete the task:

1) Do this task: Standby Hydraulic System Power Removal, AMM TASK 29-21-00-000-802.

END OF TASK

815. Hydraulic Fluid Quantity Problem - Fault Isolation

A. Description

(1) The hydraulic fluid quantity transmitter/indicators show the hydraulic fluid level in the hydraulic system A and B reservoirs.

(2) The indicator shows the fluid quantity on the mechanical gage on the reservoir in the main landing gear wheel well.

(3) The transmitter sends signal to the display electronic units (DEUs). The fluid quantity shows on the engine display.

B. Possible Causes

(1) Hydraulic fluid servicing

(2) Failure from the CDU BITE procedure

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Related Data

(1) (SSM 29-31-11)

(2) (WDM 29-31-11)

D. Fault Isolation Procedure

(1) Fill the applicable hydraulic reservoir. To fill it, do this task: Hydraulic Reservoir Servicing, AMM TASK 12-12-00-610-801.

(a) If the applicable hydraulic fluid quantity indicator shows F (100%), then you corrected the fault.

(b) If the applicable hydraulic fluid quantity indicator does not show F (100%), then continue.

(2) Do this task: CDS BITE Procedure, 31-62 TASK 801.

(a) If the BITE test fails, then do the fault isolation task for the applicable maintenance message.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details